Summary of Four-Month Internship Technical Work

Theory of Photonic Crystals and Disorder Analysis for A Line Defect Waveguide	*Learning Outcomes*	**(1)** To conduct the theoretical study of photonic crystal structures and waveguide designs using Chs. 1-8 of the books *“Photonic Crystals: Molding the Flow of Light”* **(2)** To study the problem of fabrication disorder modelling on Photonic Crystals and extracting methodology and key findings from papers on the problem. **(3)** To use the MBP (for PBSs) & MEEP (finite-difference time-domain (FDTD) based simulation tools in exploring the problem and for the reproduction of papers results **(4)** To study the PWE numerical method for obtaining the Photonic Band Structure of 2D Triangular Lattice of Air Holes Embedded in a Dielectric Background
	Research Questions Explored	-
	Practical Skills	**(1)** Summarizing research papers and extracting methodology and key findings **(2)** Familiarizing with using simulation tools, MBP and MEEP **(3)** Exploring numerical methods underlying the simulation tools **(4)** Building a unified framework across chapters and proposing deeper exploration questions and simulation setups
	Main Books and Papers Studied	**(1)** Joannopoulos, John D, et al. *Photonic Crystals Molding the Flow of Light - Second Edition*. Princeton University Press, 2011. **(2)** Savona, Vincenzo. “Electromagnetic Modes of a Disordered Photonic Crystal.” *Physical Review B*, vol. 83, no. 8, 11 Feb. 2011, https://doi.org/10.1103/physrevb.83.085301. Accessed 17 May 2026. **(3)** Saleh and Malvin Carl Teich, *Fundamentals of Photonics*.
	Link to report	
Gold Nanoparticles Localized Surface Plasmon Resonance (LSPR) and Fiber Sensing Applications	*Learning Outcomes*	**(1)** To study the LSPR phenomenon in gold nanoparticles by exploring the effects of the surrounding medium’s refractive index and the size of the gold NPs on the observed colors and resonance wavelength. **(2)** To explore the use of optical fibers and Au NPs in sensing applications and characterization. **(3)** To investigate the use of optical fibers in a classification problem by categorizing the region over which the fiber tip is positioned among various Au NPs or a clean substrate
	Research Questions Explored	**(1)** Use and characterization of a fiber with Au NPs deposited on its tip as a distance sensor. **(2)** Use of an ML model for the classification of the regions across the wafer to either be clean substrate or containg Au NPs.
	Practical Skills	**(1)** Mode FDTD simulation using Ansys Lumerical for single-mode fiber design **(2)** Familiarity with using the UV-Vis spectrophotometer **(3)** ML model training, validation, and testing **(4)** Applying theoretical knowledge to real world applications
	Main Books and Papers Studied	**(1)** M. A. Fakhri *et al.*, “A gold nanoparticles coated unclad single mode fiber-optic sensor based on localized surface plasmon resonance,” *Scientific Reports*, vol. 13, no. 1, Apr. 2023, doi: https://doi.org/10.1038/s41598-023-32852-6. **(2)** A. G. Al-Rubaye, A. Nabok, and A. Tsargorodska, “Spectroscopic ellipsometry study of gold nanostructures for LSPR bio-sensing applications,” *Sensing and Bio-Sensing Research*, vol. 12, pp. 30–35, Feb. 2017, doi: https://doi.org/10.1016/j.sbsr.2016.11.006. **(3)** Saleh and Malvin Carl Teich, *Fundamentals of Photonics*.
	Link to report	

The Fabry-Pèrot Interferometer Experiment	*Learning Outcomes*	**(1)** To study the Fabry–Pérot interferometer and its fringe profile, examining the interplay between the wavelength, angle of incidence, and cavity length parameters. **(2)** To experimentally emulate the Fabry–Pérot interferometer using optical fibres and record OSA output. To analyse experimental results and compare them against theoretical expectations **(3)** To analyse experimental results and compare them against theoretical expectations
	Research Questions Explored	-
	Practical Skills	**(1)** Usage of the optomechanical system for the alignment of the experimental fibres from the transmitter and receiver sides **(2)** Familiarity with using optical instruments
	Books Studied	Introduction to Optics by the Pedrotti's
	Link to report	
Rectangular Waveguide Theoretical Background, Design, and Simulation	*Learning Outcomes*	**(1)** To study the theoretical principles of 2D metallic and dielectric waveguides, including mode formation and wavevector discretization. **(2)** To analyze field confinement, polarization, and the impact of waveguide dimensions on effective refractive index. **(3)** To explore the differences between metallic and dielectric waveguides. **(4)** To use simulations to visualize mode patterns and polarization effects.
	Research Questions Explored	-
	Practical Skills	Mode FDTD simulation using Ansys Lumerical
	Books Studied	**(1)** Saleh and Malvin Carl Teich, *Fundamentals of Photonics*. **(2)** F. T. Ulaby and Umberto Ravaioli, *Fundamentals of Applied Electromagnetics* **(3)** D. K. Cheng, *Field and Wave Electromagnetics*. **(4)** D. Marcuse, *Theory of Dielectric Optical Waveguides*. 1991. **(5)** Lukas Chrostowski and M. Hochberg, *Silicon photonics design: from devices to systems*.
	Link to report	
The Mach-Zehnder Interferometer and Application	*Learning Outcomes*	**(1)** To study the theoretical principles of the Mach-Zehnder Interferometer (MZI) and derive its fringe/transmission profile and analyse non-idealise effects (polarization effect and non-idea Y-splitter) on the profile. **(2)** To define and study FSR and Phase change, as well as their effects on the fringe profile **(3)** To explore methods for phase and FSR changes detections, as well as derive the setup changes required for the minimal detectable change (optical Vernier effect, FFT, and cross-correlation) **(4)** To simulate, using MATLAB, the MZI fringe profile modulation on a directly acquired source spectrum **(5)** To simulate, using MATLAB, the changes on the fringe profile due to temperature changes and to compare results with the experimental measurements **(6)** To use the MZI in the application of approximating battery (soda can) temperature change, and derive FSR change and phase equations as a function of temperature under the condition of asymmetric heating **(7)** To validate theoretical plots using experimental results and draw conclusions
	Research Objectives Explored	**(1)** To Mathematically attempt decoupling and quantifying FSR Change and phase change from a recorded spectrum measurement relative to a reference spectrum **(2)** To obtain a generalized relationship between temperature change and detected changes in the MZI spectrum (FSR change) using the explored detection methods
	Practical Skills	**(1)** Fiber Splicing **(2)** Familiarity with using optical instruments **(3)** Applying theoretical knowledge to real world applications
	Books Studied	**(1)** Saleh and Malvin Carl Teich, *Fundamentals of Photonics*. **(2)** Lukas Chrostowski and M. Hochberg, *Silicon photonics design: from devices to systems*.
	Link to report	

Appendices

Appendix A – Internship Report: Theory of Photonic Crystals and Disorder Analysis for A Line Defect Waveguide

Report Title:

Theory of Photonic Crystals and Disorder Analysis for A Line Defect Waveguide

Report Objectives:

- To conduct the theoretical study photonic crystal structures and waveguide designs using Chs. 1-8 of the books *"Photonic Crystals: Molding the Flow of Light"*
- To study the problem of fabrication disorder modelling on Photonic Crystals and their modelling, extracting methodology and key findings from papers on the problem.
- To use the MBP (for PBSs) & MEEP (finite-difference time-domain (FDTD) based simulation tools in exploring the problem and reproduction of papers results
- To study the PWE numerical method for obtaining the Photonic Band Structure of 2D Triangular Lattice of Air Holes Embedded in a Dielectric Background

Shayma Alteneiji | 100063072

Internship Supervisor:

Prof. Jaime Viegas

Table of Contents

Introduction

In this report, a paper on the effect of disorder on PhC structures parameters and Bloch modes are studied and investigated, and the partial of its reproduced. For the purpose of easing the understanding of the paper, the fundamentals of photonic crystals were studied through the book *"Photonic Crystals: Modelling the Flow of Light"*. Table 1 attempts building a unified framework of the structures introduced throughout the book, summarizing the difference in (kx, ky, kz), confinement mechanisms, leakage channels, and project band structure meaning for each structure. Table 2 proposes research questions and corresponding simulation setups on MBP and MEEP to deepen the understanding of each PhC structure. The next section presents the summary of the investigated paper, *"Electromagnetic Modes of a Disordered Photonic Crystal"*.

Table 1. Summarized Framework of Various Photonic Crystal Structures

<table>
<tr><th rowspan="2">Structure</th><th rowspan="2">kx</th><th rowspan="2">ky</th><th rowspan="2">kz</th><th colspan="3">Confinement Mechanism in</th><th rowspan="2">Leakage</th><th rowspan="2">Projection meaning</th></tr>
<tr><th>x</th><th>y</th><th>z</th></tr>
<tr><td>Glass Slab

(extends in x and y, truncated in z)</td><td colspan="2">Conserved by CTS</td><td>Not conserved. Becomes discretized, donated by band index (# nodes in the z direction)</td><td colspan="2">No Confinement</td><td>Index guiding or TIR.

Confinement strength characterized heuristically by distance from the light cone.</td><td>Surrounding medium</td><td>Beginning with a glass slab that extends infinitely in the z-direction. The corresponding dispersion curve or “PBS” would be a light cone of slope c/n_glass, enclosing the continuum of allowed states within the slab. However, when the slab becomes truncated in z, symmetry is broken, kz in no longer conserved nor continuous but becomes discretized, and a function of the slab thickness. Hence the PBS, becomes the sampled form of the original PBS for the allowed values of kzs. In PBS terms, kz becomes referred to as the band index, indicating the number of nodes for the modes in z. Moreover, a light cone is also added, indicating the availability of the radiative states in air or the surrounding medium that the modes of the slab can couple to, only now that the structure is finite in z.</td></tr>
<tr><td>Point Defect in 2D PhC</td><td colspan="2">Not Conserved.</td><td>Conserved by CTS</td><td colspan="2">By bandgap. Strength characterized by the imaginary part of the wavevector, $\boldsymbol{\kappa}$</td><td rowspan="2">NC. The propagating wave can extend in the z direction, however, kz is set to zero to allow for the separation of polarizations, limiting it to in-plane propagation only.</td><td rowspan="2">No leakage</td><td>When a point defect is added, periodicity is broken in both directions, and the two Bloch wavevectors, kx and ky, become no longer conserved. Depending on the change of the defect parameter, reducing or increases radius of the hole or rod, or decreasing or increasing its dielectric constant, a localized discrete frequency mode could be pulled up or pulled down, respectively, from the continuum of extended modes. This is plotted as horizontal line at the defect mode frequency, as its k is not defined. Moreover, the DOS inside the bandgap is zero across the band gap but is a delta function at the frequency of the defect mode.</td></tr>
<tr><td>Line Defect in 2D PhC</td><td>Not Conserved.</td><td>Conserved by DTS. Restricted within the Brillouin zone. Group velocity, $\partial\omega/\partial k_y$ is defined.</td><td>Conserved by CTS</td><td>By bandgap. Strength characterized by the imaginary part of the wavevector, $\boldsymbol{\kappa}$</td><td>No Confinement</td><td>When removing or changing the parameters of a horizontal row of a 2D PhC for example, periodicity and hence DTS is broken in the y direction, and ky is no longer conserved. The waveguide PBS is then defined as the projected PBS of the 2D PhC plotted as a function of the conserved Bloch wavevector (kx) for all the values of the unlabelled and non-conserved wavevector (ky). A given mode of the waveguide is confined and does not couple to the extended modes of the PhC, is as follows. A given mode defined by (ω_o, k_y) is guided along the defect line if no k_x places the frequency ω_o on one of the bands of the 2D PhC surface, $\omega_o = \omega_n(k_x, k_y)$, such that it is not part of the projected bulk of crystal extended modes, in the same way, the mode is said to be within the projected bandgap of the unperturbed 2D PhC. The band gap is the physical mechanism ensuring the confinement of the mode transversely to the waveguide line.</td></tr>
<tr><td>2D PhC Slab</td><td colspan="2">Conserved by DTS. Restricted within the Brillouin zone.</td><td rowspan="3">Not conserved. Becomes discretized, can be donated by band index (# nodes in the z direction), however, usually the slab is made thick enough to support only the fundamental mode in z direction.</td><td colspan="2">No Confinement</td><td rowspan="3">Index guiding or TIR.

Strength characterized heuristically by distance from the light cone. Index guiding or TIR.</td><td rowspan="3">Surrounding medium</td><td>Begging with a 2D photonic infinite in the z direction, the corresponding PBS is a 3D surface with kz being continuous, not restricted to Brillouin zone. Truncating the structure in the z direction (slab), produces the projected PBS, contours of the 3D surface PBS for the new descried kzs. The density of the vertical modes is a function of the slab thickness, the larger the slab the greater the density, resulting in no band gap for the new structure in 3D (out of plane propagation allowed), and most importantly separation of polarization is not possible. However, when the slab thickness is made small, a fraction of the effective wavelength, the approximation that the modes are mostly polarized, TE or TM, holds and the band gap is restored. Moreover, as with the glass slab case, a light cone is added, indicative of the availability of radiative modes that slab guided modes can couple to.</td></tr>
<tr><td>Line Defect in 2D PhC Slab</td><td rowspan="2">Conserved by DTS. Restricted within the Brillouin zone. Group velocity, $\partial\omega/\partial k_y$ is defined.</td><td rowspan="2">Not conserved. Becomes discretized, donated by band index (# nodes in the y direction), e.g. E(o, n=2)</td><td>By bandgap. Strength characterized by the imaginary part of the wavevector, $\boldsymbol{\kappa}$</td><td>No Confinement</td><td>The difference between the PBS of 2D PhC slab and a line defect 2D PhC slab is that (1) the defect introduces a plane of mirror symmetry, y=0, allowing us to classify modes as spatially odd (TM-like) or even (TE-like with respect to the plane y = 0. Moreover, mode order or number of nodes in the y direction is donated by band index, as in E(o, n=2), read as TE-like or z-even, odd with respect to y with 2 nodes in y. (2) by defecting a line of the PhC, we effectively pull guided waveguide modes into the bandgap, localized in the transverse direction to the waveguide line.</td></tr>
<tr><td>Point Defect in a 3D Waveguide (periodic in x)</td><td>No Confinement</td><td>Index guiding or TIR. Strength characterized heuristically by distance from the light cone.</td><td>The PBS of a 3D waveguide is similar to that of a slab, only with structure being finite in y, ky becomes discretised and different values corresponds t o different mode orders or # of nodes along y. The only conserved Bloch wavevector is kx, against which the projected PBS is plotted against, bands plotted forms guided modes propagating along the waveguide and with no modes within the bandgap, since there is no defect. When a defect is added (distance between two holes or radius size), with the only kx Bloch wave vector becoming no longer conserved, this disorder causes scattering and the mixing of the guided states of the waveguide with the leaky modes within the light cone. The mode is localized to the defect but is leaky and coupled to radiative states.</td></tr>
</table>

Table 2. Research Explorations and Simulation Setups for Various Photonic Crystal Structures

Structure	Topic of Exploration	Studied quantities	Expected results and/or learning outcome	Simulation setup
Glass Slab	**Dispersion Curve and Slab Length**	The density of the produced families of vertical modes per frequency	As the slab length increases, the density of allowed states per frequency increases, filling more space (approaching the continuum of states limit for infinite thickness in z)	-Obtain the dispersion curves of a glass slab, extended in x and y, and truncated in z, for different thickness
	TIR or Index Guiding Confinement in the z-direction	For a fixed frequency, how the effective refractive index, group velocity, and concentrated energy density in the slab varies for different $k_{\parallel}$	As $k_{\parallel}$ increases, the corresponding mode band index decreases (from the $(\omega n/c)^2$ equation, the associated mode order reduces, the line is straighter, the slope is therefore higher indicating stronger confinement, hence more energy concentrated in the slab, and a greater group velocity	-For a fixed slab length and corresponding dispersion curve, plot the mode profiles with the same frequency but at various $k_{\parallel}$ values. -Note down the change in the overlap between the electric field and the surrounding as $k_{\parallel}$ is varied. -Calculate n_{eff} and $\partial\omega/\partial k_{\parallel}$ at each $(\omega, k_{\parallel})$ point. -Obtain the plot for the electric field energy density
	Intrinsic Radiation Losses	Study the variation of the quality factor (Q) parameter for different points (modes) within the light cone	*By plotting the DOS, the leakage rate (γ)* *Relation to Coupling* *Overlap between guided modes and leaky modes*	-
1D PhC	**Limit for Transverse Confinement**	Study the mode profiles as $k_{\parallel}$ is increased, reaching the limit where each plane supports an independent mode, each are very weakly coupled with its neighbour	By deriving the equation for $(\omega n/c)^2$ with respect to k_z, it can be shown that as $k_{\parallel}$ increases, the derivative approaches zero, hence the flatter the band wrt k_z, the smaller the group velocity in the z direction; resulting in weak transverse transport of energy and weak coupling.	-The overlapping tail from the horizontal mode in the vertical direction -The PBS showing the dispersion plot extended for higher $k_{\parallel}$ and for each includes the bands for various k_z values within the Brillouin zone (bounded by $k_z = 0$ and $k_z = \pi/a$).
Point Defect in 2D PhC	**Localization Strength**	How the in-plane localization strength of the defect mode varies wrt its location within the waveguide	The further the point defect mode from either band, the greater the imaginary part of the wavevector, κ, and the smaller the decay length, $1/\kappa$	Vary the location of the defect mode across the band gap by varying the radius of the defect rod (equivalent to varying its dielectric contestant). Record the resulting location of the defect mode by plotting the PBS and obtain the corresponding mode profile. Qualitatively note the observations made.
Line Defect in 2D PhC	**Smoothness of the Transition of a Localized Mode to an Extended Mode as it Enters the Band of Extended Modes**	Relate the rate of the leakage of the previously localized mode to an extended mode to the DOS at a given studied point $(\omega, k_{\parallel})$	Expectations from the book is as follows, the transition for localized to extended mode is not discontinuous, but rather its energy will leak away into the continuum of states at some leakage rate γ. The density of states available at a given frequency is proportional to the leakage rate.	-For a given line defect waveguide in a 2D PhC, obtain the mode profiles for different points of the guided mode and as it enters the band of the extended states. Qualitatively note the observations made. -Obtain the DOS and energy density plot for each corresponding frequency point.
2D PhC Slab	**Photonic Band Structure and Slab Length**	Study how varying the slab thickness changes, with allowing (out of plane propagation) the resulting PBS for quasi-TE and TM polarizations in terms of: -The density of the modes or available states per frequency -Bandgap for each polarization	-Assuming a 2D PhC, infinite in z, then removing the no in-plane propagation restriction (k_z is non-zero), then the new PhC would no longer have a bandgap and separation of polarization is not possible. Instead, we use slabs of small thickness in z (exact thickness being non-arbitrary) which as in the case of the glass slab results on the production of families of vertical mode profiles, each labelled a corresponding band index, and we obtain TE-like and TM-like polarizations which suggests the field is only mostly polarized under the restricted thickness condition, and the gaps are restored, yet are said to be incomplete, because (1) the gaps are polarization specific, and (2) the air cone modes are available for all frequencies and are available as possible modes that the crystals modes can couple to under disorder for example. Moreover, the families per frequency become denser the thicker the slab is. The band gap is polarization dependent, large enough for the fundamental mode to be confined enough, yet not thick enough with many modes jamming and removing the band gap.	- Vary the 2D PhC slab thickness, and obtain the corresponding PBS
	Confinement in z (index guiding) of a Mode wrt to its Distance from the Light Cone	Mode band index (or distance from light cone) and confinement within the slab in the z direction, determined through computing energy density within the slab relative to the surrounding	The closer the guided band to the light cone, the higher the band index, i.e. mode order, meaning the higher the transverse variation of the profile and the lesser is the effective refractive index of the mode, resulting in a larger tail in the surrounding and hence weaker confinement.	-For a slab of given length, obtain the PBS and for a fixed $k_{\parallel}$, plot the mode profiles for increasing frequency values (moving closer to the light cone) -Note observations: For a point of fixed $k_{\parallel}$, increasing frequency further moves us into the light cone, with radiative modes existing, and the trend as we approach that point.
Line Defect in 2D PhC Slab	**Horizontal Mode Confinement (band gap) of a Mode vs. its Vertical Confinement (index guiding)** ***For a Slab of rod in air***	The use of line defect parameters (e.g. radius or dielectric constant) to control its location across the band gap, accordingly, its confinement horizontally and vertically	Shrinking the radiuses of a single row of rods would pull up a mode from the dielectric band, as we continue to shrink the radius, the higher is the guided mode pulled along the band gap. Since in the case of a slab, vertical confinement is through index guiding or TIR, requiring a higher refractive index medium surrounded by a lower refractive index medium, it is therefore impossible to guide a mode in air, in the PBS, this is seen by the mode being completely within the light cone, meaning at these points modes couple strongly to radiative modes in air and no guiding is possible. Reaching up to that level, modes moving closer to the light cone, is where is weaker confinement, (why ?, mode correspond to a higher band index and hence higher transverse variations?). On the other hand, horizontal confinement is easily deduced by the distance of the mode form either band.	-Vary the defect rod parameter, obtain the PBS and plot the modes for a fixed $k_{\parallel}$. -Obtain the energy density plots. -Note down vertical and horizontal confinement for the various plots.
	Index Guided Modes in All Direction (Below the Bulk Crystal Modes) ***only in the case of a slab of holes***	Qualitative confinement analysis in the horizontal and vertical direction by TIR and relevant parameters	Hypothesis: Band index of the index guided modes in all direction indicates the mode order, an indication of variations in x and/or y direction. The higher the band index, the higher its frequency, and the closer it is to the light cone horizontally, and hence the lower the confinement.	Observe how changing the parameters of the line defect (such as dielectric constant or length) within the slab of holes affects the positing of the resulting guided mode and the corresponding change in confinement horizontally and vertically.
Point Defect in a 3D Waveguide	**Intrinsic Radiation Losses**	How does the position of the defect mode along the band gap affects the resulting intrinsic radiational losses (through calculating Q), if related at all, and possible relation to DOS	With a 3D waveguide periodic in x, adding a defect results in no longer conserving the only k Block wavevector label, this therefore results in allowing scattering and mixing of the guided modes with the radiative modes of air. I hypothesis that the higher the frequency of the defect, the greater is the number of radiative states, and potentially the higher the losses, indicated by the computed DOS of crystal modes and within light cone.	-Vary the point defect characteristics or parameters to produce structure with different corresponding frequency. -Compute Q and DOS across the band gap region for each case.

Summary of a Paper on Disordered Photonic Crystals

Paper Title:

Electromagnetic Modes of a Disordered Photonic Crystal by Vincenzo Savona*

Approach and work:

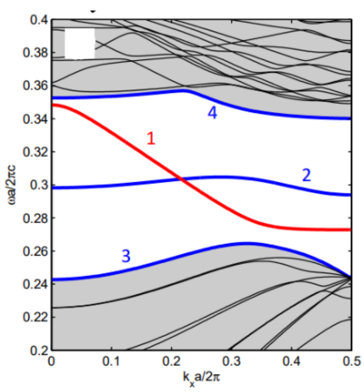
In the problem of modelling and characterizing the effects of disorder in PHCs, the use of first principles methods reliant on finite sampling of the real crystal would be intractable, for the need of a large supercell for a fairly accurate realization of the modelled disorder present, hence powerful CPU, memory, and long hours of simulation.

This paper addresses this problem by proposing simplifying the problem as follows. Based on the Guided Mode Expansion (GME) method, the eigenmodes of the perturbed PHC can expanded on the basis of the Bloch modes of the unperturbed photonic crystal, and for small perturbation or modelled disorder, the effected bands remain only in the vicinity of the guided mode considered, hence the sum can be truncated to including only a small subset of bands.

Moreover, disorder results in the mixing between guided modes and the continuum of radiative states within the light cone. The GME does not account for this radiation losses effect, instead the paper uses a perturbative approach. One of the considered structures in the application section is a W1 line defect waveguide. *Most of the* ***work and investigations*** carried out by the paper is summarized below:

- Obtained the PBS of the unperturbed PHC:

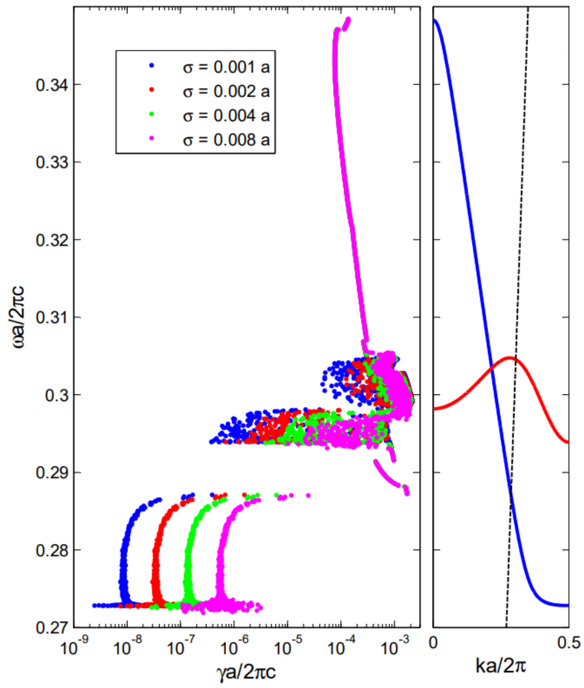
The structure is a slab of holes with a line defect using the GME method. The depth in the z is made thick enough only to support a single mode (the main guided band or spatially-even band), thereby explicitly selecting the mode's studied.



* Source [1]

- Estimate the radiation losses, separately, through a perturbative approach.

Specifically, by considering *"**overlap** integrals between modes and leaky modes of the effective slab. The continuous spectrum of leaky Bloch modes is thus accounted for by an irreversible decay of the approximately guided modes into the leaky modes of the homogeneous slab."* Also comparing the intrinsic and extrinsic radiation losses.



- ***Conclusion:***

(1) Below the light cone, the radiation loss rate is small and proportional to the disorder std

(2) Above the light cone, radiation loss rate increases significantly, with intrinsic radiation losses dominating and becoming independent of the disorder profile.

- Investigate the validity of their approximation method, through analysing the convergence of the eigenfrequency and the radiation loss rate approximation based on the number of bands (basis) included in the truncated sums given below, (μ and ν donates n).

$$\sum_{\nu} \mathcal{H}_{\mu\nu} c_{\nu} = \frac{\omega^2}{c^2} c_{\mu}$$

For each k, by diagonalization, we obtain the eigenfrequencies and eigenvectors (c_μ), which forms the coefficients of the expansion equation given below.

$$\mathbf{E}_{\beta}(\boldsymbol{r}) = \sum_{\mathbf{k}n} U_{\beta}(\mathbf{k}, n)\, \mathbf{E}_{\mathbf{k}n}(\boldsymbol{r})$$

To calculate radiation losses rate, the above equation is used in the calculation of overlap matrix equation, given below.

$$\mathcal{H}^{rad}_{\beta\mathbf{g}} = \frac{\omega_{\beta}^2}{c^2} \int d\mathbf{r} \epsilon(\mathbf{r}) \mathbf{E}^*_{\beta}(\mathbf{r}) \cdot \mathbf{E}^{rad}_{\mathbf{g}\lambda j}(\mathbf{r})$$

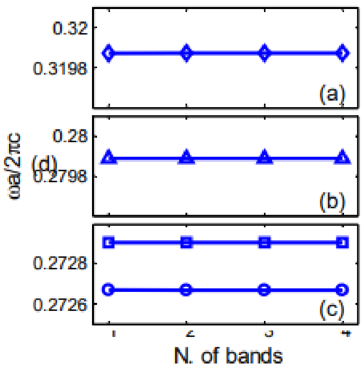
Accordingly, the radiation loss rate is computed as given below.

$$\Gamma_{\beta} = \pi \sum_{\mathbf{k},\mathbf{G}} \sum_{\lambda=\mathrm{TE},\mathrm{TM}} \sum_{j=1,3} \left|\mathcal{H}^{rad}_{\beta\mathbf{g}}\right|^2 \rho_j \left(\mathbf{g}; \frac{\omega_{\beta}^2}{c^2}\right)$$

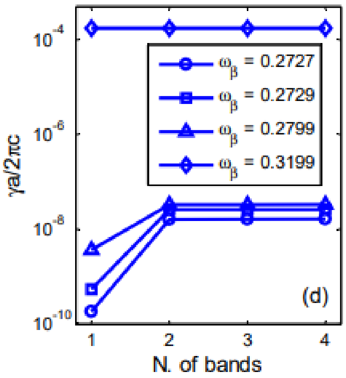
$$\gamma_{\beta} \simeq c^2 \Gamma_{\beta} / (2\omega_{\beta})$$

Conclusion:

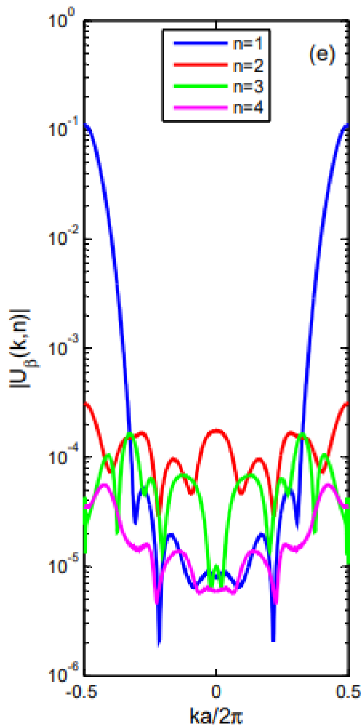
- The figure below shows the obtained eigenfrequencies converging only after including one band (basis).



As for estimation of the radiation loss rate, for the frequencies below the light, convergence occurs only after including two bands. The reason for this, as the author explains, is due to the disorder allowing the mixing of hybridization of the even band with the second spatially odd band. Resulting in the including it to fully capture the mode of the perturbed PHC.



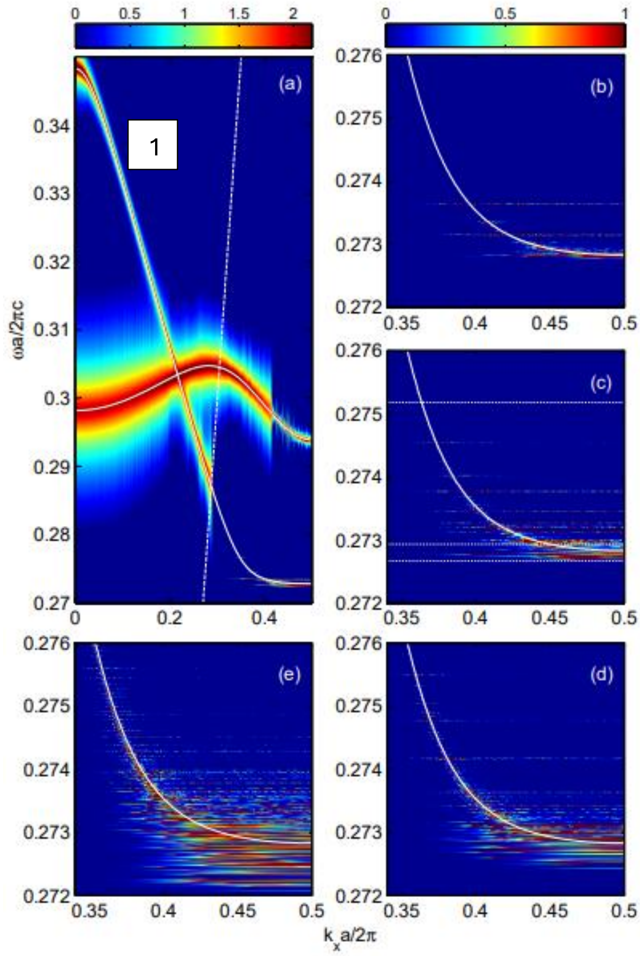
This is supported by the plot showing the coefficients of the expanded perturbed crystal mode across the k range, where the strongest contribution is by the second band (excluding the band edges). Note, however, that the approximation of the radiation loss rate when the frequency is above the light cone converges when including only one basis, this is because the intrinsic radiation losses is much strong (as was shown from the radiation loss rate plot), dominating the extrinsic radiation and the need to account for the hybridization effect



- Investigate of whether localization of modes is possible, and whether it is due to the occurrence of Anderson localization.

This was conducted through **(1)** analysing spectral properties of the system using the scalar Green function, equation and plot given below.

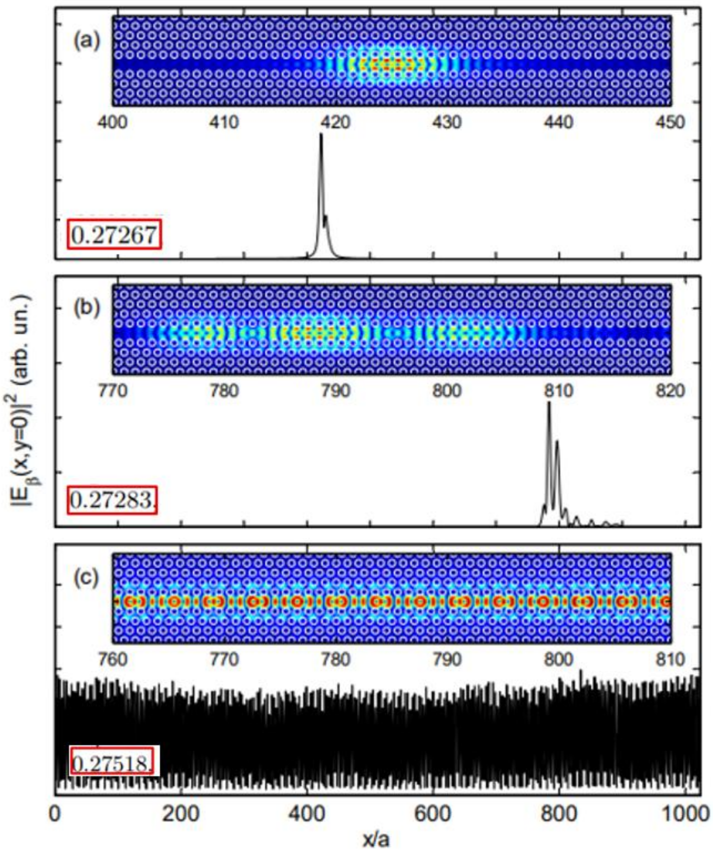
$$G_y(k,\omega) = \sum_{\beta} \frac{|\mathbf{H}_\beta(k,y)|^2}{\omega - \omega_\beta + i\gamma_\beta}$$



Conclusion (a):

- The spectral width along band 1 is much larger above the light cone, suggesting that intrinsic radiation losses are much stronger than the extrinsic radiation loses.
- As the disorder std is increased, the density of the disorder induced spatially localized modes at the lower edge of the band increases, suggesting the possibility of the occurrence of localization around this frequency range. This is investigated next.

(2) The plots for the field intensity vs. the length of the simulated structure is shown below for different frequencies for $\sigma = 0.002a$ (figure (c) of the Green function plot). The localization is strongest for the lowest frequency below the band edge. The third plot shows a delocalized mode, with the frequency being above the band edge. Next, we study attenuation length vs. localization length to conclude whether the localization observed is due to the occurrence of Anderson localization.



(3) Disorder causes multiple scattering of the wave which results in the damping of the propagating wave. This loss effect is characterized by the attenuation length, and the measure of the localization of the mode becomes a measure of its attenuation length.

In the occurrence of Anderson localization, however, localization length becomes no longer tied to the attenuation length. This occurs when the damping effect due to scattering is to be small along the plane of propagation, allowing the interference of the scattered waves before being damped by the effect of radiation losses.

The attenuation of the transmission, T, is proportional to the inverse exponential of x divided by the length parameter (L_{ω}), as given below.

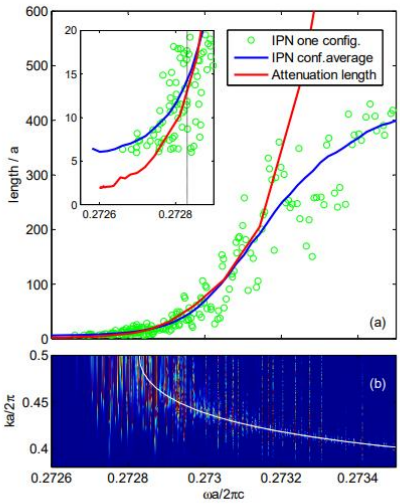
$$T(x, x', \omega) \propto \exp(-|x' - x|/L_{\omega})$$

The localization length is characterized by the inverse participation number (IPN), given below.

$$I_{\beta} = \left[\int dx |\mathbf{H}_{\beta}(x, y = 0)|^4 \right]^{-1}$$

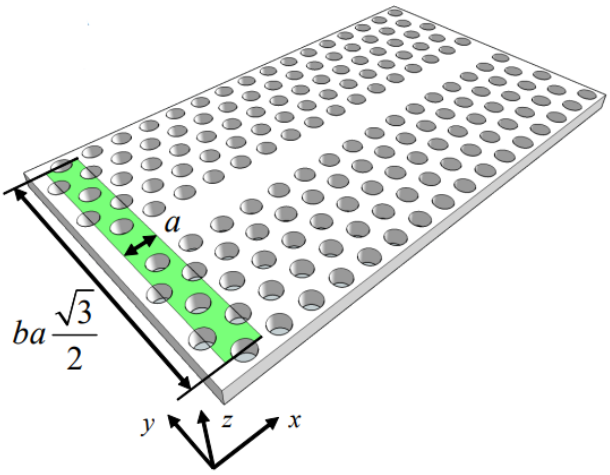
Conclusion (b):

- Based on the obtained plot, it is concluded that Anderson Localization occurs for the range of frequencies 0.2726 to 0.2728, where attenuation length ceases to follow the IPN length.
- The plotted parameter is L_{ω}, (equivalent to $1/\alpha$), for the range of frequencies 0.2726 to 0.2728, the decreasing parameter over the range indicates that the attenuation length is greater than localization length, and the condition for AL is satisfied.



Detailed Paper Methodology for the W1 Line Defect Waveguide ***(Partial)***:

1. Define the unperturbed photonic crystal structure:
   - A rectangular supercell is used of **width (a)** along x and **height (b. a. sqrt (3)/2)** along y
   - The W1 structures is of length **L = 1024a** (9216 holes) and with **b = 10**, this corresponds to 1024-k points in the interval [-π/a, π/a]
   - **d/a = 0.5**, with *d* depth in z direction
   - **r/a = 0.3**
   - Air-bridge structure with dielectric constant $\boldsymbol{\epsilon_{1,3}}$ **= 1 (cladding layers)**, and $\boldsymbol{\epsilon_2}$ = **12 for the slab material.** Hence, effective-slab dielectric constant $\bar{\boldsymbol{\epsilon}}_{\mathbf{2}}$= **8.77**



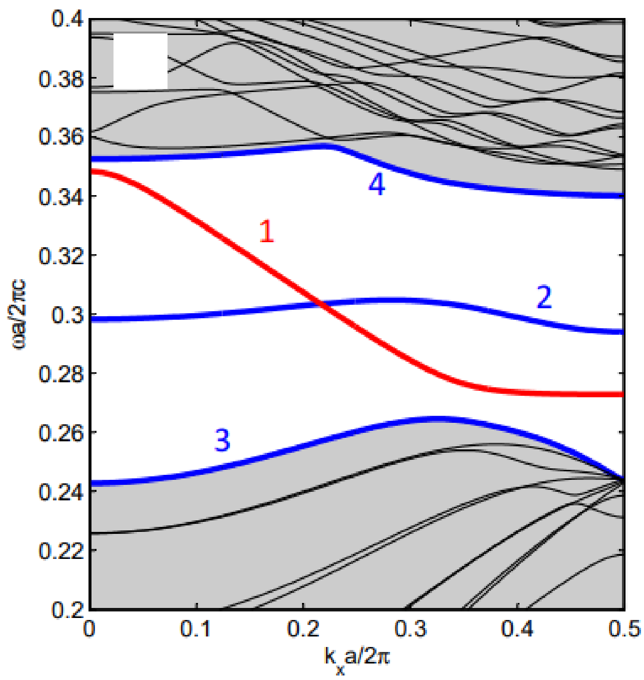
2. Obtain the photonic band structure of the W1 line defect waveguide using **guided-mode expansion method:**

Sources used: [1], [2]

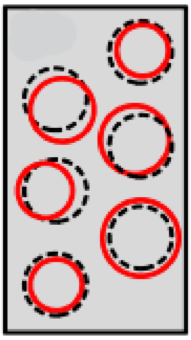
$$\mathbf{E}_{\mathbf{k}n}(\boldsymbol{r}) = \sum_{G,\alpha} c_n(\mathbf{k}+\boldsymbol{G},\alpha)\, \mathbf{E}^{(g)}_{\mathbf{k}+\boldsymbol{G},\alpha}(\boldsymbol{r})$$

- Ho's method [2] is used to obtain for the Fourier transform of the dielectric profile
- The expansion is truncated at $\mathbf{G}_{\mathbf{max}}$ **= 3**, corresponding to a basis of **229 ($\boldsymbol{\alpha}$)** guided mode
- Assumptions made (1): modes are restricted so as to having $\mathbf{k_y}$ **= 0**, (dispersion in the y direction is ~0 for large k_x values, where coupling between neighbouring rows of cylinders (along y) becomes small)
- Assumptions made (2): Guided modes are assumed to have a TE polarization, with mixing between TE and TM assumed negligible

- **Note:** "Bloch mode expansion implies periodic boundary conditions also for the modes Eβ(r). Thus, boundary effects at the side terminations of the PHC (e.g. interface with a ridge waveguide), that might give rise to additional spectral features in some structures,10 should be computed additionally, e.g. through mode-coupling or transfer-matrix approaches that go beyond the scope of the present work"



3. Define the type and parameters of the simulated disorder in the photonic crystal
   - The paper models the disorder in the hole radii $\mathbf{r_m}$ and positions $\mathbf{(x_m, y_m)}$ as following a Gaussian distribution.
   - In the way that, $r_m = r + \delta r_m$, $x_m = x_{m,0} + \delta x_m$, $y_m = y_{m,0} + \delta y_m$, where r is the radius of the holes of the regular photonic crystal, and $(x_{m,0}, y_{m,0})$ are the coordinates of the mth hole of the regular photonic crystal. $\delta r_m, \delta x_m$, and δy_m are RVs of Gaussian distribution of mean (μ) zero and equal standard deviations, $\sigma_x = \sigma_y = \sigma_r = \sigma$
   - The convergence of Maxwell solution as well as the radiation loss analysis is studied with the standard deviations set in the range from 0.001a to 0.008a.
   - The new dielectric constant function is described as: $\epsilon'(\boldsymbol{r}) = \epsilon(\boldsymbol{r}) + \delta\epsilon(\boldsymbol{r})$
   - Disorder introduced by hole's contour surface roughness is not modelled



4. Obtain the mode spatial profile (Electric field) of the perturbed photonic crystal:

Sources used: [1], [3]

- Modes of the perturbed crystal can be expanded on the basis of the Bloch modes of the regular photonic crystal, as shown below:

$$\mathbf{E}_\beta(\boldsymbol{r}) = \sum_{\mathbf{k}n} U_\beta(\mathbf{k}, n)\, \mathbf{E}_{\mathbf{k}n}(\boldsymbol{r})$$

- Under the assumptions: (1) $\delta\epsilon(\boldsymbol{r}) = 0$ in the claddings, (2) the structure being structure symmetric with respect to a mirror reflection $\hat{\sigma}_{xy}$.
- Using the (GME) method, $\mathbf{E}_{\mathbf{k}n}(\boldsymbol{r})$ is expressed as given below. Where μ = (k + G, α) is the global index for the guided mode.

$$\mathbf{E}_{\mathbf{k}n}(\mathbf{r}) = \sum_{\mu} c_n(\mu) i \frac{\omega_\mu}{c} \boldsymbol{\epsilon}_{\mathbf{g}} \frac{e^{i\mathbf{g}\cdot\boldsymbol{\rho}}}{\sqrt{S}} 2\mathrm{Re}(A_{2\mu}) \cos(q_\mu z)$$

- This equation is then substituted into the equation for what is referred to as the disorder matrix (equation given below),

$$V_{\mathbf{k}n,\mathbf{k}'n'} = \int d\mathbf{r} \delta\epsilon(\mathbf{r}) \mathbf{E}^*_{\mathbf{k}'n'}(\mathbf{r}) \cdot \mathbf{E}_{\mathbf{k}n}(\mathbf{r})$$

- Yielding the expression:

$$V_{\mathbf{k}n,\mathbf{k}'n'} = \sum_{\mu,\mu'} c^*_n(\mu) c_{n'}(\mu') \delta\epsilon(\mathbf{g}' - \mathbf{g}) M_{\mu\mu'}$$

Where:

$$\begin{aligned} M_{\mu\mu'} &= 2\frac{\omega_\mu \omega_{\mu'}}{c^2} \boldsymbol{\epsilon}_{\mathbf{g}} \cdot \boldsymbol{\epsilon}_{\mathbf{g}'} \left[A^*_{2\mu} A_{2\mu'} I_{2-} + \mathrm{Re}(A_{2\mu} A_{2\mu'}) I_{2+}\right] \\ I_{2\pm} &= 2\frac{\sin\left[(q_\mu \pm q_{\mu'})\frac{d}{2}\right]}{q_\mu \pm q_{\mu'}}. \end{aligned}$$

- The expression for $\delta\epsilon(\boldsymbol{g}' - \boldsymbol{g})$ is substituted for with its Fourier representation, given below:

$$\begin{aligned} \epsilon'(\mathbf{g}) &= \frac{1}{S} \int_S d\boldsymbol{\rho} e^{i\mathbf{g}\cdot\boldsymbol{\rho}} \epsilon'(\boldsymbol{\rho}, 0) \\ &= \bar{\epsilon}_2 \delta_{\mathbf{g},0} + (1 - \bar{\epsilon}_2) \sum_m \frac{1}{S} e^{i\mathbf{g}\cdot\boldsymbol{\rho}_m} \frac{2\pi r_m}{g} J_1(g r_m) \end{aligned}$$

To understand what the expression for $V_{\mathbf{k}n,\mathbf{k}'n'}$ outputs are used for, we consider the formulism presented in source [3] (original work on the GME method).

- The expression for the magnetic field mode profile of the perturbed crystal, similar to the expression for the electric field mode profile.

$$\mathbf{H}(\mathbf{r}) = \sum_{\mu} c_\mu \mathbf{H}_\mu(\mathbf{r})$$

- Substituting this equation into the master equation, and under the assumption that the Bloch modes of the unperturbed crystal are orthonormal, the master equation becomes:

$$\sum_{\nu} \mathcal{H}_{\mu\nu} c_\nu = \frac{\omega^2}{c^2} c_\mu$$

  Where $\mathcal{H}_{\mu\nu}$ is the disorder matrix when expressed as a function of the magnetic field mode. The integral form of the disorder matrix is given below.
- The matrix is computed numerically using the equation given below. (neglecting the contribution due to the mixing of the TM-TE polarizations)

$$\begin{aligned}
&\mathcal{H}_{\mu\nu}^{\mathrm{TE-TE}} = \left(\frac{\omega_\mu}{c}\right)^2 \left(\frac{\omega_\nu}{c}\right)^2 \hat{\epsilon}_{\mathbf{g}} \cdot \hat{\epsilon}_{\mathbf{g}'} \\
&\times \left\{ (\bar{\epsilon}_1)^2 \eta_1(\mathbf{G}, \mathbf{G}') B_{1\mu}^* B_{1\nu} I_1 \right. \\
&+ (\bar{\epsilon}_3)^2 \eta_3(\mathbf{G}, \mathbf{G}') A_{3\mu}^* A_{3\nu} I_3 + (\bar{\epsilon}_2)^2 \eta_2(\mathbf{G}, \mathbf{G}') \\
&\times \left. \left[(A_{2\mu}^* A_{2\nu} + B_{2\mu}^* B_{2\nu}) I_{2-} + (A_{2\mu}^* B_{2\nu} + B_{2\mu}^* A_{2\nu}) I_{2+} \right] \right\}
\end{aligned}$$

  Where:

$$\begin{aligned}
I_3 &\equiv \int_{d/2}^{\infty} e^{-(\chi_{3\mu} + \chi_{3\nu})(z - d/2)}\, \mathrm{d}z = (\chi_{3\mu} + \chi_{3\nu})^{-1} \\
I_{2\pm} &\equiv \int_{-d/2}^{d/2} e^{i(q_\mu \pm q_\nu) z}\, \mathrm{d}z = \frac{\sin((q_\mu \pm q_\nu) d/2)}{(q_\mu \pm q_\nu)/2} \\
I_1 &\equiv \int_{-\infty}^{-d/2} e^{(\chi_{1\mu} + \chi_{1\nu})(z + d/2)}\, \mathrm{d}z = (\chi_{1\mu} + \chi_{1\nu})^{-1}
\end{aligned}$$

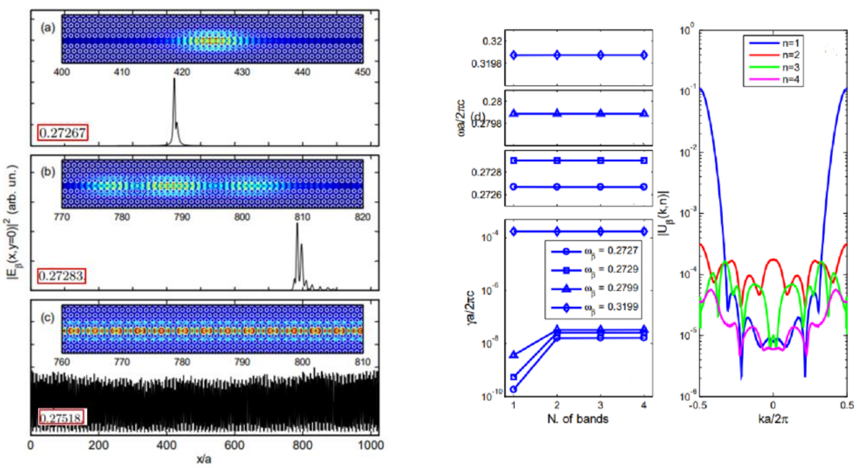
\

- The parameter ν (which is equal to μ) in the below equation sets the number of bases of the Bloch mode used for the reconstruction of the mode profile of the perturbed crystal, which is also the number of indices for the expansion equation.

$$\sum_{\nu} \mathcal{H}_{\mu\nu} c_{\nu} = \frac{\omega^2}{c^2} c_{\mu}$$

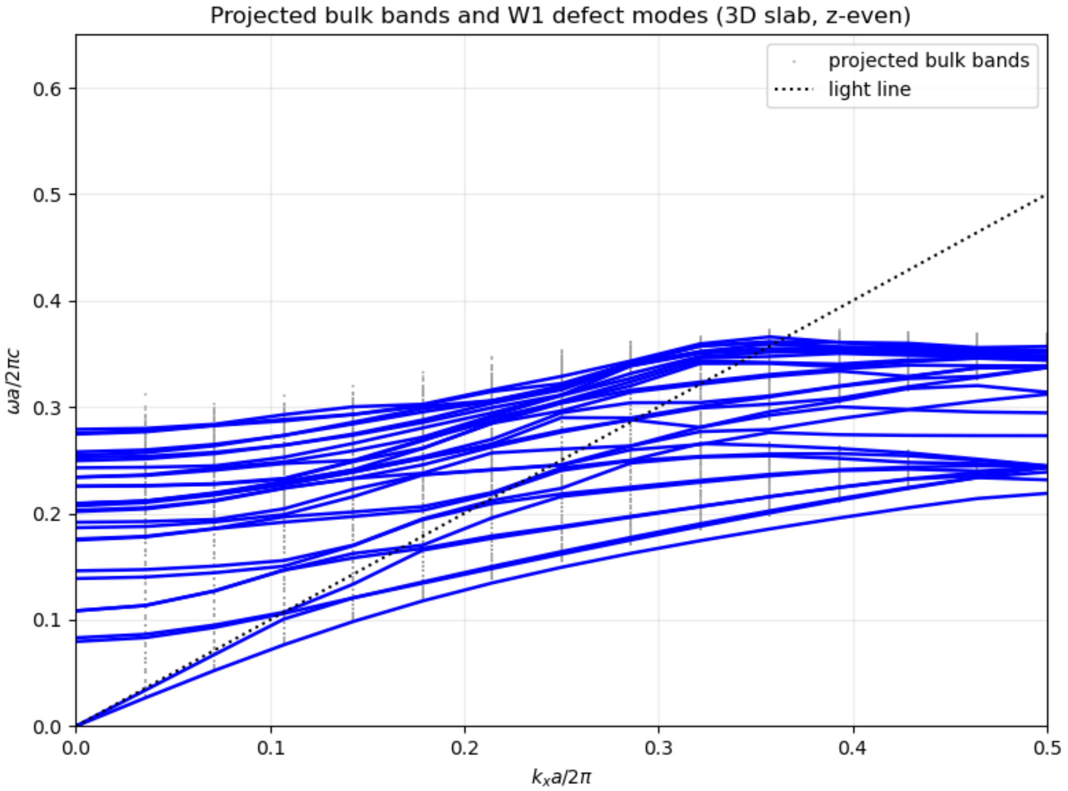
- We obtain the vector mode c_μ and the eigen frequencies by diagonalizing the disorder matrix, re-expressing it in the form PDP^T.
- The electric field mode profile can be obtained from the magnetic field mode profile through the equation given below.

$$\mathbf{E}(\mathbf{r}) = \frac{ic}{\omega\epsilon(\mathbf{r})} \nabla \times \mathbf{H}(\mathbf{r})$$

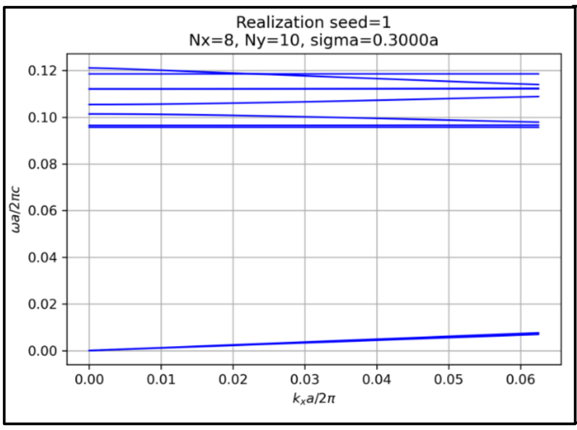


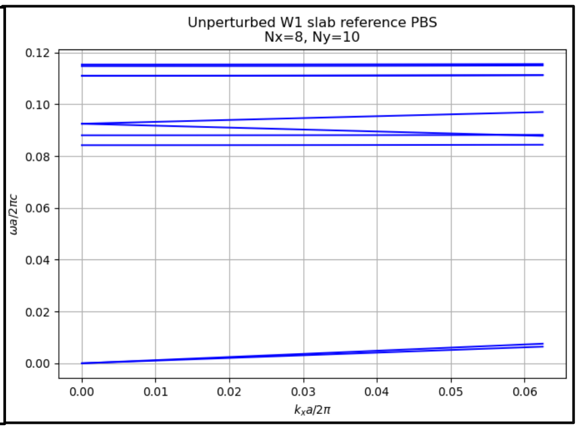
Partial Reproduction of Paper Results

(1) Waveguide guided modes and projected PBS

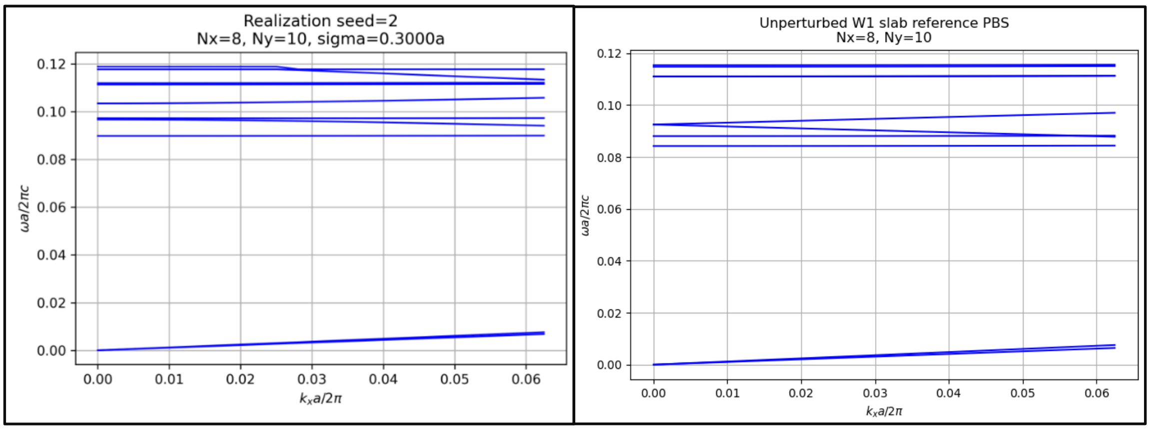


(2) Disorder effects on PBS when σ = 0.3a (left) 1st realization or seed, and unperturbed PBS (right)

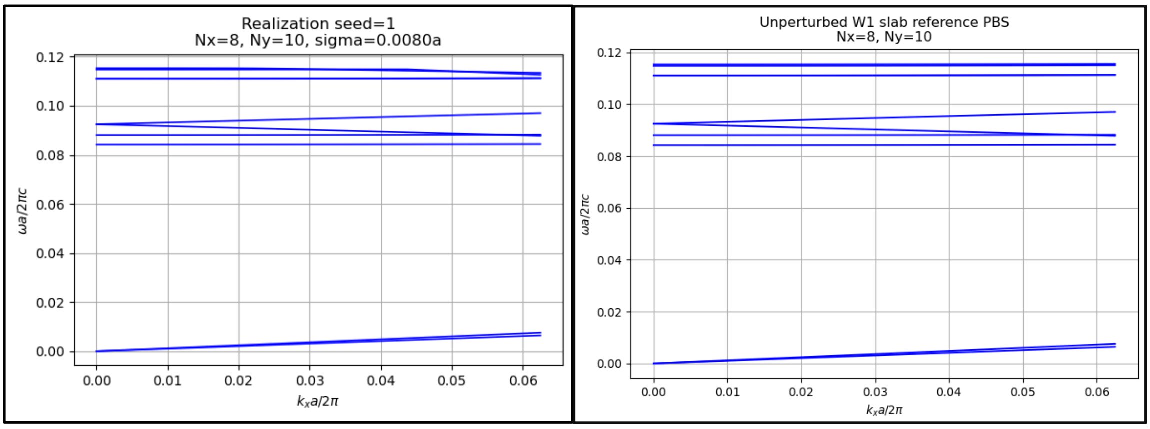




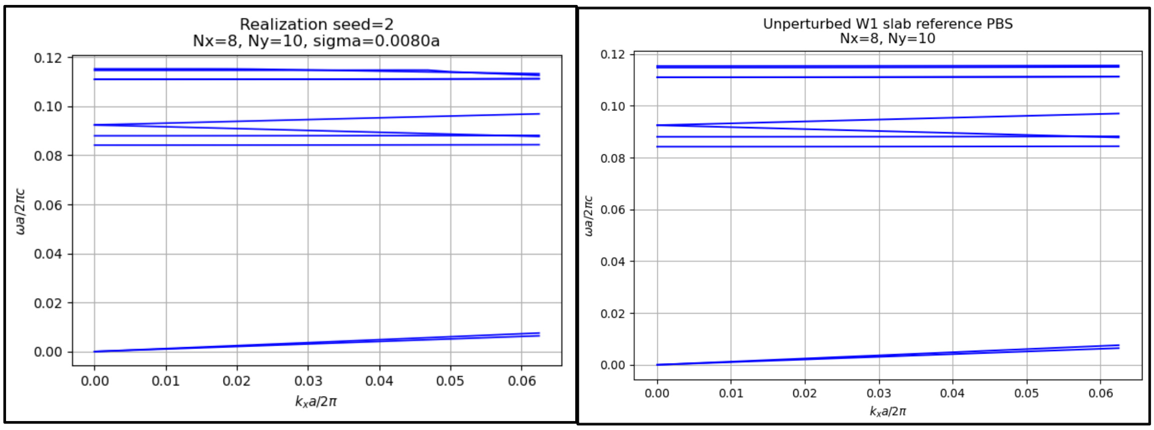
(3) Disorder effects on PBS when σ = 0.3a (left) 2nd realization or seed, and unperturbed PBS (right)



(4) Disorder effects on PBS when σ = 0.008a (left) 1st realization or seed, and unperturbed PBS (right)



(5) Disorder effects on PBS when σ = 0.008a (left) 2nd realization or seed, and unperturbed PBS (right)



Observation:

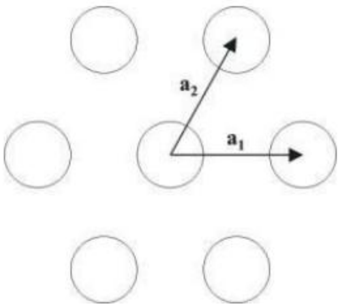
As expected, as the perturbation strength or disorder std increases, the number of affected bands increases, and the Bloch modes of the Perturbed PhC require including more basis or bands in the expansion sum of the unperturbed Bloch modes. As explained, this is referred to as hybridization effect, naturally, bands forbidden from mixing due to *symmetry* become mixed due to disorder and breakig of symmetry.

Theoretical Outline and Steps for Numerically Obtaining the PBS using PWE Method

Obtaining the Photonic Band Structure of 2D Triangular Lattice of Air Holes Embedded in a Dielectric Background, ***Numerically***

Source used: [4], [5]

- Define the lattice (triangular lattice), and primitive lattice vectors, **a1** and **a2**



$$\overline{a_1} = a\hat{x}$$

$$\overline{a_2} = a\left(\frac{1}{2}\hat{x} + \frac{\sqrt{3}}{2}\hat{y}\right)$$

$$\overline{b_1} = \frac{2\pi}{a}\left(\hat{x} - \frac{1}{\sqrt{3}}\hat{y}\right)$$

$$\overline{b_2} = \frac{2\pi}{a}\frac{2}{\sqrt{3}}\hat{y}$$

- Find the corresponding reciprocal lattice vectors, **b1** and **b2**

$$\overline{b_1} = \frac{2\pi}{a}\left(\hat{x} - \frac{1}{\sqrt{3}}\hat{y}\right)$$

$$\overline{b_2} = \frac{2\pi}{a}\frac{2}{\sqrt{3}}\hat{y}$$

- Substitute **b1** and **b2** in the expanded Bloch mode form for the field and Fourier composition of the inverse of the dielectric function

$$E_{x,y,or\,z} = \sum_n \sum_q K_{n,q}^{E_{x,y,or\,z}} e^{-j\frac{2\pi n}{a}x} e^{-j\frac{2\pi(-n+2q)}{a\sqrt{3}}y} e^{-jk_x x - jk_y y - jk_z z}$$

$$\frac{1}{\varepsilon_r} = \sum_m \sum_p K_{m,p}^{\varepsilon_r} e^{-j\frac{2\pi m}{a}x} e^{-j\frac{2\pi(-m+2p)}{a\sqrt{3}}y}$$

- The previous equations are then substituted into the generalized master equation in terms of the electric field intensity, as given below.

$$\begin{cases} \dfrac{1}{\varepsilon_r}\left(-\dfrac{\partial^2 E_x}{\partial y^2}-\dfrac{\partial^2 E_x}{\partial z^2}+\dfrac{\partial^2 E_y}{\partial x\partial y}+\dfrac{\partial^2 E_z}{\partial x\partial z}\right)=\dfrac{\omega^2}{c^2}E_x \\ \dfrac{1}{\varepsilon_r}\left(-\dfrac{\partial^2 E_y}{\partial x^2}-\dfrac{\partial^2 E_y}{\partial z^2}+\dfrac{\partial^2 E_z}{\partial y\partial z}+\dfrac{\partial^2 E_x}{\partial x\partial y}\right)=\dfrac{\omega^2}{c^2}E_y \\ \dfrac{1}{\varepsilon_r}\left(-\dfrac{\partial^2 E_z}{\partial x^2}-\dfrac{\partial^2 E_z}{\partial y^2}+\dfrac{\partial^2 E_x}{\partial x\partial z}+\dfrac{\partial^2 E_y}{\partial y\partial z}\right)=\dfrac{\omega^2}{c^2}E_z \end{cases}$$

- Yielding the form given below,

$$\sum_m \sum_p \kappa^{\varepsilon_r}_{n-m,q-p}\,\overline{\mathbf{M}}\begin{pmatrix}\kappa^{E_x}_{m,p}\\ \kappa^{E_y}_{m,p}\\ \kappa^{E_z}_{m,p}\end{pmatrix}=\frac{\omega^2}{c^2}\begin{pmatrix}\kappa^{E_x}_{m,p}\\ \kappa^{E_y}_{m,p}\\ \kappa^{E_z}_{m,p}\end{pmatrix}$$

With,

$$\overline{\mathbf{M}}=\begin{pmatrix} \left(\frac{2\pi s}{a}+k_y\right)^2+k_z^2 & -\left(\frac{2\pi m}{a}+k_x\right)\left(\frac{2\pi s}{a}+k_y\right) & -k_z\left(\frac{2\pi m}{a}+k_x\right) \\ -\left(\frac{2\pi m}{a}+k_x\right)\left(\frac{2\pi s}{a}+k_y\right) & \left(\frac{2\pi m}{a}+k_x\right)^2+k_z^2 & -k_z\left(\frac{2\pi s}{a}+k_y\right) \\ -k_z\left(\frac{2\pi m}{a}+k_x\right) & -k_z\left(\frac{2\pi s}{a}+k_y\right) & \left(\frac{2\pi m}{a}+k_x\right)^2+\left(\frac{2\pi s}{a}+k_y\right)^2 \end{pmatrix}$$

And,

$$s=\frac{-m+2p}{\sqrt{3}}$$

- The above expressions, in the same way, can be expressed as given below.

$$\sum_{\mathbf{G}}\left[-\varepsilon^{-1}_{\mathbf{G}'-\mathbf{G}}\cdot(\mathbf{k}+\mathbf{G}')\times(\mathbf{k}+\mathbf{G})\times\right]\mathbf{c}_{\mathbf{G}}=\frac{\omega^2}{c^2}\mathbf{c}_{\mathbf{G}'}$$

- Finally, the coefficients that the field profiles are expanded by are solved by diagonalizing the matrix on the left side of the equation into PDP^{T}. The coefficients of FD of the invers dielectric function are solved for using the expressions below, for the described structure.

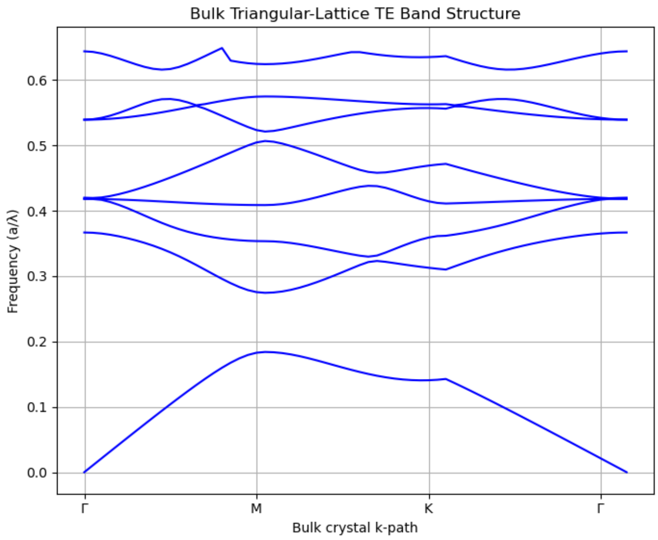
$$\kappa_{m,p}^{\varepsilon_r} = \frac{1}{Area}\iint \frac{1}{\varepsilon_r(x,y)} e^{j\frac{2\pi m}{a}x} e^{j\frac{2\pi(-m+2p)}{\sqrt{3}a}y} dxdy$$

$$\kappa_{m,p}^{\varepsilon_r} = \frac{1}{\sqrt{3}a^2}(1-\frac{1}{\varepsilon_r})2\pi d \frac{J_1\left(\frac{d}{2}\sqrt{\left(\frac{2\pi m}{a}\right)^2 + \left(\frac{2\pi(-m+2p)}{\sqrt{3}a}\right)^2}\right)}{\sqrt{\left(\frac{2\pi m}{a}\right)^2 + \left(\frac{2\pi(-m+2p)}{\sqrt{3}a}\right)^2}} + \frac{1}{\varepsilon_r}\delta_{m,p}$$

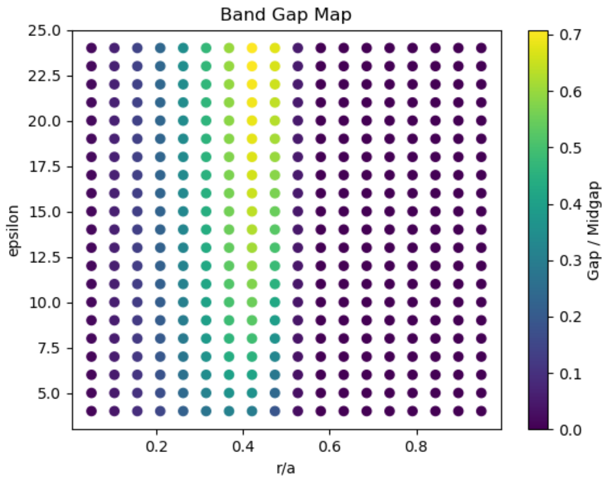
- Moreover, the assumptions are: (1) For the separation of the polarizations of TM and TE, $k_z = 0$, (2) Confinement of the guided mode is assumed strong such that there is little coupling, hence $k_y = 0$

Simulation Explorations Using the MBP and MEEP Tools

(1) Obtain the Photonic Band Structure of 2D Triangular Lattice of Air Holes Embedded in a Dielectric Background ***using MPB***



(2) Identifying the optimal r/a ratio for maximum gap-to-mid-gap ratio.



Simulation results:

-eps = 30.0

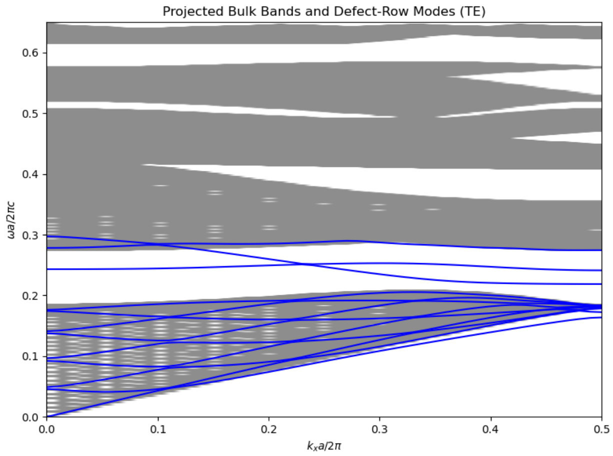
-r/a = 0.45

-gap ratio = 0.763230654828772

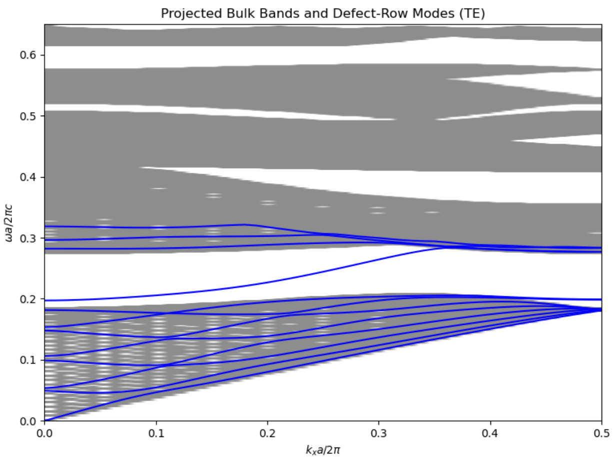
The r/a changed with the maximum of the range of structure dielectric constant, and with gap ratio always maximized for that constant, yet remaining within the vicinity of 0.45.

(3) Obtain the Band Structure of A 2D Photonic Crystal with A Line Defect waveguide for various line defect configurations

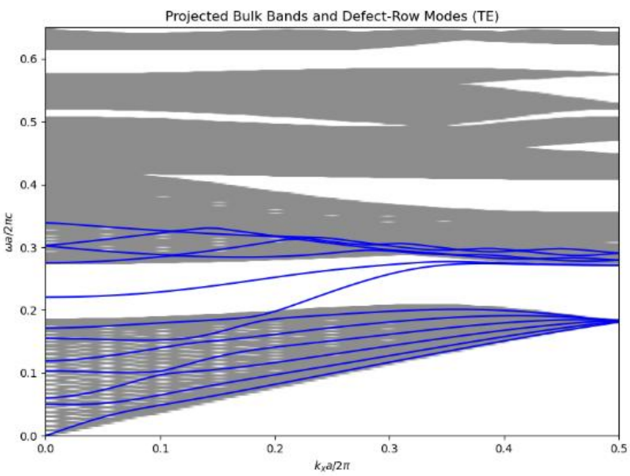
(a) Bulk r/a is 0.3, and line defect r/a is 0.1



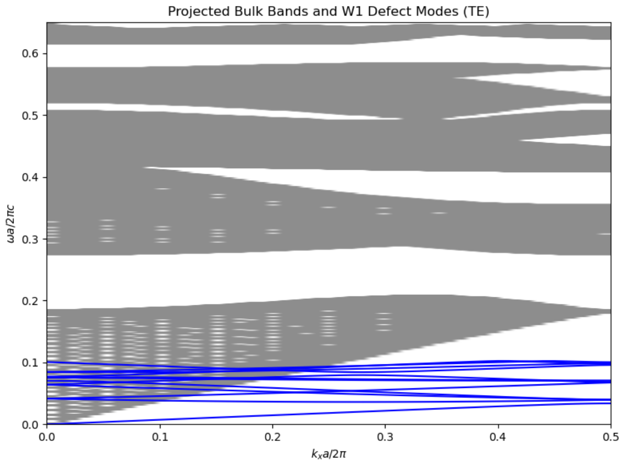
(b) Bulk r/a is 0.3, and line defect r/a is 0.5



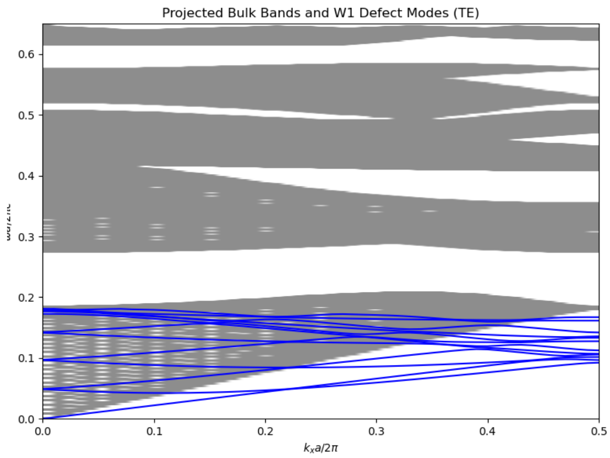
(c) Bulk r/a is 0.3, and line defect r/a is 0.9



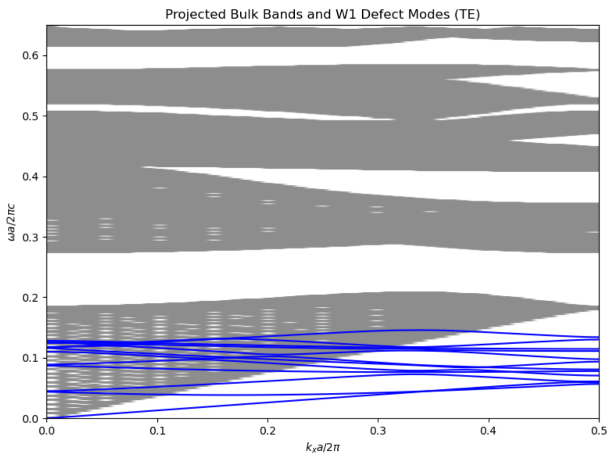
(d) Periodicity of holes is a in the bulk crystal, and 0.1a along the defect line



(e) Periodicity of holes is a in the bulk crystal, and 0.5a along the defect line

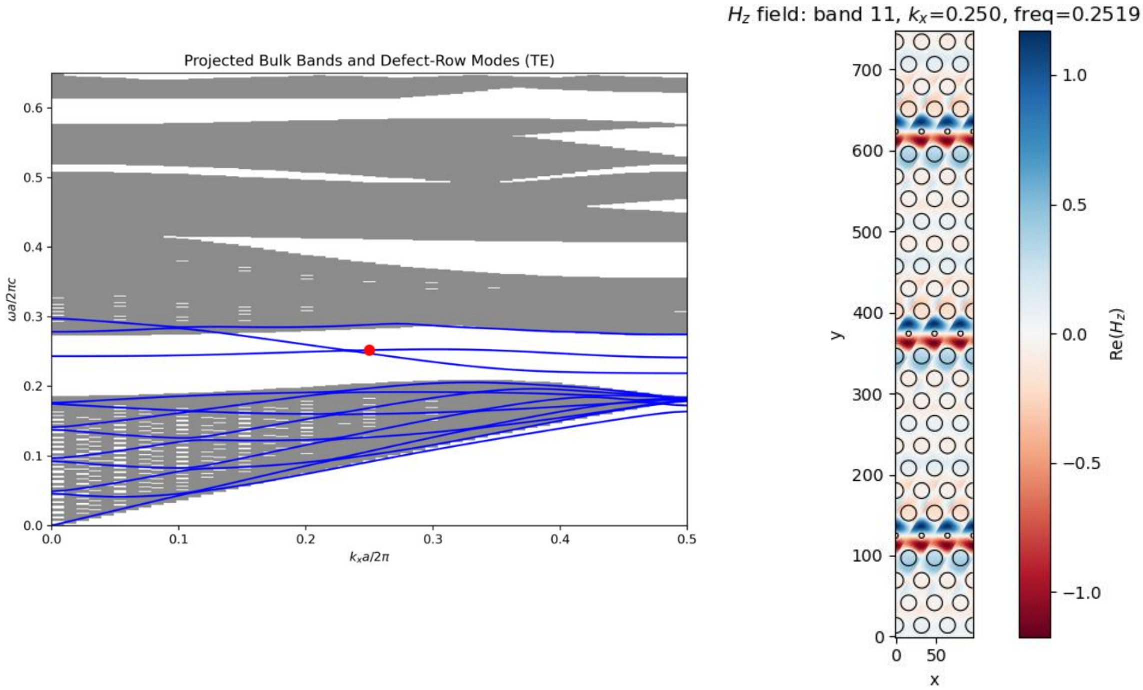


(a) Periodicity of holes is a in the bulk crystal, and 3a along the defect line

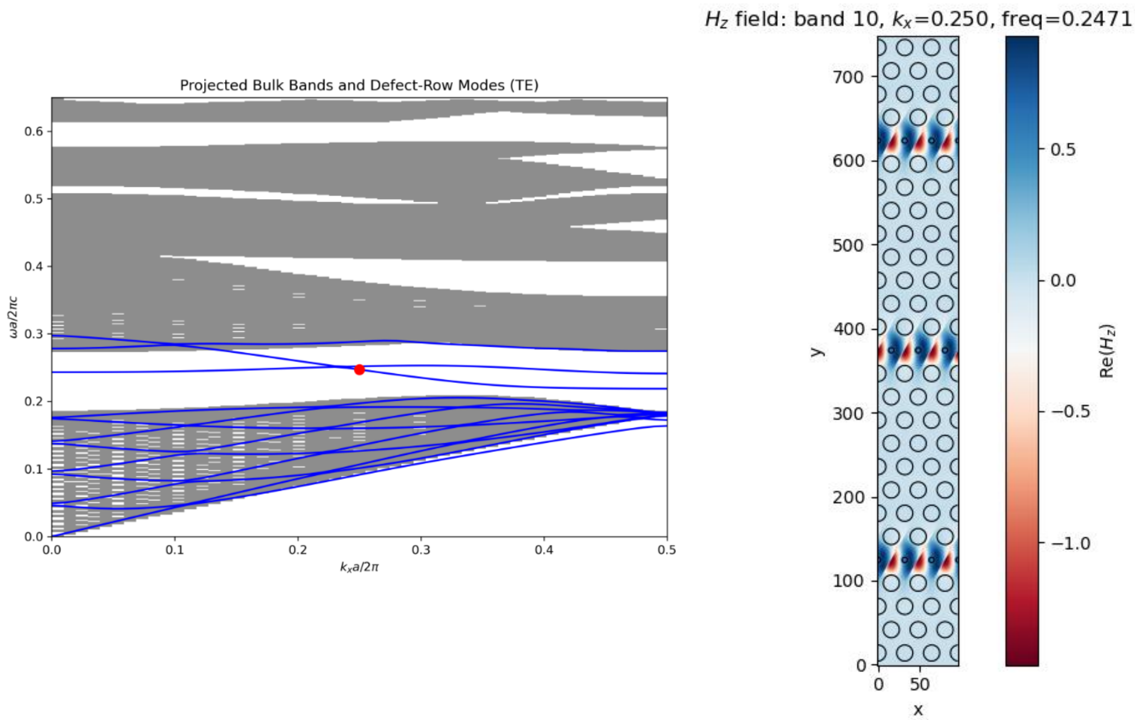


(4) Plot the mode filed profiles for different (k_x, ω) points

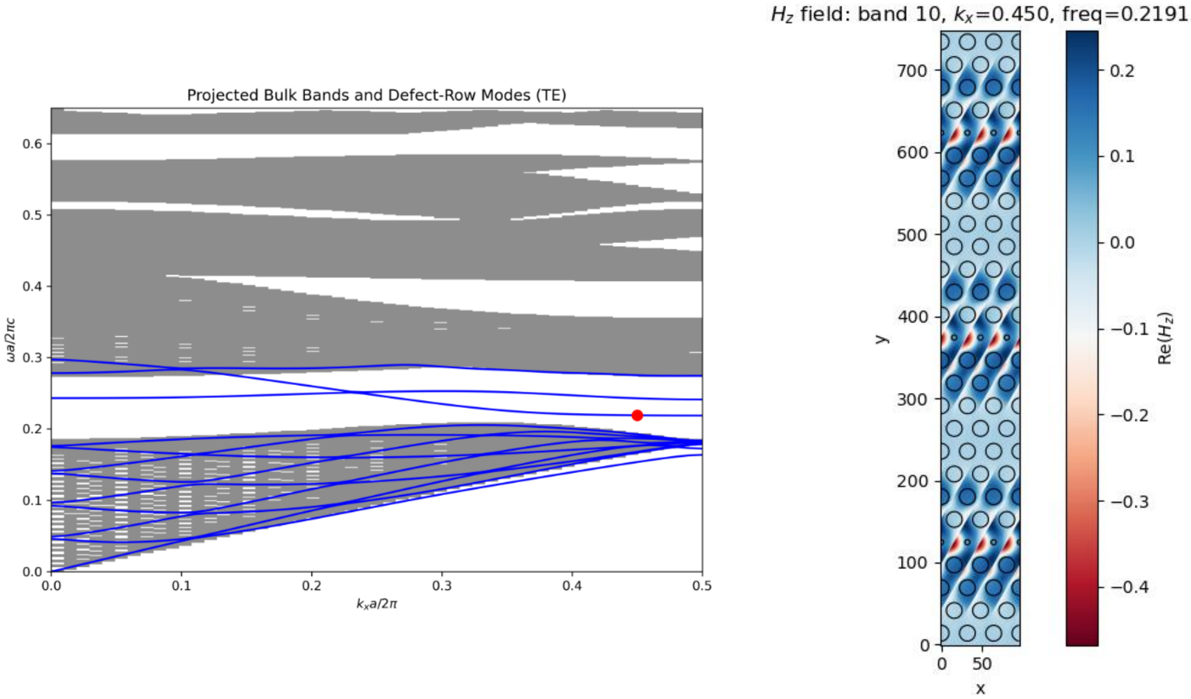
(a) PBS for the Case of bulk r/a is 0.3, and line defect r/a is 0.1, red dot showing the plotted field profile on the right. **(band = 11, kx = 0.25)**



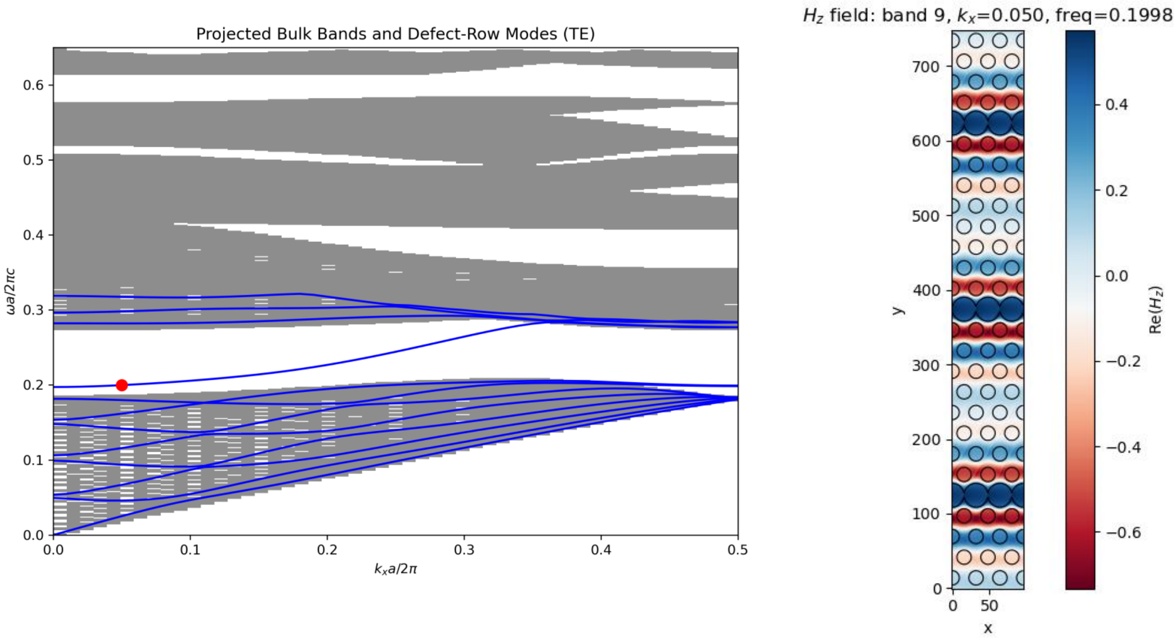
(b) PBS for the Case of bulk r/a is 0.3, and line defect r/a is 0.1, red dot showing the plotted field profile on the right. **(band = 10, kx = 0.25)**



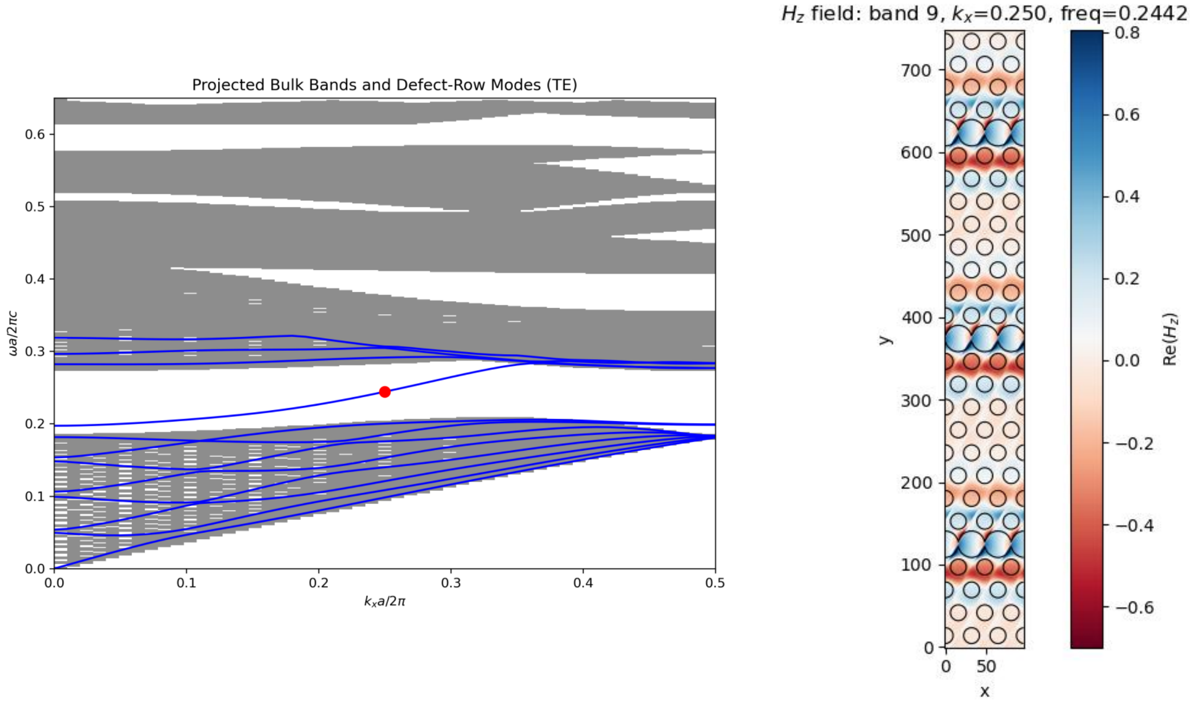
(c) PBS for the Case of bulk r/a is 0.3, and line defect r/a is 0.1, red dot showing the plotted field profile on the right. **(band = 10, kx = 0.45)**



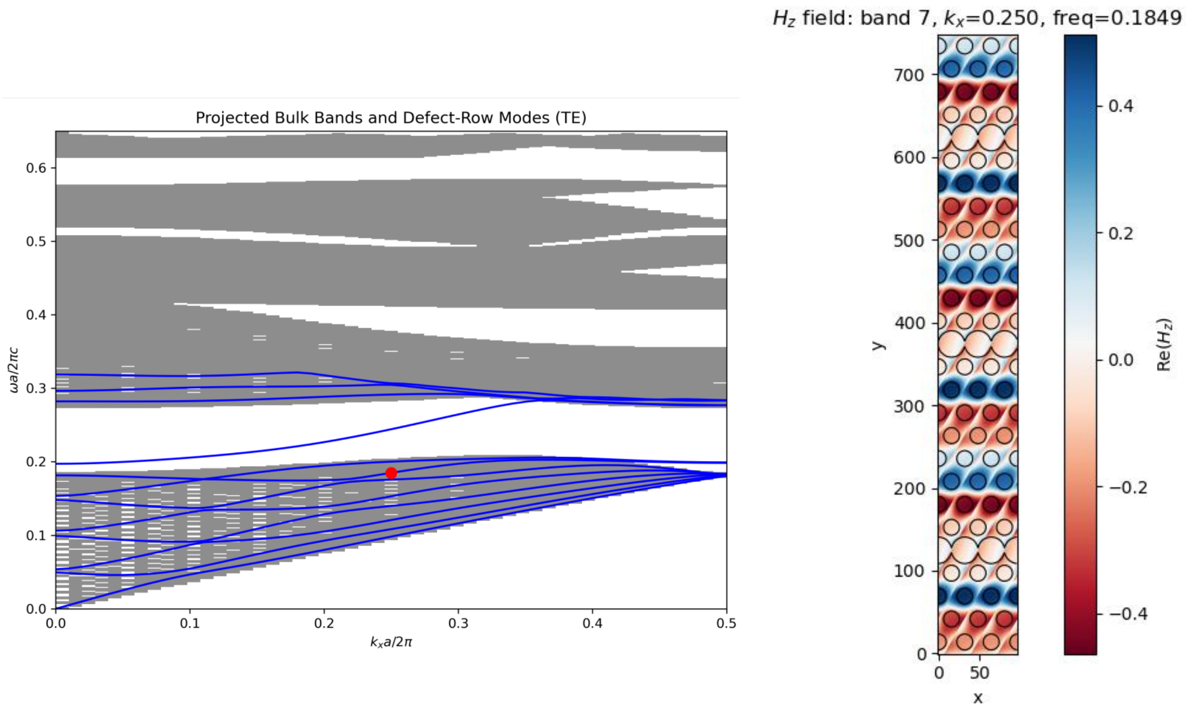
(d) PBS for the Case of bulk r/a is 0.3, and line defect r/a is 0.5, red dot showing the plotted field profile on the right. **(band = 9, kx = 0.05)**



(e) PBS for the Case of bulk r/a is 0.3, and line defect r/a is 0.5, red dot showing the plotted field profile on the right. **(band = 9, kx = 0.25)**



(f) PBS for the Case of bulk r/a is 0.3, and line defect r/a is 0.5, red dot showing the plotted field profile on the right. **(band = 7, kx = 0.25)**



References

[1]. V. Savona, “Electromagnetic modes of a disordered photonic crystal,” *Physical Review B*, vol. 83, no. 8, Feb. 2011, doi: https://doi.org/10.1103/physrevb.83.085301.

[2]. Henri Benisty, “Modal analysis of optical guides with two-dimensional photonic band-gap boundaries,” vol. 79, no. 10, pp. 7483–7492, May 1996, doi: https://doi.org/10.1063/1.362419.

[3]. L. C. Andreani and D. Gerace, “Photonic-crystal slabs with a triangular lattice of triangular holes investigated using a guided-mode expansion method,” *Physical Review B*, vol. 73, no. 23, Jun. 2006, doi: https://doi.org/10.1103/physrevb.73.235114.

[4]. S. Guo, “Modul 2 Plane wave method (PWM).” Accessed: May 18, 2026. [Online]. Available: https://www.phys.ubbcluj.ro/~emil.vinteler/nanofotonica/PWM/Modul2_PWM.pdf

[5]. S. Guo and S. Albin, “Simple plane wave implementation for photonic crystal calculations,” *Optics Express*, vol. 11, no. 2, p. 167, Jan. 2003, doi: https://doi.org/10.1364/oe.11.000167.

Appendix B – Internship Report: Gold Nanoparticles Localized Surface Plasmon Resonance (LSPR) and Fiber Sensing Applications

Draft

Report Title:

Fiber Sensor Using Gold Nanoparticles Localized Surface Plasmon Resonance (LSPR)

Report Objectives:

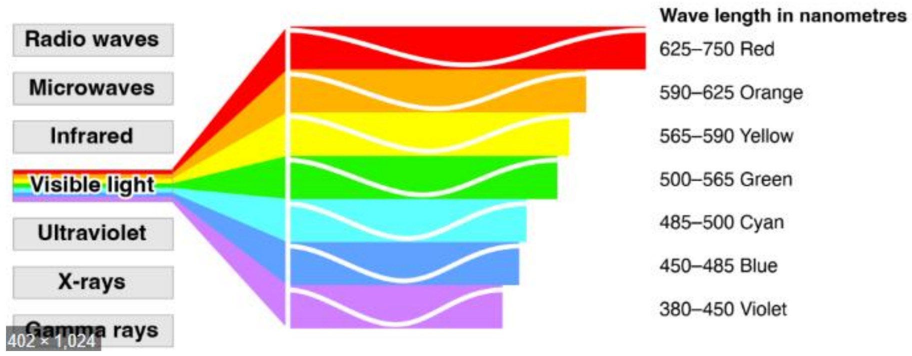
- To study the LSPR phenomenon in gold nanoparticles by exploring the effects of the surrounding medium's refractive index and the size of the gold NPs on the observed colors and resonance wavelength.
- To explore the use of optical fibers and Au NPs in sensing applications and characterization.
- To investigate the use of optical fibers in a classification problem by categorizing the region over which the fiber tip is positioned among various Au NPs or a clean substrate

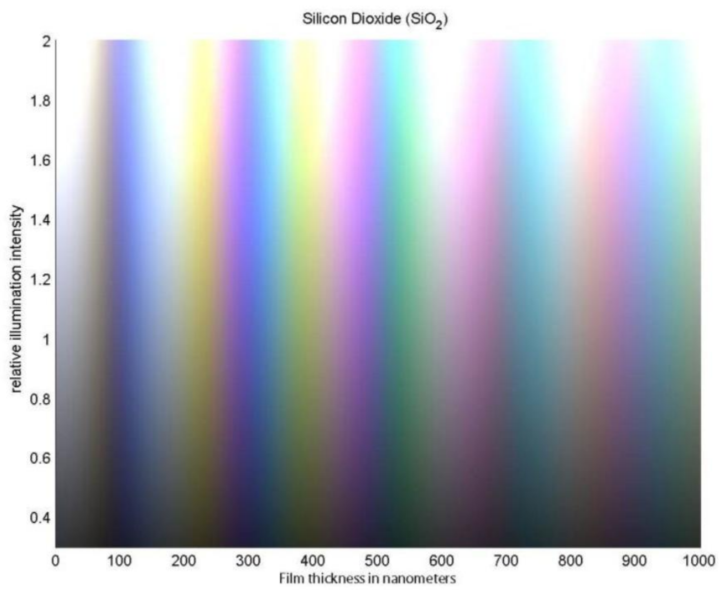
Shayma Alteneiji | 100063072

Internship Supervisor:

Prof. Jaime Viegas

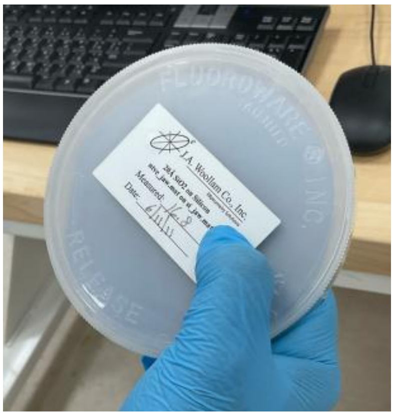
Introduction

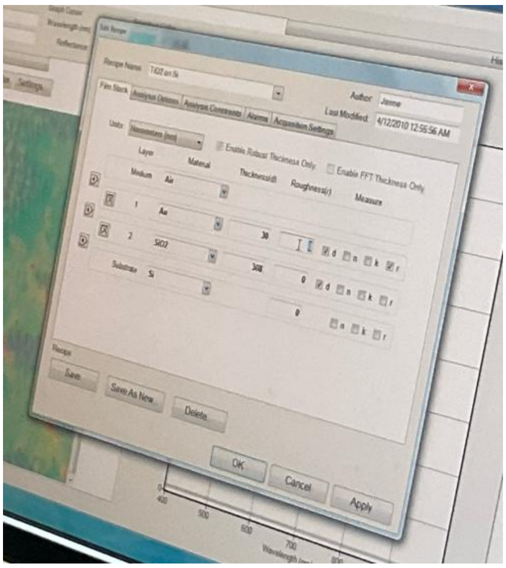


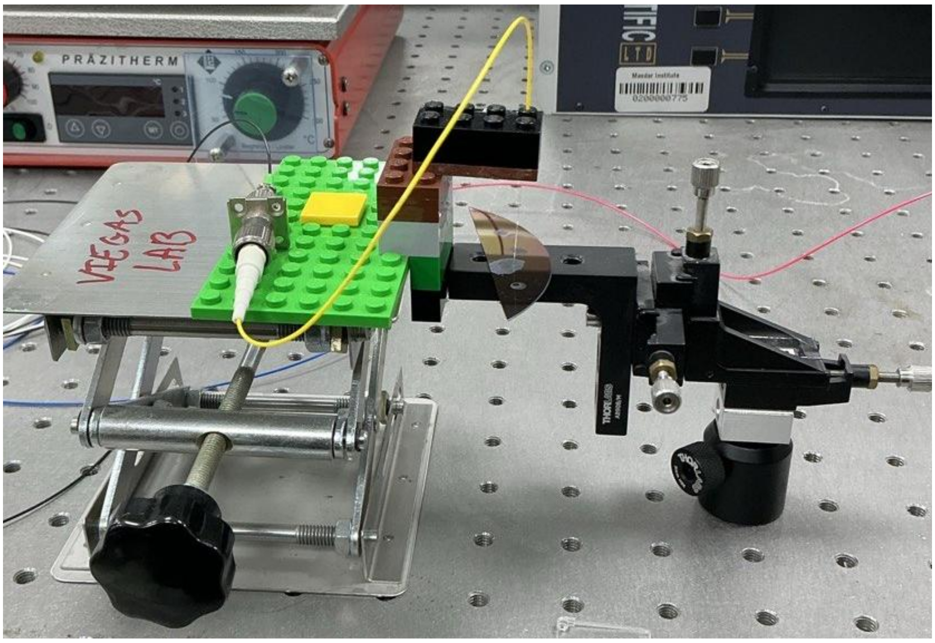


Detailed Experimental Procedure

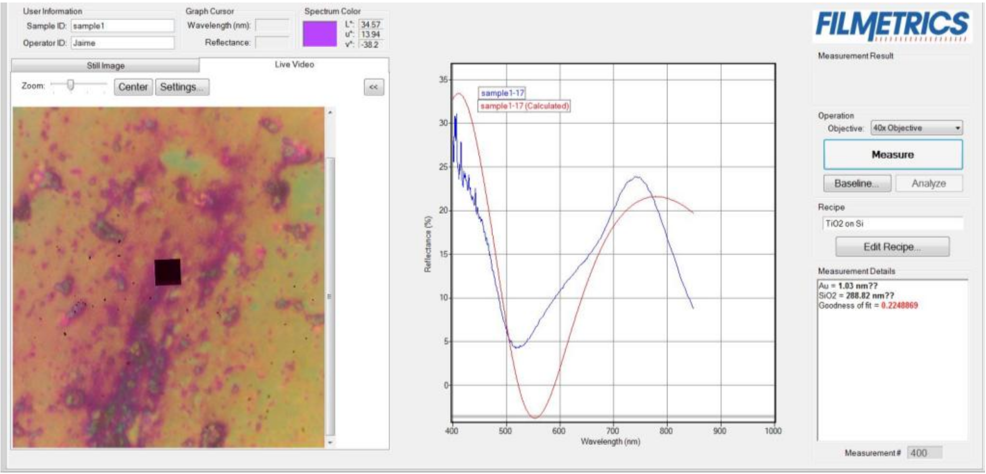

KRÜSS

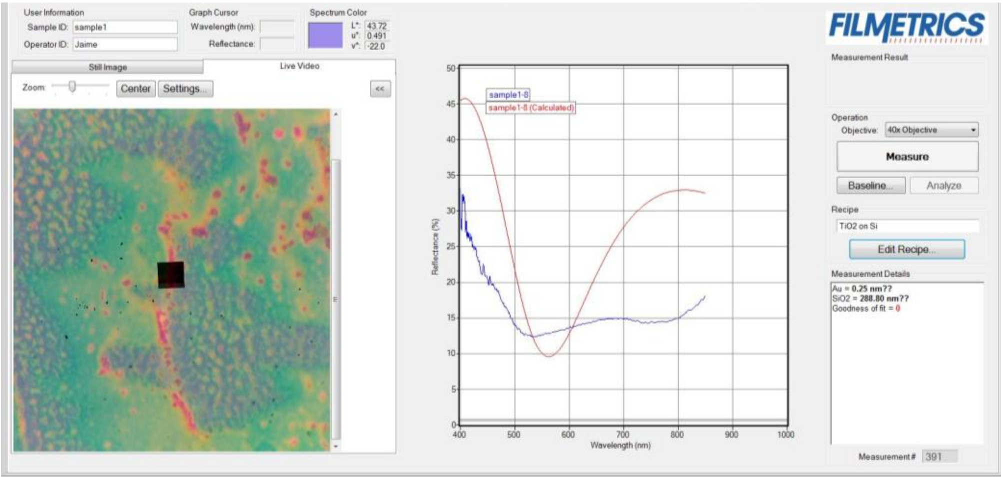

FLUOROWARE
J.A. Woollam Co. Inc.
Measured:
Date:
RELEASE

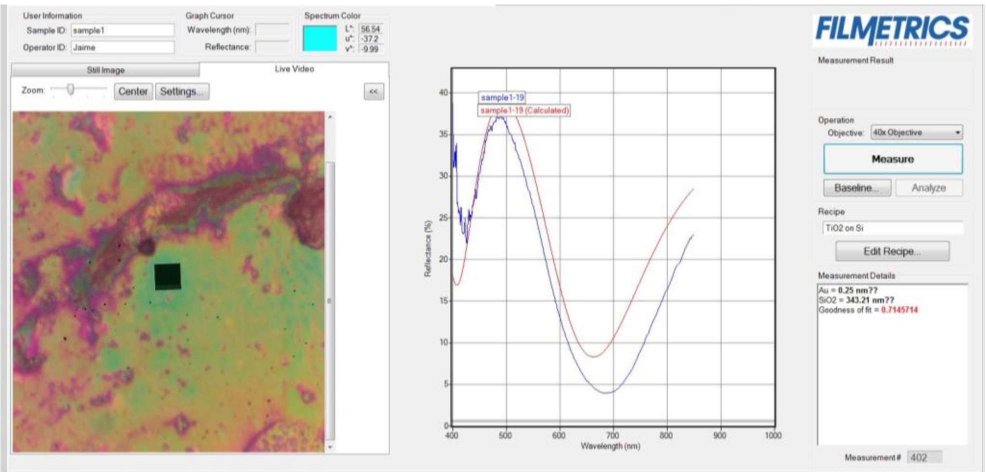

Author
Last Modified
Enable FFT Thickness Only
Save
Save As New
Delete
OK
Cancel
Apply

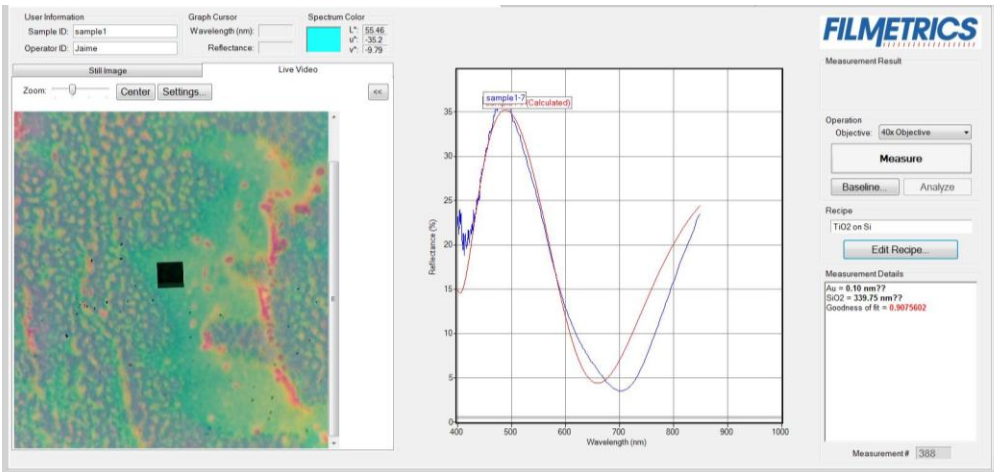

PRÄZITHERM
VIEGAS
LAB
TIFIC
LTD

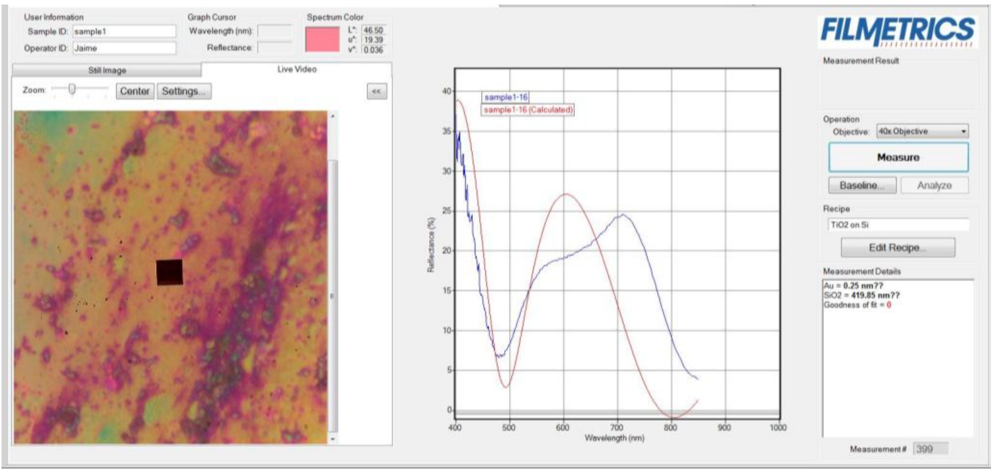
Spectrophotometer Results

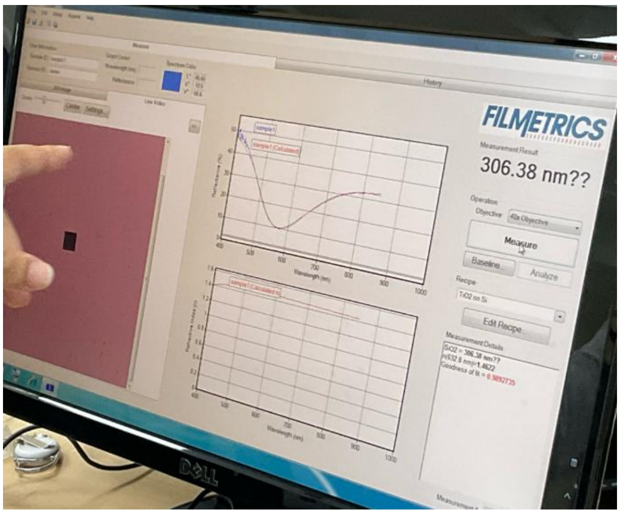







User Information
Sample ID: sample1
Operator ID: Jaime
Graph Cursor
Wavelength (nm):
Reflectance:
Spectrum Color
L*: 55.46
u*: -35.2
v*: -9.79
Still Image
Live Video
Zoom:
Center
Settings...
<<
FILMETRICS
Measurement Result
sample1-7
(Calculated)
Reflectance (%)
Wavelength (nm)
Operation
Objective: 40x Objective
Measure
Baseline...
Analyze
Recipe
TiO2 on Si
Edit Recipe...
Measurement Details
Au = 0.10 nm??
SiO2 = 339.75 nm??
Goodness of fit = 0.9075602
Measurement # 388


User Information
Sample ID: sample1
Operator ID: Jaime
Graph Cursor
Wavelength (nm):
Reflectance:
Spectrum Color
L*: 46.50
u*: 19.39
v*: 0.036
Still Image
Live Video
Zoom:
Center
Settings...
<<
FILMETRICS
Measurement Result
sample1-16
sample1-16 (Calculated)
Reflectance (%)
Wavelength (nm)
Operation
Objective: 40x Objective
Measure
Baseline...
Analyze
Recipe
TiO2 on Si
Edit Recipe...
Measurement Details
Au = 0.25 nm??
SiO2 = 419.85 nm??
Goodness of fit = 0
Measurement # 399


FILMETRICS
Measurement Result
306.38 nm??
Operation
Measure
Baseline
Analyze
Recipe
Edit Recipe
Measurement Details
DELL

Analytical Study and Detailed Design Specifications

***(1)** Describe the experimental setup in detail. Discuss of other potential setup design and potential necessity of using Faraday rotators (optical insulator).*

The experimental setup is that of two cascaded Fabry-Perrot interferometers integrated within a Michelson interferometer. For a Michelson interferometer, the interfering lights result from the following paths: for light entering from the source side, a portion gets transmitted entering the FPs cavities side, and a portion gets reflected entering the mirror side.

The light at the FP's side then reflects back; the reflected portion contributes to the fringe profile at the output side, and the transmitted portion goes back to the source. As for the mirror side, the transmitted part of the reflected light reaches the output, while the reflected part also goes back to the source. Since in this setup, some reflections may reach the source, optical insulators or Fraday rotators are usually used, to protect the source.

This setup can also be emulated by using a coupler instead of a beam splitter; with the source and optical spectrum analyser connected at the (In) side of the couplers and the FP at the other side. The other arm of the (Out) side of the coupler is left as is, the "mirror" is simply the glass(silica)-air interface.

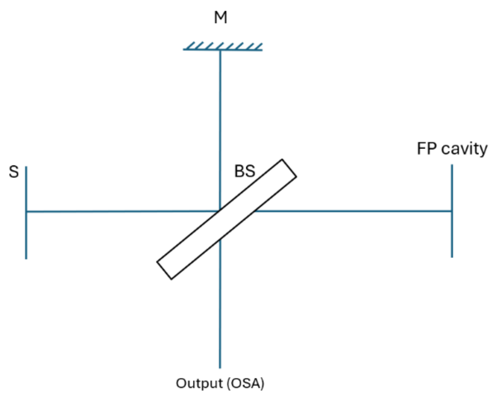


Figure 1.

For the two reasons, the Michelson interferometer fringes are ignored in the simulation and experimental analysis. (1) the OPD set by the extra arm length at (FPIs) side results in an FSR of extremely small periods relative to the resolution of the OSA, making them unabsorbable. (2) the reflections from the mirror side are relatively small (4% reflection), making the inference pattern of very low contrast. For that, we focus our attention to the FP's cavities only in the derivation of the transmission or fringe profile equation, discussed in the next section.

(2) *Derive the transmission spectrum equation of the setup and discuss the non-idealises considerations.*

The diagram below, Figure 2, shows the two FP cavities, air and the silica on silicon. The resulting fringe profile of the structure can be determined using the transfer matrix method (TMM). The resulting wave-transfer or transmission matrix is given below.

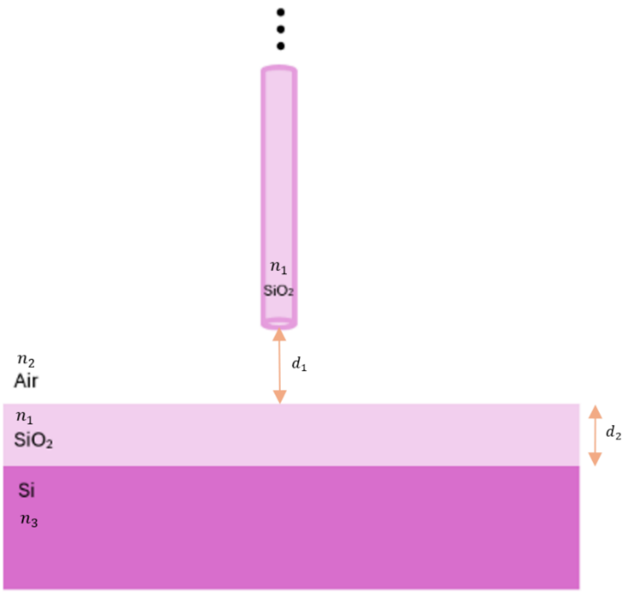


Figure 2.

$$M = \frac{1}{4n_1n_2}\begin{bmatrix} n_1+n_2 & n_1-n_2 \\ n_1-n_2 & n_1+n_2 \end{bmatrix} \times \begin{bmatrix} e^{-j\varphi_1} & 0 \\ 0 & e^{j\varphi_1} \end{bmatrix} \times \begin{bmatrix} n_1+n_2 & n_2-n_1 \\ n_2-n_1 & n_1+n_2 \end{bmatrix} \times \begin{bmatrix} t_{13}t_{31}-r_{13}r_{31} & r_{31} \\ -r_{13} & 1 \end{bmatrix} \times \begin{bmatrix} e^{-j\varphi_2} & 0 \\ 0 & e^{j\varphi_2} \end{bmatrix}$$

Were,

$$\varphi_1 = \frac{4\pi n_2}{\lambda} \cdot d_1$$

$$\varphi_2 = \frac{4\pi n_1}{\lambda} \cdot d_2$$

The reflection coefficient and reflectance are obtained from the M matrix using the equations below.

$$r = \frac{\mathrm{M}_{2\times 1}}{\mathrm{M}_{1\times 1}}$$

$$R = |r|^2$$

The equation can be simulated in MATLAB and verified against the calculations done by *Filmetrics Spectral Reflectance Calculato,* whose results are shown in Figure 3.

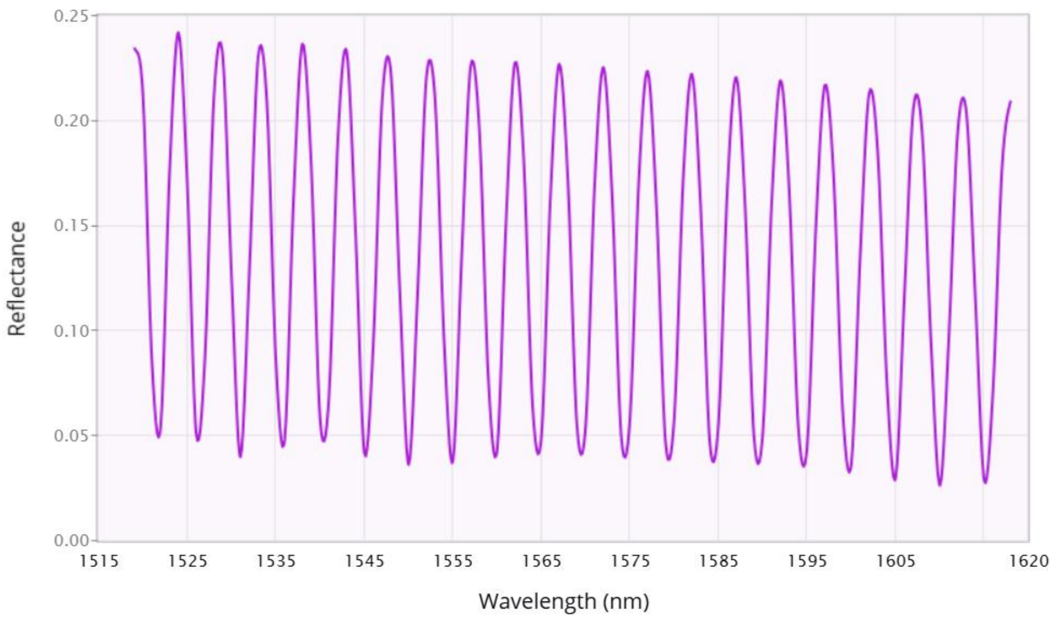


Figure 3.

The fringe profile modulating the source spectrum is obtained using the equation below. The simulated results are shown in an upcoming section.

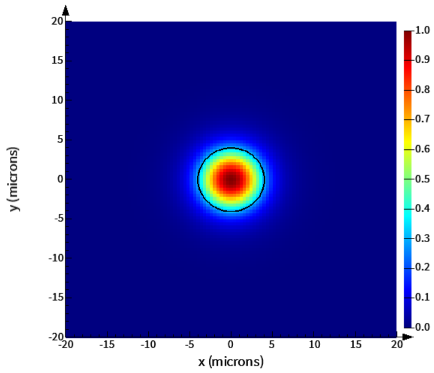
$$I(\lambda) \ = \ R(\lambda) \times S(\lambda)$$

(3) *Design and parameterize single mode fibres, SMF-28, for future FDTD simulations.*

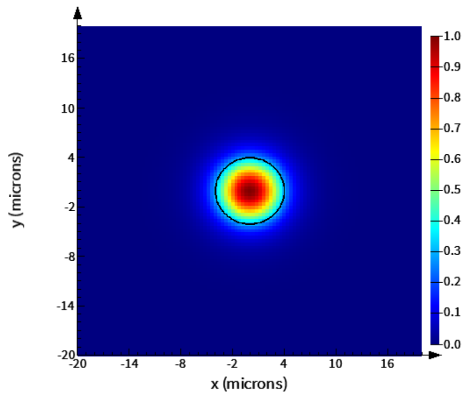
Wavelength = 1550 nm

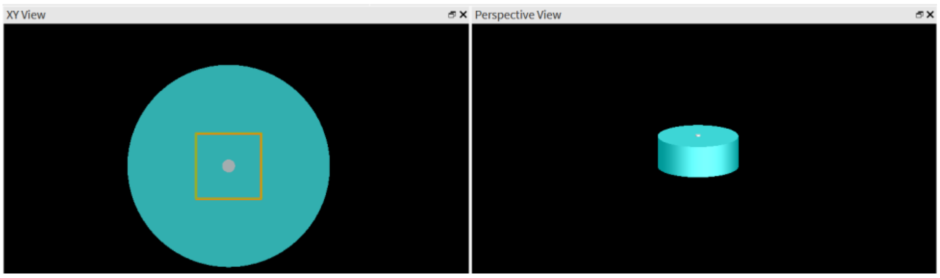
Parameter	*Value*
Core refractive index	1.444
Cladding refractive index	1.44
Core radius	9 µm
Cladding diameter	125 µm
Effective area	96.496 μm^2

TM

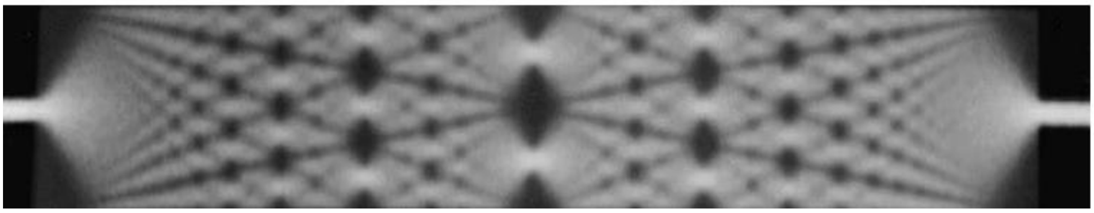


TE





(4) *Describe how single mode field, leaving the fibre tip, extends and interfere, forming multimode profiles on the silica surface.*



(5) *Explain how the optical power emitted at the fibre tip can move the nano particles closer, effectively depositing or attaching them to the fibre.*

"The square of the magnitude of the electric field is equal to the intensity of the beam as a function of position. Therefore, the result indicates that the force on the dielectric particle, when treated as a point dipole, is proportional to the gradient along the intensity of the beam. In other words, the gradient force described here tends to attract the particle to the region of highest intensity."

$$\mathbf{F} = \frac{1}{2}\alpha\nabla E^2 = \frac{2\pi n_1 a^3}{c}\left(\frac{m^2-1}{m^2+2}\right)\nabla I(\mathbf{r})$$

(6) *State purpose or expectations from the designed sensor:*

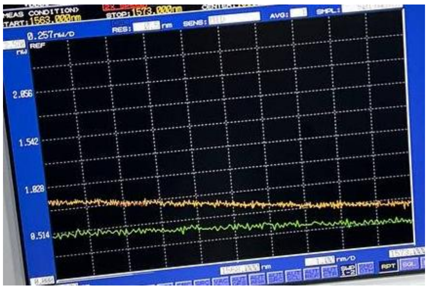
With Au deposited onto the fibre tip, we may obtain a more sensitive sensor to the distance from the silica on silicon wafer. The new sensor's response can be characterized with respect to the distance to obtain a generalizable relation.

(7) *How to confirm whether the Au NPs where deposited successfully?*

One way to confirm it, is to compare the spectrum measurements, for some distance from the silica on silicon wafer, before the Au particles were deposited. Noticeable changes could indicate the successful deposition.

(8) *What Data can be obtained from the system?*

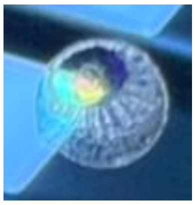
a) To check whether Au particles where deposited, compare the spectrum at some distance from the wafer (orange) with that before having immersed fibre into the gold NPs water solution (green).



<u>*Observation:*</u>

No noticeable consistent difference is observed. It could be concluded that the Au particles were not deposited successfully.

b) Given the unsuccessful usage the fibre as a distance sensor, instead the setup could be used; to design a classifier code informing us of the region we are in across the wafer. Dried gold NPs formed the pattern shown below, with different regions of colours based on size and concentration of the Au NPs.



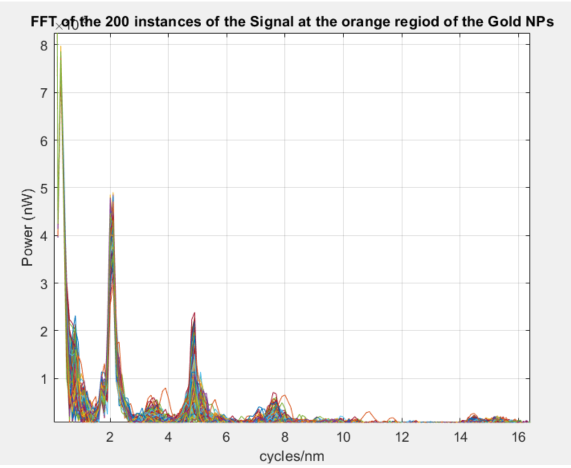
This can be achieved either through spectral and physics informed analysis of the spectrum results, using MATLAB, or through using a machine learning model, discussed next.

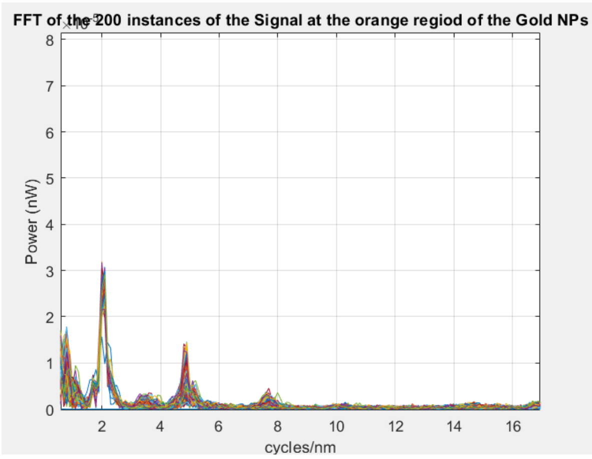
(9) *1st Proposition for making use of the obtained Data: Physics Based Analysis using MATLAB*

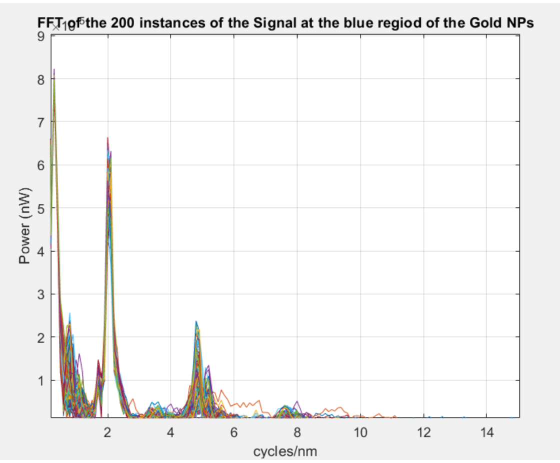
- *Data:* for three fixed distances from the wafer, obtain the spectrum results in the orange, blue, and grey area of the dried gold NPs and of silica.

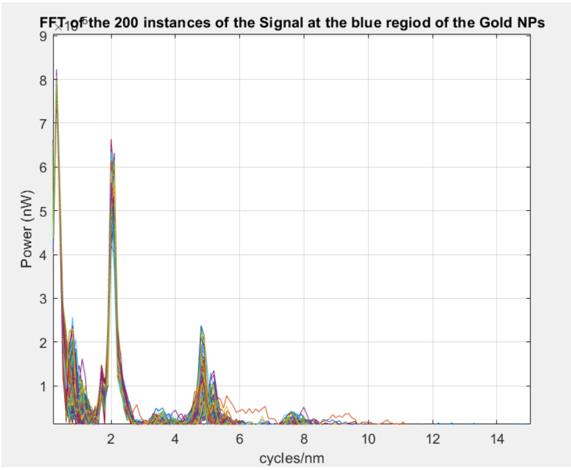
- *FFT & waterfall functions:* Use the FFT to determine the Fabry-Perrot cavity length (d_{FP}) and Michelson interferometer mismatch distance (d_M), and the waterfall function to observe for the stability.

For fixed ***d1*** *(distance from wafer)*,

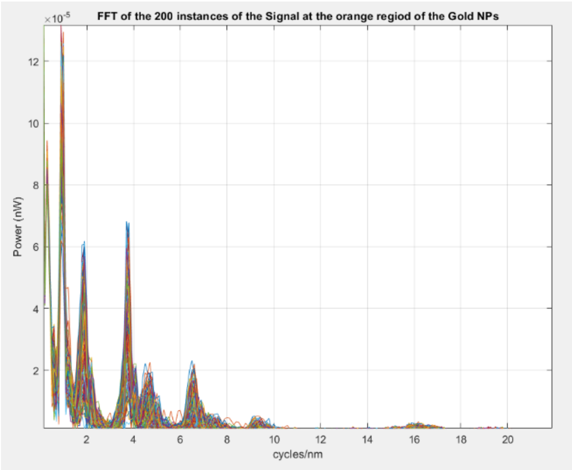


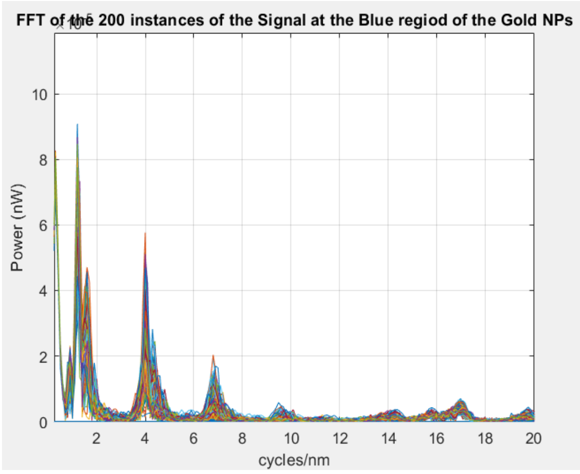






For fixed ***d2*** *(distance from wafer)*,





Observation:

Since the Michelson interferometer mismatch length results in fringe patterns of FSR exceeding the resolution of the OSA, it was expected to observe a single peak in the FFT spectrum, accounting of the FP cavity, however, multiple peaks are detected. This could be due to reflections of the emitted light off of other parts of the wafer, which are weak. Hence, the strongest peak should be used for the calculation of the FP cavity length.

For the results obtained for ***d1***, the computed FP cavity length is as given below:

$$d_{FP} \;=\; \frac{\lambda_o^2 \times f_{cycles/nm}}{2n}$$

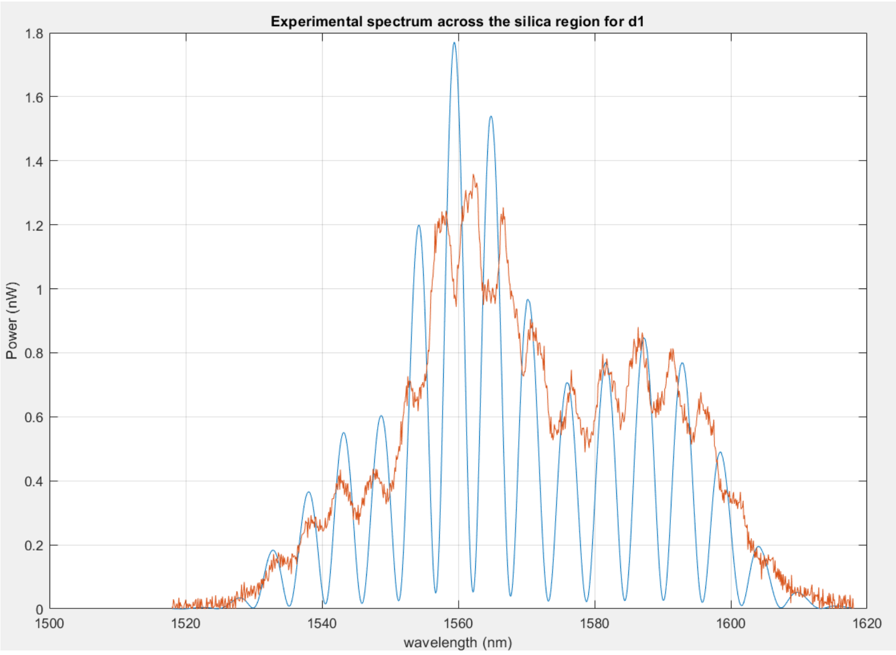
$$d_{FP} \;=\; \frac{(1550\times10^{-9})^2\times(0.2098\times10^9)}{2\times1.444} \;=\; \mathbf{174.5\ \mu m}$$

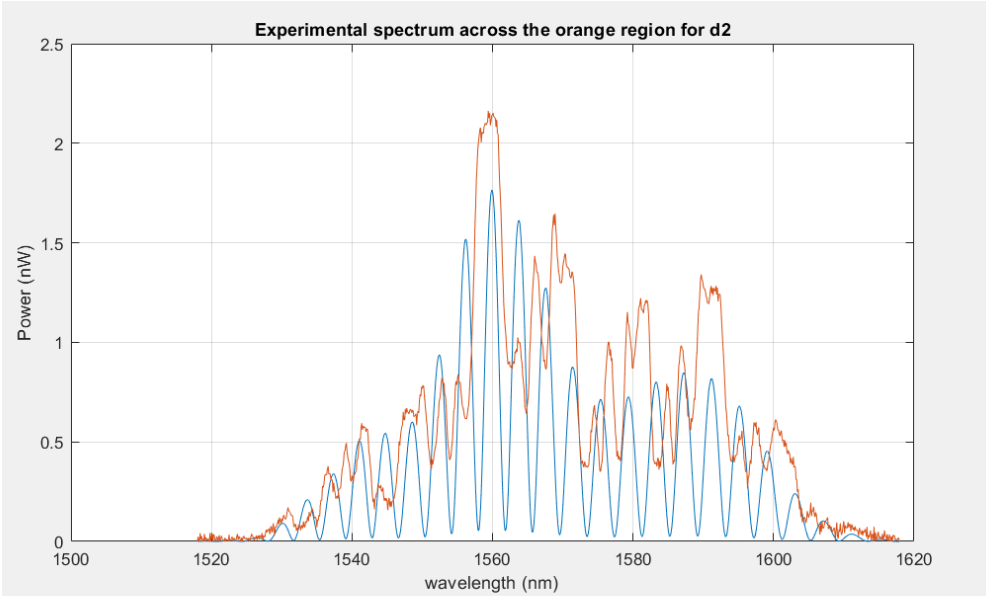
Moreover, the Michelson interferometer mismatch distance (d_M) was measured to be:

$$d_M \;\approx\; \mathbf{25\ cm}$$

- *Simulation:* Write the simulation code of the transmission spectrum for the setup.

The figure below shows the simulated spectrum superimposed on the experimental spectrum across silica.





<u>*Observation:*</u>

- *Compare (a):* Compare the experimental spectrum across the regions.

- *Compare (b):* Compare the simulated spectrum to the experimental results for any noticeable and characterizable differences.

(10) *2nd Proposition for making use of the obtained Data: ML-Classification Problem*

For their ability to capture complex and nonlinear relation between features and output, it was decided to use the **artificial neural network (ANN)** for the classification problem.

Features

Label

	Spectrum				*FFT*				*d*	*Region Category*
	λ_1	λ_2	λ_3	....	λ_1	λ_2	λ_3	....		
Cases 1										
…										

Observations

d (as a numerically encoded data)

Conclusion

References

https://www.jeppix.eu/wp-content/uploads/2025/03/Part-3-Chapter-19-MMI-Couplers-DRAFT.pdf

https://en.wikipedia.org/wiki/Optical_tweezers

https://en.wikipedia.org/wiki/Radiation_pressure

Appendices

Appendix A – ML Collected Data

Row	d class	Spectrum file	Label
1	0	AQ6370C_trace_D_timeseries_20260512_052728.csv	1
2	0	AQ6370C_trace_D_timeseries_20260512_053858.csv	1
3	0	AQ6370C_trace_D_timeseries_20260512_054359.csv	1
4	0	AQ6370C_trace_D_timeseries_20260512_054841.csv	1
5	0	AQ6370C_trace_D_timeseries_20260512_055906.csv	1
6	0	AQ6370C_trace_D_timeseries_20260512_061618.csv	1
7	0	AQ6370C_trace_D_timeseries_20260512_062041.csv	2
8	0	AQ6370C_trace_D_timeseries_20260512_062701.csv	2
9	0	AQ6370C_trace_D_timeseries_20260512_063101.csv	2
10	0	AQ6370C_trace_D_timeseries_20260512_063825.csv	2
11	0	AQ6370C_trace_D_timeseries_20260512_064748.csv	2
12	0	AQ6370C_trace_D_timeseries_20260512_065419.csv	2
13	0	AQ6370C_trace_D_timeseries_20260512_065931.csv	2
14	0	AQ6370C_trace_D_timeseries_20260512_070640.csv	3
15	0	AQ6370C_trace_D_timeseries_20260512_071044.csv	3
16	0	AQ6370C_trace_D_timeseries_20260512_072003.csv	3
17	0	AQ6370C_trace_D_timeseries_20260512_072704.csv	3
18	0	AQ6370C_trace_D_timeseries_20260512_073419.csv	3
19	0	AQ6370C_trace_D_timeseries_20260512_074325.csv	3
20	0	AQ6370C_trace_D_timeseries_20260512_074828.csv	3
21	0	AQ6370C_trace_D_timeseries_20260512_075402.csv	4
22	0	AQ6370C_trace_D_timeseries_20260512_081932.csv	4
23	0	AQ6370C_trace_D_timeseries_20260512_082957.csv	4
24	0	AQ6370C_trace_D_timeseries_20260512_083339.csv	4
25	0	AQ6370C_trace_D_timeseries_20260512_090034.csv	4
26	0	AQ6370C_trace_D_timeseries_20260512_090752.csv	4
27	0	AQ6370C_trace_D_timeseries_20260512_092125.csv	4
28	1	AQ6370C_trace_D_timeseries_20260512_094521.csv	1
29	1	AQ6370C_trace_D_timeseries_20260512_100347.csv	1
30	1	AQ6370C_trace_D_timeseries_20260512_101455.csv	1
31	1	AQ6370C_trace_D_timeseries_20260512_101846.csv	1
32	1	AQ6370C_trace_D_timeseries_20260512_103735.csv	1
33	1	AQ6370C_trace_D_timeseries_20260512_105730.csv	1
34	1	AQ6370C_trace_D_timeseries_20260512_110416.csv	1
35	1	AQ6370C_trace_D_timeseries_20260512_111122.csv	2
36	1	AQ6370C_trace_D_timeseries_20260512_111733.csv	2
37	1	AQ6370C_trace_D_timeseries_20260512_112136.csv	2
38	1	AQ6370C_trace_D_timeseries_20260512_114150.csv	2
39	1	AQ6370C_trace_D_timeseries_20260512_114602.csv	2
40	1	AQ6370C_trace_D_timeseries_20260512_115122.csv	2
41	1	AQ6370C_trace_D_timeseries_20260512_120240.csv	2
42	1	AQ6370C_trace_D_timeseries_20260512_120737.csv	3
43	1	AQ6370C_trace_D_timeseries_20260512_121318.csv	3

44	1	AQ6370C_trace_D_timeseries_20260512_123317.csv	3
45	1	AQ6370C_trace_D_timeseries_20260512_124105.csv	3
46	1	AQ6370C_trace_D_timeseries_20260512_124617.csv	3
47	1	AQ6370C_trace_D_timeseries_20260512_125250.csv	3
48	1	AQ6370C_trace_D_timeseries_20260512_125910.csv	3
49	1	AQ6370C_trace_D_timeseries_20260512_130944.csv	4
50	1	AQ6370C_trace_D_timeseries_20260512_131438.csv	4
51	1	AQ6370C_trace_D_timeseries_20260512_131927.csv	4
52	1	AQ6370C_trace_D_timeseries_20260512_132400.csv	4
53	1	AQ6370C_trace_D_timeseries_20260512_133335.csv	4
54	1	AQ6370C_trace_D_timeseries_20260512_134207.csv	4
55	1	AQ6370C_trace_D_timeseries_20260512_134549.csv	4
56	2	AQ6370C_trace_D_timeseries_20260512_135100.csv	1
57	2	AQ6370C_trace_D_timeseries_20260512_135554.csv	1
58	2	AQ6370C_trace_D_timeseries_20260512_140025.csv	1
59	2	AQ6370C_trace_D_timeseries_20260512_140831.csv	1
60	2	AQ6370C_trace_D_timeseries_20260512_141445.csv	1
61	2	AQ6370C_trace_D_timeseries_20260513_072553.csv	1
62	2	AQ6370C_trace_D_timeseries_20260513_073133.csv	1
63	2	AQ6370C_trace_D_timeseries_20260513_073912.csv	2
64	2	AQ6370C_trace_D_timeseries_20260513_073912.csv	2
65	2	AQ6370C_trace_D_timeseries_20260513_074952.csv	2
66	2	AQ6370C_trace_D_timeseries_20260513_075526.csv	2
67	2	AQ6370C_trace_D_timeseries_20260513_082815.csv	2
68	2	AQ6370C_trace_D_timeseries_20260513_080753.csv	2
69	2	AQ6370C_trace_D_timeseries_20260513_082009.csv	2
70	2	AQ6370C_trace_D_timeseries_20260513_090315.csv	3
71	2	AQ6370C_trace_D_timeseries_20260513_091702.csv	3
72	2	AQ6370C_trace_D_timeseries_20260513_092532.csv	3
73	2	AQ6370C_trace_D_timeseries_20260513_093722.csv	3
74	2	AQ6370C_trace_D_timeseries_20260513_094406.csv	3
75	2	AQ6370C_trace_D_timeseries_20260513_095729.csv	3
76	2	AQ6370C_trace_D_timeseries_20260513_100305.csv	3
77	2	AQ6370C_trace_D_timeseries_20260513_101605.csv	4
78	2	AQ6370C_trace_D_timeseries_20260513_102229.csv	4
79	2	AQ6370C_trace_D_timeseries_20260513_103018.csv	4
80	2	AQ6370C_trace_D_timeseries_20260513_103833.csv	4
81	2	AQ6370C_trace_D_timeseries_20260513_110038.csv	4

Appendix B – Summary of the Paper on Gold NPs LSPR for Biosensing Applications

Paper Title:

Spectroscopic Ellipsometry Study of Gold Nanostructures for LSPR Bio-Sensing Applications

Objectives and Approach:

This work demonstrated the relationship between the size of the Au Nps and the shift on the position of the LSPR band on the observed absorption and spectroscopic ellipsometry (SE) spectra. This work also compared sensitivity of these analysis methods in detecting the change as function of the change on refractive index of the medium surrounding the AU NPs. Moreover, total internal reflection ellipsometry (TIRE) was also tested in detecting the attachment of in a biosensing application.

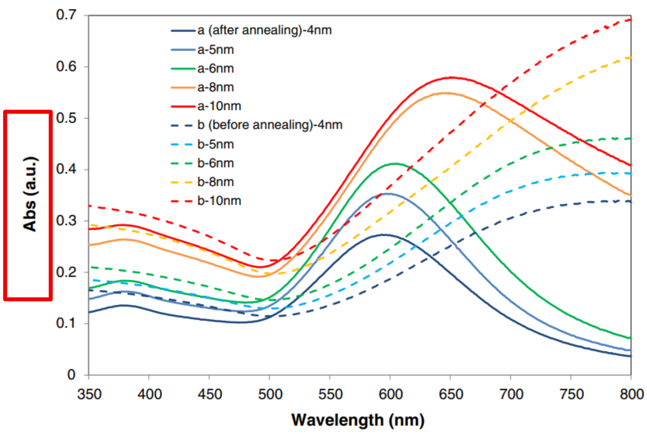
Findings (*partial*):

Au Size and Synthesis Specification

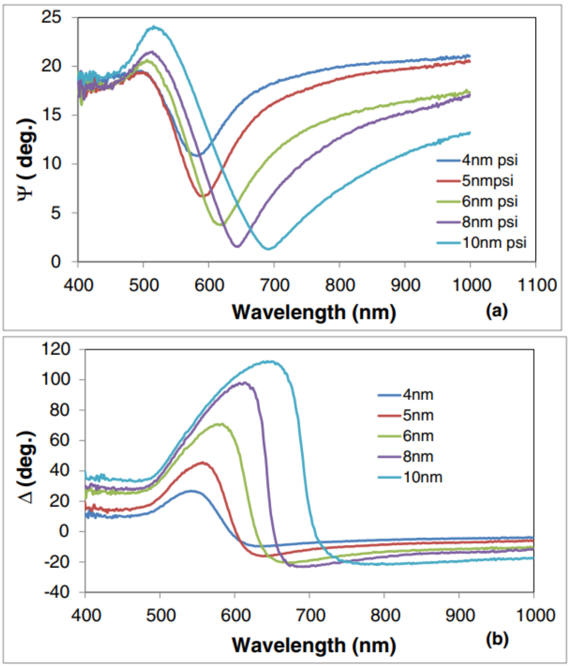
- The mean size of Au nano-islands increases with the increase of the nominal thickness of the annealed Au films.
- Au nano-islands formed at higher temperature 550 °C and longer annealing time (10h) are larger than those annealed at lower temperature and shorter time.

Optical Measurements

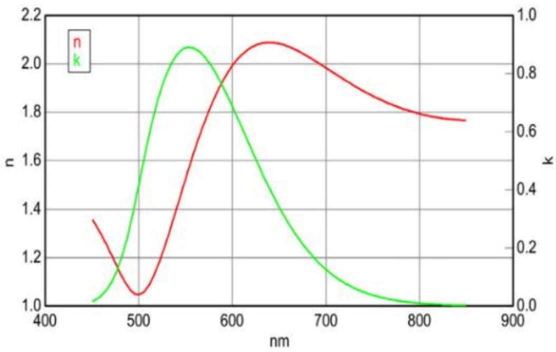
- Absorption Spectra: The position of LSPR band shifts to higher wavelength with the increase in the mean size of Au nano islands.



- SE Spectra: The position of LSPR features shift towards high wavelengths when the size of Au nano-islands increases.

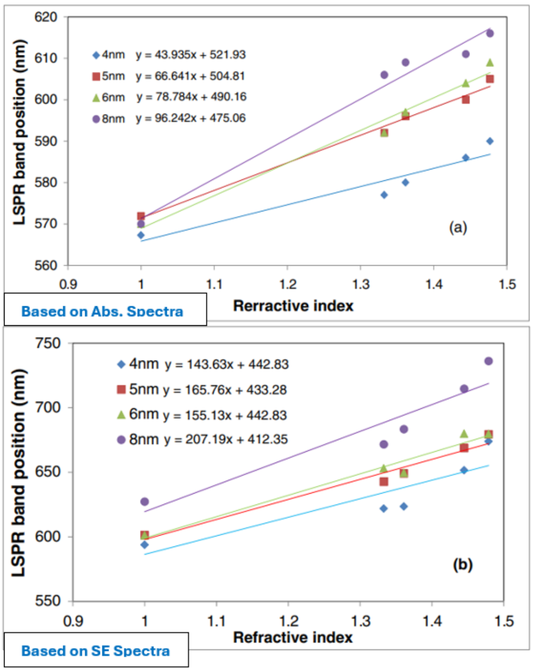


- Using a dedicated J.A. Woollam software for Ellipsometry data fitting, resulting in the dispersion characteristics of refractive index (n) and extinction coefficient (k), typical plot shown below.



Evaluation of refractive index sensitivity

- RIS increases with the increase in the size of Au nano-islands
- SE measurements gave higher RIS values as compared to that obtained by absorption spectroscopy



Proposed Next Steps: *FDTD Simulation*

Setup Specifications

- Array of Au particles with non-ideal cylinder contours.
- **Thickness** is 4 nm and **radius** is 30 nm.
- Let particles be separated by a **gap** of 20 nm.
- Particles are sitting on **silica substrate of thickness 340 nm**
- Silica's refractive index and wavelength are related by **Sellmeier e**quation.
- **Drude-Lorentz** model employed for dispersion relation of Au
- Broadband plane wave **source** is used spanning the wavelength range (300 nm – 780 nm)
- Uses periodic **boundaries**, enclosing a simulation domain of 5um x 3um x 2 um (width, length, thickness)
- Uses fine **mesh** of size 0.5nm in the xy direction and 1 nm laterally
- Let the **polarization** be s-polarization and direction of propagation be normal incidence
- Monitor **output** includes the Electric field intensity plot of the modes and the absorption and the reflection spectra (normalized with respect to structure without Au NPs)

Appendix C – Summary of the Paper on usage of Gold NPs LSPR Deposited on Fiber for Sensing Applications

Paper Title:

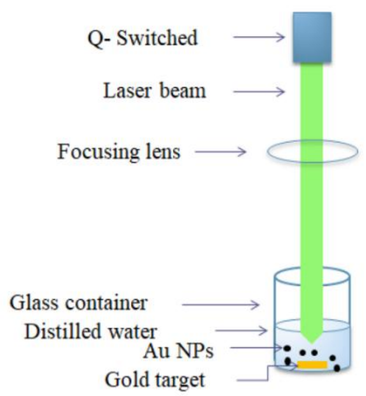
A Gold Nanoparticles Coated Unclad Single Mode Fibre-Optic Sensor Based on Localized Surface Plasmon Resonance

Objectives and Approach:

This paper proposes and analyses the design of a bio-sensing device which uses silica glass optical fibre unclad over a small potion where Au NPs are deposited. Hence making use of the shift of the LSPR band for the detection of the change in the refractive index of the analyte. The NPs are generated using pulsed radar ablation in liquid (PLAL) technique, with the ablation energy controlling the concentration and sizes of the synthesized Au NPs. The paper demonstrates an improved RIS compared to previous work, including those using plastic and photonic crystal fibres.

Methodology and Findings (***partial***):

- Synthesis of Au NPs using pulsed radar ablation in liquid (PLAL), demonstrated below.

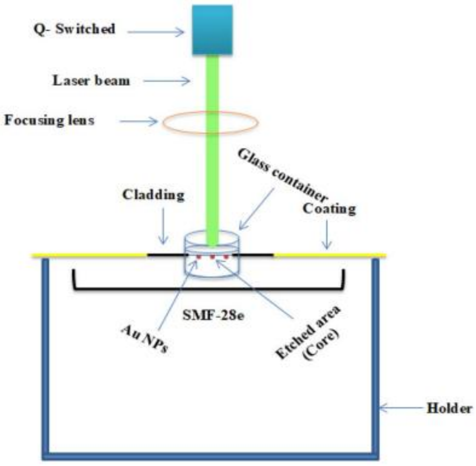


- Preparation of the **Analyte** samples.

A magnetic stirrer was used to mix three samples with de-ionized (DI) water at 25 °C for 10 min. The resultant solutions are **NaCl-DI** water, **Sucrose-DI** water, and **glycerol-DI** water.

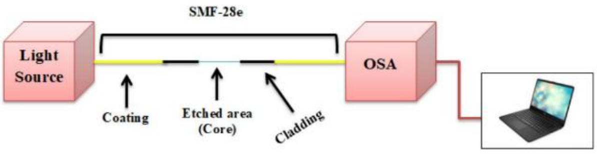
- Unclad/etch a standard single-mode fibre SMF-28e
- Deposition of the Au NPs on the unclad core, demonstration figure below, where:

A laser beam with a wavelength of 415 nm and a power of 5 mw was used to deposit the gold NPs on the fibre core using a laser-induced method for a duration of 4 hours. The applied laser beam was expanded along the etched section using a beam expander

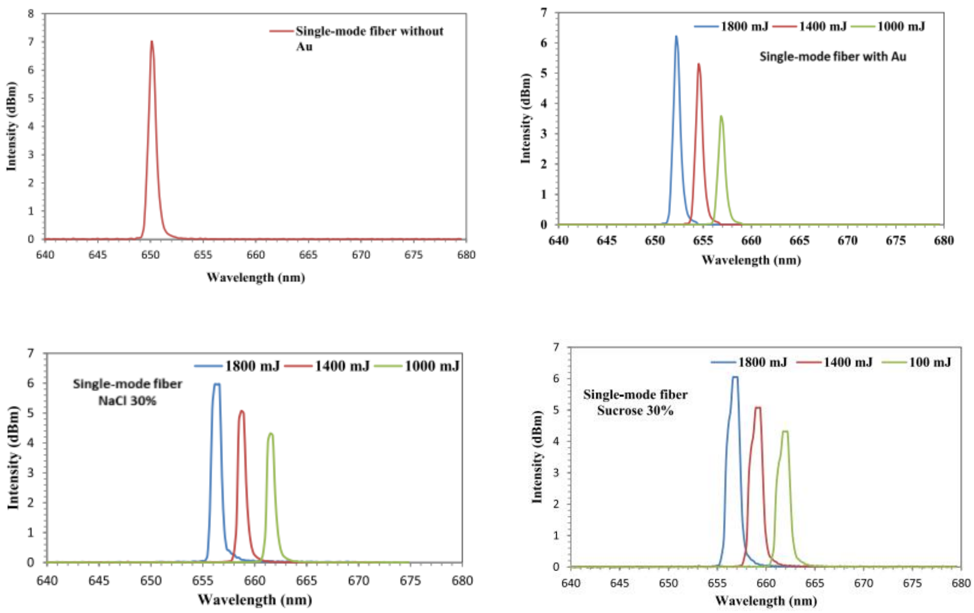


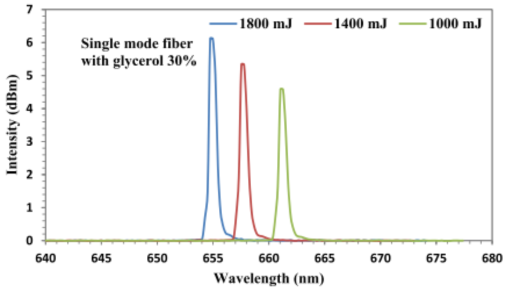
- Testing the sensor using the OSA (setup shown below)

The OSA analyser results were obtained for the sensor without the Au NPs, with the Au NPs with air as analyte, and finally for the three prepared analytes solutions.



OSA results:





Conclusion drawn from OSA results:

- As expected, the results demonstrate that as the analyte is varied (the effective refractive index of medium outside the core), the LSPR wavelength undergoes and red shift. The underwent spectral shift is specific to the Au NPs size (ablation energy)
- *It was expected that the paper would summarize the results using a plot, with each analyte solution being a refractive index point, and the calculated resonance wavelength for each, enabling a plot showing the sensitivity for each ablation energy (Au NPs size and concentration)*

Overall, it seems like the paper is not well organized, and conclusions made are not accurate nor consistent with the abstraction results

Appendix C – Internship Report: The Fabry-Pèrot Interferometer Experiment

Report Title:

The Fabry-Pèrot Interferometer Experiment

Report Objectives:

- To study the Fabry–Pérot interferometer and its fringe profile, examining the interplay between the wavelength, angle of incidence, and cavity length parameters.
- To emulate the Fabry–Pérot interferometer using optical fibres.
- To analyse experimental results and compare them against theoretical expectations

Report Summary:

This report studies the background theory of the Fabry-Pérot interferometer, the Airy function, and the interplay between the wavelength, cavity length, and angle parameters on the Fabry-Pérot fringe profile. Finally, the experimental setup, procedure, and specifications of the components and equipment used are discussed. The analysed results and future work are also presented.

Shayma Alteneiji | 100063072

Internship Supervisor:

Prof. Jaime Viegas

Table of Contents

Introduction

In the book *Introduction to Optics* by Pedrottis, the authors define an interferometer as an instrument that uses interference patterns, or fringe profiles, resulting from optical path differences between interfering waves for various applications, ranging from high-resolution wavelength measurements and determining the refractive indices of glasses to measuring the Moon's tidal effect on Earth. An interferometer encompasses two types: a wavefront-division interferometer, such as Young's double-slit experiment, and an amplitude-division interferometer, such as the one studied in this report, the Fabry-Pérot interferometer. The interferometer's theoretical background is discussed in the next section.

Theoretical Background

This section details the theoretical background of the Fabry–Pèrot interferometer, including a discussion of the optical path length, the standard setup of the Fabry–Pèrot interferometer, the interferometer's transmittance as a function of the round-trip phase shift, and finally, the interactions between the wavelength, distance, and angle of incidence parameters in the produced interference patterns.

Optical Path Length & Constructive and Destructive Interferences

The optical path length equation and method of derivation for the interfering waves in the Michelson and Fabry–Pèrot interferometers, as well as other interferometers, are the same, with only minor differences. The interferometer shown in Figure 1 resembles the Fabry–Pèrot interferometer, with the region between the two mirrors forming the Fabry-Pèrot cavity.

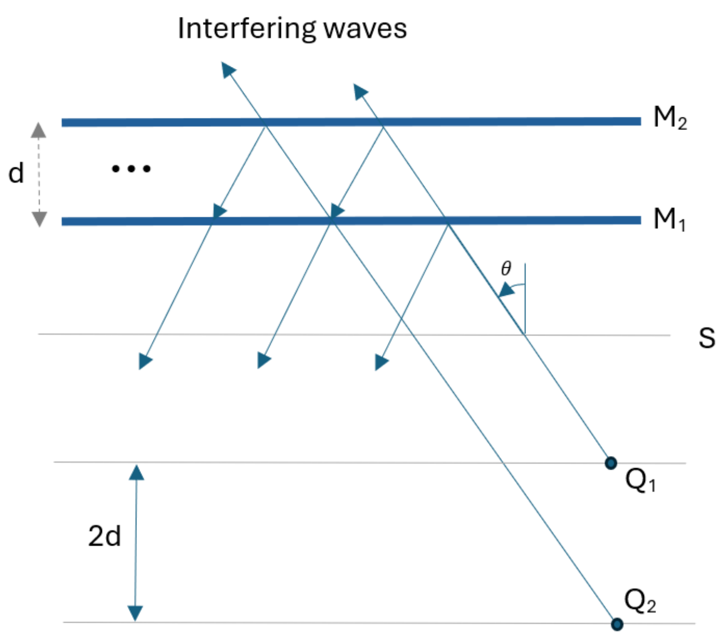


Figure 1. The Fabry- Pèrot Interferometer

Assuming, a ray from a source S was incident at an angle of θ into the first mirror of the cavity, a fraction of the incident wave is reflected, while the remaining fraction is transmitted. A portion of the transmitted wave then transmits through the second mirror (M2) and is captured at the photodetector, the remaining reflected fraction gets then reflected back towards the first mirror, then reflected back towards the second mirror, with the transmitted fraction now interfering with the first transmitted fraction, producing the interference patter we view. The remaining reflected fractions continue contributing through higher-order reflections. Considering the first two rays, it can be shown that the optical path difference (OPD) between them is as given below.

$$OPD = 2 \cdot d \cdot cos(\theta)$$

Constructive interference occurs when the OPD is an integer multiple of the wavelength, as expressed below.

$$m\lambda = 2 \cdot d \cdot \cos(\theta)$$

Note that the π phase shift introduced when light propagates from a low-index (air) to a high-index (mirror) medium is countered by a second π phase shift due to intersection by the second mirror, before transmission to the photodetector. Destructive interference occurs when the OPD equals an odd multiple of half the wavelength, as shown below.

$$m\lambda + \frac{\lambda}{2} = 2 \cdot d \cdot \cos(\theta)$$

$$\frac{\lambda}{2} \cdot (2m + 1) = 2 \cdot d \cdot \cos(\theta)$$

The corresponding phase shift as a function of the optical path length is as given below. Constructive interference occurs when the phase shift is an integer multiple of 2π, while destructive interference occurs when the phase shift is an odd multiple of π.

$$\delta = k \cdot OPD = \frac{4\pi \cdot d \cdot cos(\theta)}{\lambda}$$

Considering a monochromatic wave (constant wavelength) and fixed cavity length, d, (an etalon Fabry-Pèrot cavity), then if the interference intensity is measured spatially, concentric circular fringes are produced (Figure 2), with each m-order fringe (bright or dark) corresponding to some angle of incidence that satisfies the interference condition (bright or dark).

t can also be shown that as the optical path length decreases (larger wavelength, smaller cavity length, or larger angle of incidence from the optical axis), the angular separation between consecutive fringes increases with the separations decreasing as the fringe m-order increases. In the next section, the OPD and phase-shift equations are used to interpret the Fabry–Pèrot interferometer Airy function.

Figure 2. Alternate ordering fringes patterns

The Fabry-Pèrot Interferometer and the Airy Function

Considering the setup shown in Figure 1, the Fabry-Pèrot cavity transmittance (T) can be derived as the ratio of the squared magnitude of the transmitted wave to that of the incident wave. The transmittance equation can be expressed in the form of the Airy function as given below:

$$T = \frac{1}{1 + F \cdot \sin^2\left(\frac{\delta}{2}\right)}$$

Where F is referred to as the coefficient of finesse, given by:

$$F = \frac{4r^2}{(1 - r^2)^2}$$

Figure 3 shows a plot of the transmittance, which varies from 0 to 1, as a function of the phase shift, δ, for various reflection coefficients, r. The greater the reflectivity of the cavity mirrors, the higher the coefficient of finesse and the more visible the observed interference pattern, or fringes. For this reason, the transmittance plot is also referred to as the fringe profile.

Moreover, the relation between the reflection coefficient and the fringe visibility can be explained as follows. As the reflection coefficient decreases, the reflectance (r^2) decreases, and the more rapidly does higher-order reflections power decreases per reflection. This reduces the number of non-negligible interfering signals reaching the detector, resulting in fringes with lower visibility.

To physically intuit the Fabry- Pèrot cavity transmittance (T) function, we restate the delta, (δ), expression given below.

$$\delta = 2 \cdot k \cdot d \cdot \cos(\theta) = \frac{4\pi d \cos(\theta)}{\lambda}$$

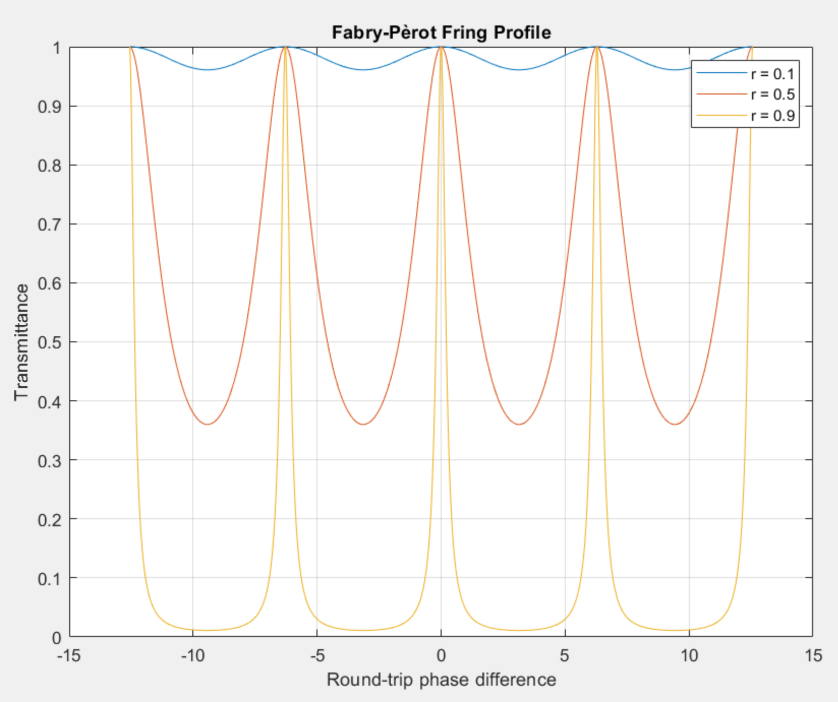


Figure 3. The Fabry- Pèrot Transmittance (T) as a function of the round-trip phase difference for various reflection coefficient values., r = 0.1, 0.5, and 0.9.

From the plot (Figure 3), constructive interference occurs when δ is an integer multiple of 2π. Therefore, the axis against which the Airy function is plotted can be re-expressed in terms of the parameters d, λ, and θ. By fixing two of these parameters at a time and varying the third, the values at which constructive interference occurs correspond to conditions under which the Fabry–Pèrot cavity produces strong transmission. While the values at which destructive interference occurs indicate combinations of parameters that are suppressed by the Fabry-Pèrot cavity. In the next section, we discuss how varying these parameters affects the transmittance plot.

The Airy Function Against the Parameters d, λ, and θ

To reliably study the interplay of the parameters affecting the produced fringe profile, in the following subsections, we examine and discuss each parameter independently, with the others held fixed.

Varying the Cavity Length (d) and the Source Wavelength (λ)

The parameter d controls both the OPL as well as the phase shift between the interfering waves. The equation below solves for (d) under the condition of bright fringes.

$$d = \frac{m \cdot \lambda}{2\cos(\theta)}$$

Considering only rays from a monochromatic source incident at the same angle from the optical axis, the equation suggests that when the cavity is made an integer multiple of the incident source half-wavelength, constructive interference is observed (provided that d does not exceed the source coherence length). The problem becomes more complicated when considering a broadband source.

Each constitutive wavelength or frequency component making up the spectrum of the broadband source can be treated independently. For a given fixed cavity length and angle of incidence, wavelengths that can be expressed as given below interfere constructively. The expression is derived from the round-trip phase equation.

$$\lambda = \frac{2d\cos{(\theta)}}{m}$$

If we plot the wavelength as a function of the discrete integer (m), the known inverse function would result. As (m) increases, the spacing between each consecutive wavelength satisfying the constructive interference decreases.

If we consider two cavities of different lengths (d), and the mapped m values for each wavelength is always larger for the larger cavity length. Hence the larger the cavity, the more do we translate along the m axis were spacing between the constructive pattern wavelength is progressively decreasing, resulting in more closely spaced the fringes and making them less likely to be observed in a spectrum analyser of finite resolution.

Figures 4 and 5 show the fringe profile against the wavelength for two cavity lengths with a controlled angle of incidence. This analysis suggests that the cavity can be thought of as a filter, with one of its parameters being cavity length (d), allowing the suppression of certain wavelengths and the enhancement of others. The effect of varying the angle of incidence is discussed in the next subsection.

Varying the Source Angle (θ)

As for the effect of varying the angle of the incident rays, the same idea applies as when varying the cavity length (d). The cosine function varies from 0 to 1 for angles from 90° to 0°, respectively. The larger the angle of incidence, the smaller the value of m required to represent a given wavelength. Hence, as expected, in Figures 6 and 7, the plots for the fringe profile against the wavelength show that when $\theta = 0°$ and $60°$ with $d = 1$ mm, the fringe profile is more spread for the latter.

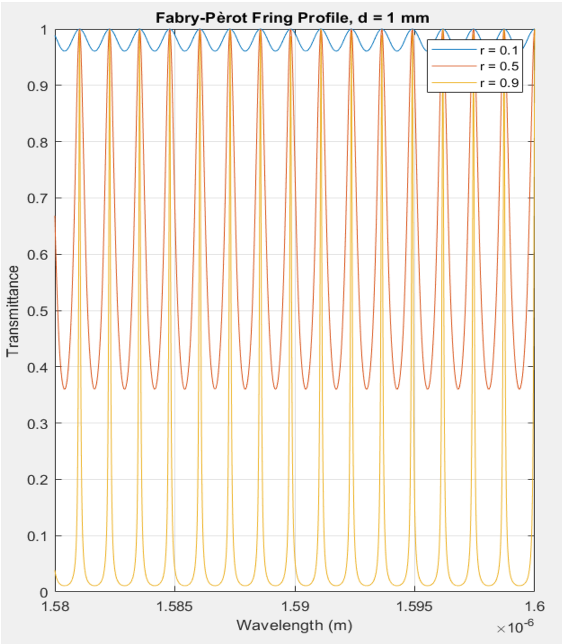


Figure 4. The Fabry- Pèrot Transmittance (T) as a function of wavelength when d = 1 mm, for various reflection coefficients r = 0.1, 0.5, and 0.9.

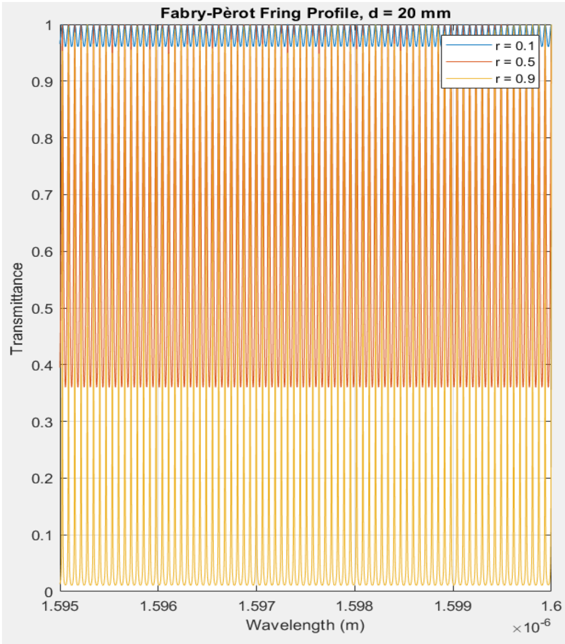


Figure 5. The Fabry- Pèrot Transmittance (T) as a function of wavelength when d = 20 mm, for various reflection coefficients r = 0.1, 0.5, and 0.9.

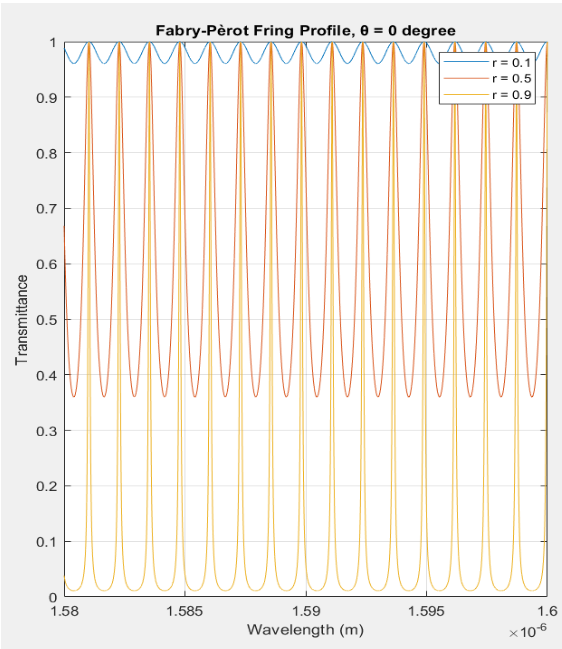


Figure 6. The Fabry- Pèrot Transmittance (T) as a function of wavelength when $\theta = 0°$, for various reflection coefficients r = 0.1, 0.5, and 0.9.

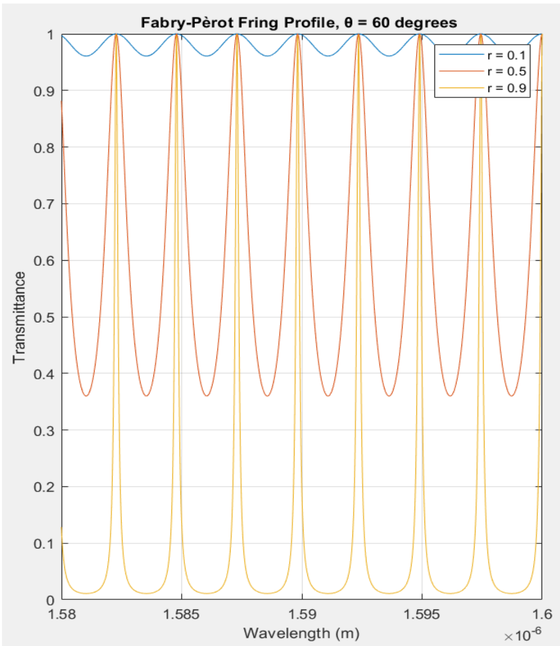


Figure 7. The Fabry- Pèrot Transmittance (T) as a function of wavelength when $\theta = 60°$,, for various reflection coefficients r = 0.1, 0.5, and 0.9.

The Optical Fiber Experimental Setup and the Likeness to the Fabry-Pèrot Interferometer

The optical fibre experimental setup emulates the Fabry-Pèrot interferometer, in that the glass, which is the optical fibre cladding, forms the mirrors from which the source signal is reflected. The source is directly fed through the optical fibre from the transmitter end, and the optical fibre at the receiver end collects the interfering waves. The captured waves are then fed to a spectrum analyser, allowing the view of the resulting interference pattern as a function of wavelength. Moreover, for an air-to-glass interface, the reflection coefficient is calculated as shown below:

$$r = \frac{n_{air} - n_{glass}}{n_{glass} + n_{air}}$$

$$r = \frac{1.5 - 1}{1 + 1.5} = -0.2$$

Hence, the observed fringe patterns are expected to have only mild visibility. Note that the negative sign indicates a phase reversal. In the experiment, a broadband source was used, and the distance between the two optical fibres was varied to study its effect on the visibility and contrast of the fringes. Figure 8 shows a drawn diagram of the experimental setup. The infrared source expands out of a 10 μm diameter circle of the optical fibre, forming approximately a 1 mm diameter cone. Since at the receiving end the optical fibre only captures the interfering waves over a spatial length of 10 μm diameter, we may assume that the captured waves have the same angle of incidence, and hence the angle parameter is assumed constant.

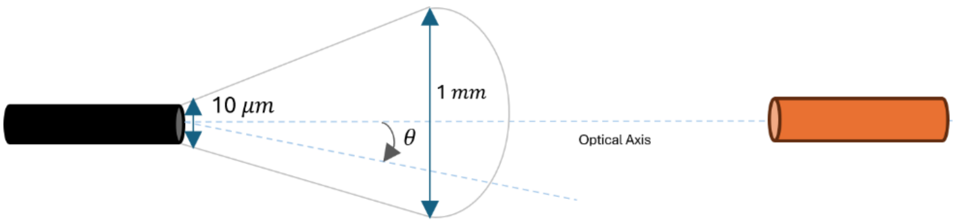


Figure 8. Optical fibre experimental setup diagram

As mentioned above, the cavity can be thought of as a filter, with the transmitter Airy function being its transfer function in the wavelength domain. Accordingly, each wavelength component making up the source spectrum, $S(\lambda)$, gets modulated in amplitude by an amount specified by the cavity's transmittance Airy function, which is based on its length. Hence, the resulting output signal spectrum measured at the receiver side is expressed as shown below. In the next section, we discuss the experimental methodology.

$$I(\lambda) = S(\lambda) \cdot T(\lambda)$$

Methodology

In this section, we discuss the methodology of conducting the experiment. Including the specifications of the components and equipment used and the procedure followed.

Summary of Experimental Components and Equipment Specifications

Listed below are the components and equipment used and their specifications:

- Optical Spectrum Analyzer: YOKOGAWA AQ6370C (Figure 9)

Operates in the telecom range of 600 nm to 1700 nm, has a wavelength resolution of 0.02 nm, and a sensitivity of -90 dBm, hence can detect signals in the pico-W range.

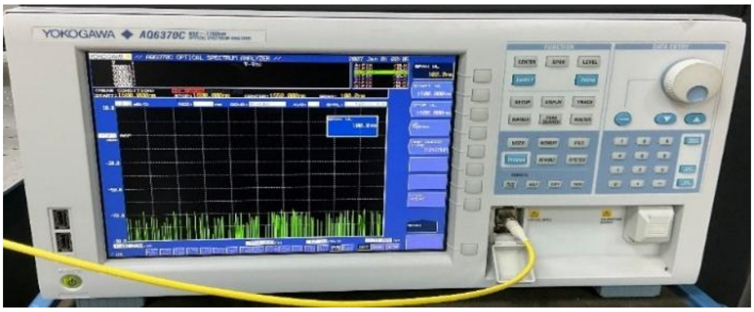


Figure 9. Optical spectrum analyzer, YOKOGAWA AQ6370C

- C & L Band ASE730 Light Source (Figure 10)

Operates in the wavelength range of 1530 nm to 1610 nm and delivers + 15 dBm over the operating wavelength range.

Figure 10. C & L Band ASE730 Light Source

- Polarizer Controller (Figure 11)

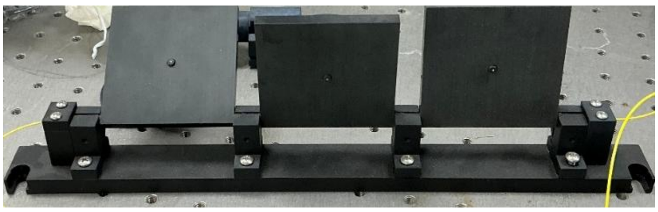


Figure 11. Polarizer controller

- Opto-mechanical System Set (Figure 12)

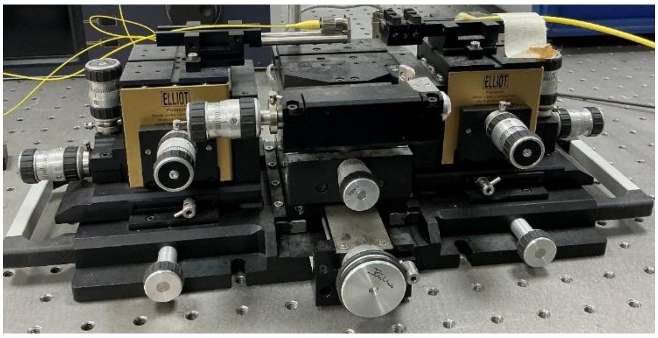


Figure 12. Opto-mechanical System Set

- Ferrule Connecter/ Physical Contact Fiber Connector (Figure 13)

Figure 13. FC/PC fibre connector, polished at 90 degrees.

- Optical Fibre

Experimental Setup Description and Procedure

From the receiver end, an optical fibre was connected to the C & L Band ASE730 Light Source, then through the polarizer controller, and finally connected to the optical fibre FC/PC connector, which is screwed into the opto-mechanical system, as shown in Figure 14. From the transmitter end, an exposed fibre was placed and taped in alignment with the receiver optical fibre (Figure 14). The end of the exposed fibre was connected to the YOKOGAWA AQ6370C optical spectrum analyser.

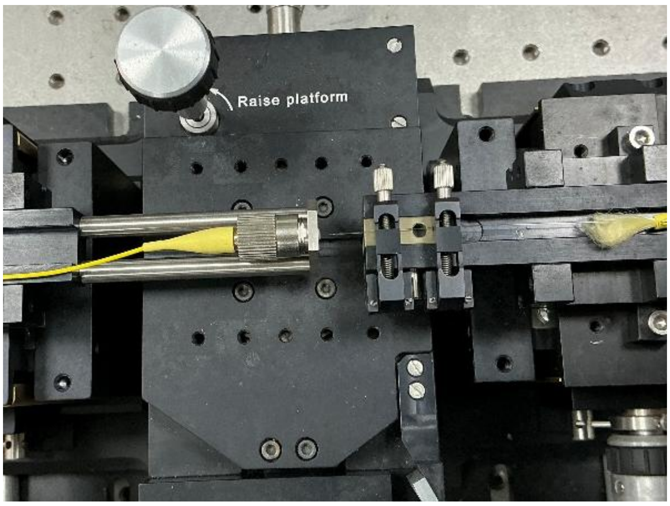


Figure 14. Experimental setup, showing the receiver and transmitter sides.

Before recording the spectrum analyser results, it was crucial to precisely align the receiver and transmitter fibres using the three rotational controllers of the opto-mechanical setup. This was done carefully, as shown in Figure 15. In the next section, we present and discuss the spectrum analyser results captured for different cavity lengths.

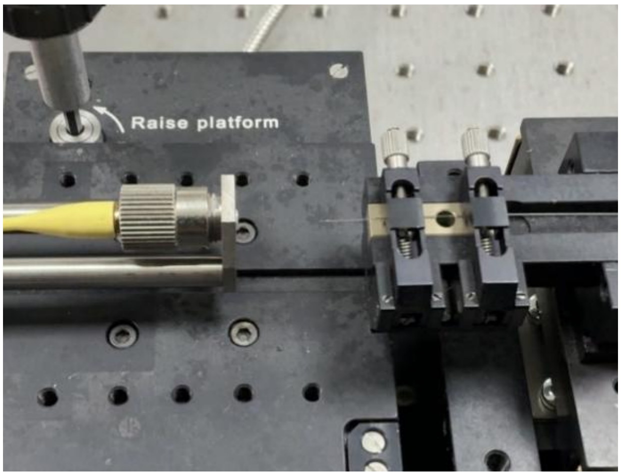


Figure 15. Experimental setup, showing the receiver and transmitter fibre optics after careful alignment of both sides.

Results and Analysis

Figures 16 and 17 show the fringe patterns on the spectrum analyser for two different cavity lengths. Figure 16 shows the fringes more spread out compared to those of Figure 17, indicating that the cavity length was smaller for the former. The observed higher power measurement in Figure 17 compared to Figure 16 could be due to improved alignment between the receiving and transmitting optical fibres.

The measured power was in the range of nW, several orders of magnitude lower than the source signal power, which is delivered in the mW range. This is expected due to losses along the optical fibre line and as the signal crosses the gap to the other fibre.

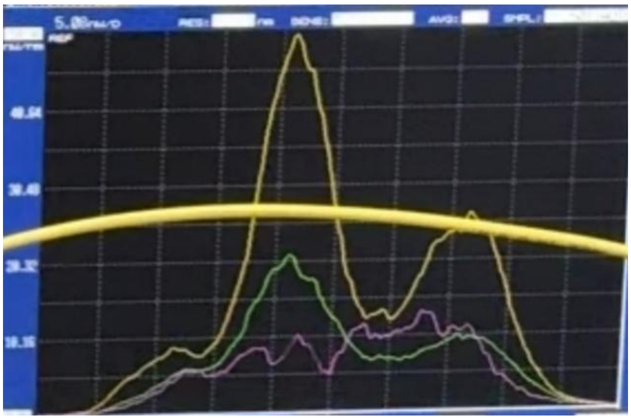


Figure 16. Recorded optical spectrum analyser measurement one results. Pink trace showing the fringe patterns for a given cavity length.

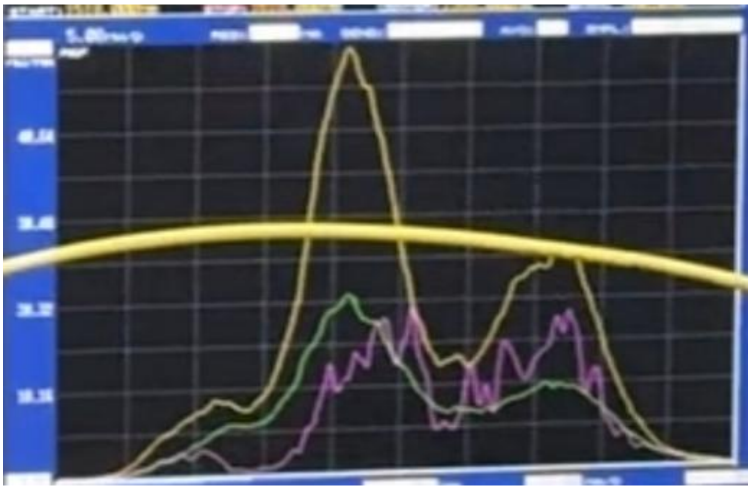


Figure 17. Recorded optical spectrum analyser measurement two results. Pink trace showing the fringe patterns for a given cavity length.

Future Work

Future work will involve:

- Simulation of closely spaced Fabry-Pèrot cavities to study the power distribution in 1-to-N port configurations and determine separation thresholds that minimize leakage.
- Exporting the measured traces as .csv files to extract the coefficient of finesse and approximate the cavity's absolute length.

Reference

F. L. Pedrotti, Leno Matthew Pedrotti, and L. S. Pedrotti, *Introduction to optics*. Cambridge, United Kingdom: Cambridge University Press, 2018.

Appendix D – Internship Report (incomplete): Rectangular Waveguide Theoretical Background, Design, and Simulation using Ansys Lumerical

Report Title:

Rectangular Waveguide Theoretical Background, Design, and Simulation using Ansys Lumerical

Report Objectives:

- To study the theoretical principles of 2D metallic and dielectric waveguides, including mode formation and wavevector discretization.
- To analyze field confinement, polarization, and the impact of waveguide dimensions on effective refractive index.
- To explore the differences between metallic and dielectric waveguides.
- To use simulations to visualize mode patterns and polarization effects.

Shayma Alteneiji | 100063072

Internship Supervisor:

Prof. Jaime Viegas

Introduction

A waveguide is a structure which guides light along its path. Through ray optics simplifications, this is modelled by two rays undergoing total internal reflection with the boundary. The usage of ray optics is however limiting in analysing waves traveling through waveguides [1], instead wave optics is adopted, as we shall be introduced. This report discusses a simple 2D metallic waveguide discussing the definition of modes and wavevector discretization as well introduces the metallic waveguide properties. The following section then distinct dielectric waveguides from metallic waveguides and discusses their properties. Generalization to 3D waveguides is also presented.

Following that, simulation results explore the confinement of the modes polarization (TE and TM), modal and group velocity dispersion as a function of waveguide thickness and width, and finally the final section utilizes the plots of the different appearing modes at different waveguide lengths for the design of a single mode waveguide.

Simple 2D Metallic Waveguide

Consider a waveguide constrained in the y-direction by two perfect mirrors or metal plates separated by distance **d**, and with the plate being infinite in the x and z directions. When this waveguide is excited, two things differ from the known plane wave propagation. (1) The waveguide cannot, mathematically, support TEM modes, only either TE or TM modes [2, 3]; with TM mode defined as when the magnetic field intensity (**H**) is transvers to k, while the electric field intensity (**E**) is not ($E_z \neq 0$), and vice vera for TE mode.

The reason for that is because waveguides are only single conductor structures, by Ampere's law, circulations of the transverse magnetic field are induced by current flowing through a second conductor as with coaxial cables which support TEM, the absence of that current suggest there are no closed contour, and transverse $\mathbf{k} \perp \mathbf{H} \perp \mathbf{E}$ mode cannot be supported [3].

(2) The propagating waves must also satisfy the boundary condition set by the material at the interface in the y-direction. For a metal or a perfect mirror interface, the tangential component of the electric field must go to zero at the surface. For that, the phase of the reflected wave electric field is shifted by a π.

The second observation made renders useful when studying how a given incident wave can propagate sustainably throughout the guide, which is what is referred to as the self-consistency condition. It is where the interfering reflections of the wave come together at a given wavefront plane while all having the same phase [4]. A given bounce angle dissociates the wave's wavevector into two components, transverse ($k_y = k \cdot \sin \theta_m$) and longitudinal ($\beta = k \cdot \cos \theta_m$), which is Considering the phase shift introduced by OPD transversely and the that introduced by the metal boundary (-2π) (comment on)assuming an x-polarized wave, the phase shift equation for self-consistency to be satisfied is as given below.

$$2k_y d - 2\pi = 2\pi m$$

$$2\left(\frac{2\pi}{\lambda}\right)d \cdot \sin\theta_m - 2\pi = 2\pi m$$

θ_m is the bounce angle from the line parallel to the metal plane. Since the transverse wavevector describes the rate of the spatial variation of phase along the y-direction. The equation therefore suggests that for given a waveguide length d, only a discrete number of wavevectors produce sufficient to fit within the waveguide and compatible phase variations for a given mode pattern. $k_y$is referred to as the eigenvalue or the characteristic value of the waveguide.

Hence for two different of frequencies within the allowed range, the same geometrical patterns are produced, with the wavevector scaled by the bounce angle, given by the equation below, to produce the same transverse variation. Where discretization of the wavevector leads to discretization of the bounce angle.

$$\sin\theta_m = m\frac{\lambda}{2d}$$

Moreover, when two waves of opposite bounce angles interfere together, the transvers pattern forms become standing and does not vary along the waveguide axis [1], as illustrated in Figure 1. Therefore, the formation of those field patters, which are referred to as *mode*, is a direct consequence of the self-consistency rule.

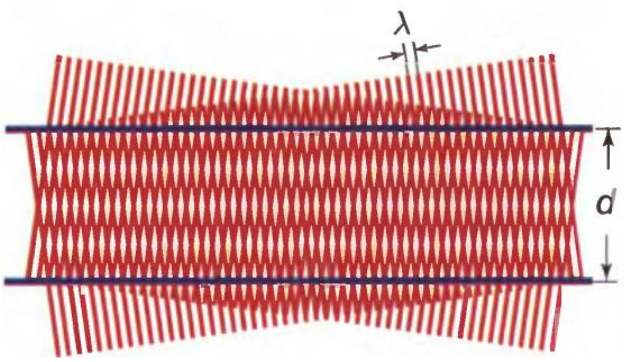


Figure 1. Two interfering waves of opposite bouncing angles, forming unvarying field distribution along the direction of propagation. [1]

Thus far, ray optics was used to visualize the formation of the modes, wave optics, can be used to derive the same conclusions and expand in wave propagation behaviour. Source [2] considers a 3D rectangular waveguide, leading to the consideration of extra E and H components along x, as well as the extra transverse wavevector component along x; it derives the conclusion of the discretization of the wavevector and the mode patterns as follows.

Using the conclusion that TEM cannot be supported, the TE and TM modes were defined as given. Using the curl of (E) and (H), equations for Ex, Ey, Hx, and Hy where derived as functions of Hz and Ez. Then the homogenous wave equation's longitudinal component (z) of the fields is only considered.

With the assumption that the longitudinal expression varies exponentially with z, when constraining the resulting differential equation is solved using separation of variables, the resulting solutions of field patterns are (1) independent of (z), hence form standing patterns in z.

(2) By applying the boundary condition where the tangential components are equal to zero at the metal surface, the solution must be of the form of sines and cosines, and the transverse wavevector was enforced to be only discrete to satisfy the boundary conditions. The equation for Ez, which is used to determine the expressions of E_x, E_y, H_x, and H_y as a function E0, is given below. a is the waveguide length along the x direction, and b is the waveguide length along the y direction.

$$\tilde{E}_z = E_0 \cdot \sin(k_x \cdot x) \cdot \sin(k_y \cdot x) \cdot e^{-j\beta z} \text{ , where } k_x = \frac{m\pi}{a} \text{ and } k_y = \frac{n\pi}{b}$$

m and n describe the mode orders or the number of half wavelengths along the x and y directions, respectively. Moreover, the spatial rate of variation of the wave phase longitudinally (β) to the transverse wavevector through the relation shown below. This can be derived from the equation introduced in the ray optics.

$$k^2 = \beta^2 + k_x^2 + k_y^2$$

$$\beta^2 = \left(\omega\sqrt{\mu\varepsilon}\right)^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2$$

For a given wavelength, higher order modes propagate at lower speeds, oscillations in the z direction. β becomes imaginary the transverse wavevector demands more variation than can be provided by the wavevector (k), hence for each mode (m,n) there is a defined cutoff frequency, below which waves are referred to as evanescent, and the electric field decays exponentially along z. The cutoff equation, given below, for 3D rectangular waveguide can be derived by setting $\beta = 0$.

$$f_{mn} = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

The calculations of the number of modes allowed for a given waveguide depends on the different combinations of m and n that produce kx and ky combinations within the constraint set by the k^2 equation above. This can be done by drawing in the first quarter a quarter of a circle corresponding to $k^2 = n \cdot k_o^2$, all points within the quarter circle encompasses all the different discrete combinations of kx and ky allowed and hence the number of modes. Hence, the larger the slab dimensions are, the more closely spaced the consecutive kx and ky values are, and the greater the number of modes.

Moreover, β can be expressed as given below, with the concept of n_{eff} used to convey how much of the total wavevector lies in the longitudinal direction, hence influencing confinement.

$$\beta = n_{eff} \cdot k_o$$

This relation will be discussed more elaborate in the upcoming section. However, it is used in the kx-ky plots, by drawing circular contours connecting kx-ky components having the same β, n_{eff}, and hence confinement. The intuition for why the contours is circular can be understood as given by the as shown below. Hence points of (kx, ky) that lies in the same contour have the same β.

$$k^2 = \beta^2 + k_x^2 + k_y^2$$

$$c = \left(n_{eff} \cdot k_o\right)^2 + k_x^2 + k_y^2,$$

Since $c - \left(n_{eff} \cdot k_o\right)^2 > 0$, then,

Let $r = \sqrt{c - \left(n_{eff} \cdot k_o\right)^2}$, therefore,

$$r^2 = k_x^2 + k_y^2$$

Note that the calculation of the number of modes for 2D waveguides is simpler and is derived from the ray optics bounce angle condition as the (m) integers satisfying the condition of $\sin \theta_m < 1$.

Dielectric Waveguide

The treatment of dielectric waveguides differs from that of metal waveguides on several ways. Symmetric dielectric waveguides are discussed here. (1) TIR can only be possible when the outer material "cladding" has a lower refractive index relative the inner material "core", such that $n_1 > n_2$, and with the condition that the angle of incidence from the normal to the interface is higher than the critical angle. In waveguides, it is more convenient to consider the complementary of the incident angle, as shown in Figure 2. Snell's law is then expressed as shown below [4].

$$n_1 \cdot \cos \alpha_i = n_2 \cdot \cos \alpha_t$$

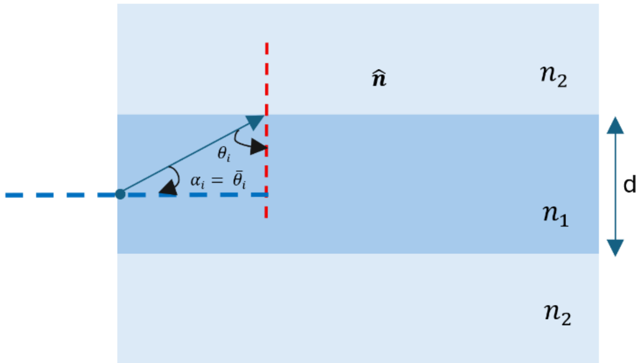


Figure 2. Symmetric dielectric slab showing an incident ray at the critical angle.

For incident angles, α_i, below the complementary of the critical angle (or θ_i above the critical angle), Snell's law cannot be satisfied by real values of $\cos \alpha_t$. This can also be shown through the equation of $\cos \alpha_t$ expressed as shown below [1]:

$$\cos(\alpha_t) = \sqrt{1 - \sin^2(\alpha_t)} = \sqrt{1 - \left(\frac{n_1}{n_2}\right)^2 \cdot \sin^2(\alpha_i)}$$

The cruciality of this conclusion shows when discussing the discretization of the wavevector by the dielectric boundary conditions. Not all rays satisfying TIR continue propagating indefinitely in the lossless core. Similarly to how it was derived that only a discrete number wavevectors satisfy the self-consistency condition, the same applies here, only that the boundary modulates magnitude and phase of the reflected wave by the reflection coefficient. Since $\cos \alpha_t$ for $\alpha_i < \bar{\theta}_c$ is complex [1, 4], so is the reflection coefficient, and the reflected wave phase is dependent on the polarization and α_i, contrary to the metallic waveguide, where for both polarizations, TE and TM, the reflection coefficient was always real and the phase was either 0 or π.

The equation for self-consistency used, shown below, is similar to the metallic waveguide equation, only now the phase shift introduced by the double reflection from the boundary is donated by 2φ.

$$2k_y d - 2\varphi = 2\pi m$$

As stated, φ is polarization dependent, the self-consistency equation is solved graphically and solutions of $\sin\theta_m$ are the intersection points between the right- and left-hand equations. It is important to clarify that it is through TIR, confinement in the core in the core in ensured by allowing the bounce angles, $\cos\theta_m$, *to range* from $\cos\theta_m < 1$ for $\theta_m < 0°$ to $\cos\bar{\theta}_c = \frac{n_2}{n_1}$ [1].

Discretization by the self-consistency only selects a number of bounce angles from the range. Togather, this leads to (1) a range of allowed longitudinal and transverse wavevector components, outside which confinement in the transverse direction is not maintained. (2) a discretised set of longitudinal and transverse wavevector outside which a wave does not propagate sustainability longitudinally. The range of allowed longitudinal wavevector is determined as shown below [1].

$$k_z = \beta = n_1 \cdot k_o \cdot \cos\theta_m$$

$$\beta_{upper} = n_1 \cdot k_o \cdot (1) = n_1 \cdot k_o$$

$$\beta_{lower} = n_1 \cdot k_o \cdot \left(\frac{n_2}{n_1}\right) = n_2 \cdot k_o$$

Therefore, effective refractive index is limited by:

$$n_2 < n_{eff} < n_1$$

For a 3D waveguide, the transverse wavevectors are also restricted in range by TIR, only rays of angles lower than $\bar{\theta}_c$ are accepted by the dielectric waveguide, hence contrary to the Metallic waveguide, $k_x, k_y \leq n_1 \cdot k_o \cdot \sin\bar{\theta}_c$. There lower bound is when $\theta_m = 0°$, resulting in zero transverse variation.

The concept applies similarly here in metallic waveguides, where the higher the mode order, the lower β, the lower the n_{eff}. In metallic waveguide, however, a lower effective refractive index indicates a lower longitudinal confinement of energy and leads to the concept of cutoff, relating to the high in the dielectric waveguide, it describes the transverse confinement of energy. The effects of $n_{eff} < n_2$, is clearer when discussing the field distribution within the dielectric waveguide materials in the next subsection.

Field Distributions

The field pattern within the core follows the form derived for the metallic waveguide, forming half wavelengths of sines or cosine as a function of the mode order, with $k_x = \frac{m\pi}{a}$ and $k_y = \frac{n\pi}{b}$. The field, however, does not equal to zero at the interface between the core and the cladding, it is however evanescent [1], the tail of the field into the cladding is instead a function of the mode order. The external field into the cladding in the transverse y direction, is expressed as shown below [1, 5].

$$E \propto e^{-\gamma_m y} \text{ for } |y| > \frac{d}{2}$$

Were,

$$\gamma_m^2 = \beta_m^2 - n_2^2 k_o^2, \text{ or,}$$

$$\gamma_m = \frac{2\pi}{\lambda}\sqrt{n_{eff}^2 - n_2^2}$$

The higher the mode order, the lower the effective refractive index, and the more does the mode extends into the cladding, suggesting lower confinement. Moreover, when $n_{eff} < n_2$, γ_m becomes imaginary, TIR is not satisfied, and the mode becomes guided in the cladding rather than the core. For that, cutoff is not a function of frequency as with metal waveguides but instead relates to waves of any frequency which satisfy TIR for sustainable mode propagation. For that, for a symmetric dielectric slab with n1 > n2, the fundamental mode is always present for all frequencies.

Moreover, the NA, given below, is the numerical aperture, which informs us of the contrast between the refractive index of the cladding and the core. It can be shown that NA can be expressed as a function of $\sin\bar{\theta}_c$, hence the larger the NA, the $\sin\bar{\theta}_c$ (which informs the range of allowed bounce angles), and hence the greater the number of allowed modes. Moreover, although larger NA increases the possibility of exciting higher modes at bents or other imperfections, it also means larger γ_m, and reduced losses in the cladding for higher order modes.

$$NA = \sqrt{n_1^2 - n_2^2}$$

The relation between the transverse and longitudinal wavevectors, confinement, and the NA can be viewed more clearly using the k_x and k_y plot.

As stated above, k_x and k_y are limited by $\sin\bar{\theta}_c$ factor forming a quarter of a circle of radius $r = n_1 \cdot k_o \cdot \sin\bar{\theta}_c$ within which allowed combinations exists.

$$k^2 = \beta^2 + k_x^2 + k_y^2$$

$$(n_1 \cdot k_o)^2 = \left(n_{eff} \cdot k_o\right)^2 + k_x^2 + k_y^2,$$

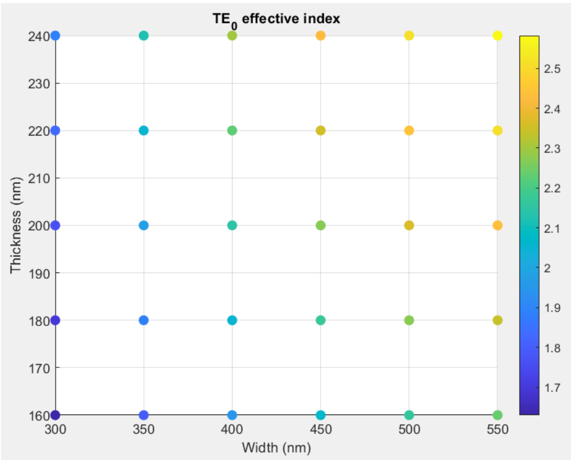
$$k_o^2 \cdot \left(n_1^2 - n_{eff}^2\right) = k_x^2 + k_y^2, \text{ therefore,}$$

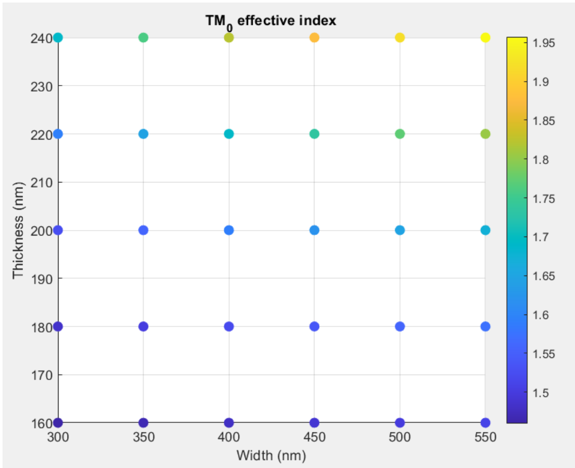
$$r = k_o\sqrt{n_1^2 - n_{eff}^2}$$

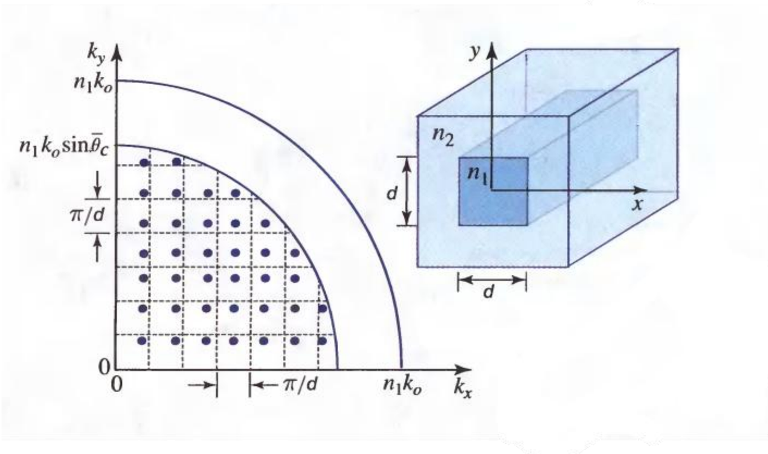
As with the metallic waveguide, points that lie at the same distance from the origin of the plot (same circular contour), have the same effective refractive index. The higher the NA, the higher the maximum possible r, the higher is the area, and the greater the number of modes which could be allowed. To design single-mode waveguides, the slab width and thickness can be made close such that distance between each consecutive transverse wavevector is greater, and the NA (and hence $\sin\bar{\theta}_c$), should be made smaller.

Ansys Lumerical Simulations of Dielectric Waveguides and Advanced Parameters Analysis

Polarization and Confinement





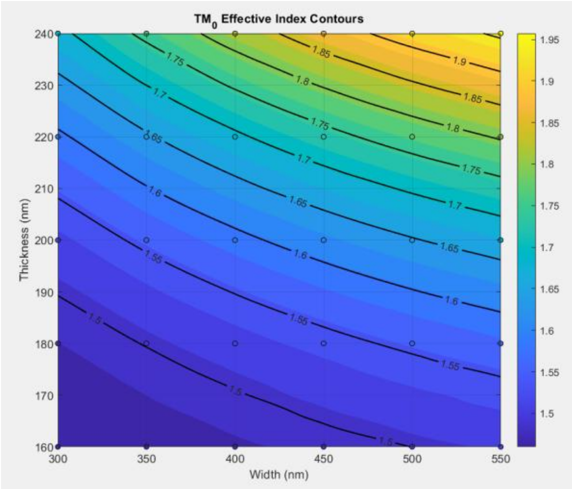


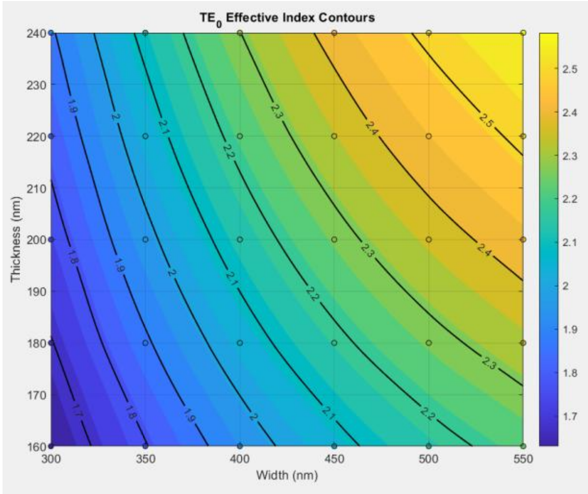
For TE (10):

$$(n_1 \cdot k_o)^2 = \left(n_{eff} \cdot k_o\right)^2 + \left(\frac{\pi}{W}\right)^2 + \left(\frac{\pi}{T}\right)^2$$

For TM:

$$c = \left(n_{eff} \cdot k_o\right)^2 + \left(\frac{\pi}{W}\right)^2 + \left(\frac{\pi}{T}\right)^2$$





- We are simulating a single point in the kx-ky plot, corresponding to (1,1)
- Increasing the width and thickness, the kx-ky points becomes more packed together, suggesting a greater number of allowed modes, with each simulated combinations of W and T has its own kx, ky plot
- When increasing the width or thickness, n effective also increases, (as also satisfied by the equation) but also, higher order modes begin appearing. As more points can be fitted in the quarter of the circle.
- The plot produced with the produced contours only informs us of the W, T slab dimension combinations that produce the same effective refractive index, with greater dimensions producing higher n _ eff , as expected, but this comes at the cost of enabling the excitement of higher unwanted order modes, with similar n effective, resulting in potential coupling and unwanted power transfer.

Dominant mode and dominant polarization

Discuss polarization and hence why te has greater n effective, why also dielectric has te like while metal has only purses,

Show plots of TE 0 and TM0 showing slab length and n effective,

Contours , differentiate it from that of metals.

Application: Single-Mode Waveguide Design

Discuss prefiely higher length, cross offer, coupling , and instability , and why even if source, what so on , so on ,

Lightly touch on orthogonality

Future Work

References

[1]. Saleh and Malvin Carl Teich, *Fundamentals of Photonics*. John Wiley & Sons, 2019.

[2]. F. T. Ulaby and Umberto Ravaioli, *Fundamentals of Applied Electromagnetics, Global Edition*. Pearson Higher Ed, 2022.

[3]. D. K. Cheng, *Field and Wave Electromagnetics*. Addison Wesley Publishing Company, 1983.

[4]. D. Marcuse, *Theory of Dielectric Optical Waveguides*. 1991.

[5]. Lukas Chrostowski and M. Hochberg, *Silicon photonics design: from devices to systems*. Cambridge: Cambridge University Press, 2016.

Appendix E – Internship Report (incomplete): The Mach-Zehnder Interferometer and Application

Report Title:

The Mach-Zehnder Interferometer and Application

Report Objectives:

- To present the theoretical principles of the Mach-Zehnder Interferometer (MZI) and derive its fringe/transmission profile.
- To analyse the impact of thermal fluctuations on MZI fringe patterns, including phase shifts and changes in free spectral range (FSR).
- To explore methods for detecting phase and FSR changes.
- To compare experimental measurements with simulations and calculations to draw conclusions.

Shayma Alteneiji | 100063072

Internship Supervisor:

Prof. Jaime Viegas

Introduction

This report serves as an extension to the Fabry-Pérot interferometer, where due to the low reflectance of the glass-air interface (r = 0.2, R = 0.4, at normal incidence) and the setup loses, where reflected waves at the second mirror are less likely to be reflected back towards the first mirror and participating into higher order reflections, and simplifications such as ignoring diffraction effects, lead to the observed fringes being with less visibility and contrast compared to the theoretically expected results. Extractions of setup parameters from the fringes such as cavity length (d), and cavity reflectance yielded highly erroneous values.

For that, in this report, the Mach-Zander interferometer using fibres is described and built. Moreover, this report also aims to investigate the underlying causes on the changes in the detected fringes pattern of the interferometer per sweeping cycle. The two possible hypothesis (1) FSR change, (2) phase change are studied, and tools used for examining these two mechanisms also presented. The experiment parameter, optical path difference, L, is also extracted from the fringe patterns.

Theoretical Background

In this section, we discuss the Mach-Zander Interferometer, derive its fringe patters or transmission spectrum, present its FSR equation, as well as study the methods for detectability of FSR and phase change, as well as the extraction method of the FSR of a given spectrum.

Mach-Zehnder Interferometer

The structure for the Mach-Zander Interferometer is shown in Figure 1. It is composed of a Y-splitter, two arms of different OPLs, and another Y-combiner/splitter at the output end. Assuming a perfect Y-splitter, then an input light intensity of I_i gets divided equally between the two arms.

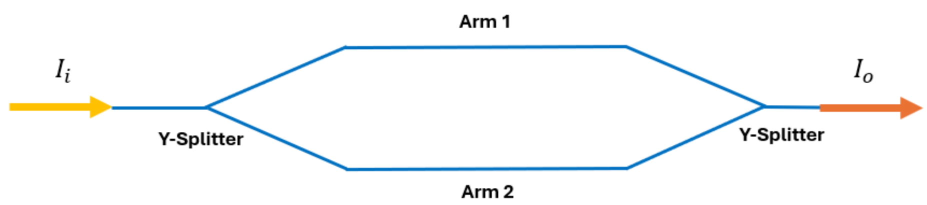


Figure 1. Mach-Zander Interferometer [1]

Intensity is a scalar value and describes the time average of the wave's optical power per unit area [2]. Phase accumulation, therefore, is tracked through the Electric field, which is related to the intensity through the expression given below [2, 3].

$$I \propto |E|^2$$

The total intensity, however, of the interfering waves, includes the interference terms, due to the superposition property of waves. The total intensity of two waves is described as shown below [2]:

$$I_T = |\boldsymbol{E}_1 + \boldsymbol{E}_2|^2 = (\boldsymbol{E}_1 + \boldsymbol{E}_2) \cdot (\boldsymbol{E}_1 + \boldsymbol{E}_2)^*$$

$$I_T = |\boldsymbol{E}_1|^2 + |\boldsymbol{E}_2|^2 + \boldsymbol{E}_1^* \boldsymbol{E}_2 + \boldsymbol{E}_2^* \boldsymbol{E}_1$$

The interference terms are $\boldsymbol{E}_1^* \boldsymbol{E}_2$, and $\boldsymbol{E}_2^* \boldsymbol{E}_1$ [2], as their magnitude is scaled by the relative phase differences between them. Returning to the Mach-Zander setup, assuming arm two is longer than arm one by some length, then the phase accumulated by the electric fields going through both arms, are given as shown below. Note that the refractive index of both arms is assumed to be equal.

$$OPL_1 = n \cdot \mathrm{L}$$

$$OPL_2 = n \cdot (\mathrm{L} + \Delta L)$$

$$\Delta\varphi_1 = k \cdot OPL_1 = \frac{2\pi n \mathrm{L}}{\lambda}$$

$$\Delta\varphi_2 = k \cdot OPL_2 = \frac{2\pi n \cdot (\mathrm{L} + \Delta L)}{\lambda}$$

Moreover, if both arms have equal lengths, the Mach-Zander fringe patterns can be produced by inserting a material of length (d_f) and of a refractive index (n_f) in one arm. The OPD is then expressed as given below, and accordingly the relative phase difference between the two arm's electric field can be known.

$$OPD = n_f d_f - n d_f = d_f \cdot (n_f - n)$$

Recurring back to the assumed setup, at the output end, the Y-combiner/splitter divides the intensity in halves once more, and the combined electric fields becomes scaled down by $\sqrt{2}$. The resulting output electric field.

$$\boldsymbol{E}_o = \frac{1}{\sqrt{2}} \cdot \left(\frac{E_i}{\sqrt{2}} \cdot e^{-j \cdot \Delta\varphi_1} + \frac{E_i}{\sqrt{2}} \cdot e^{-j \cdot \Delta\varphi_2} \right)$$

The resulting output intensity is determined as given below using I_T equation given above.

$$I_o = |\boldsymbol{E}_o|^2$$

$$I_o = \frac{1}{4} \cdot \left(E_i^2 + E_i^2 + E_i \cdot E_i \cdot e^{-jkn \cdot \Delta L} + E_i \cdot E_i \cdot e^{jkn \cdot \Delta L} \right)$$

And,

$$I_o = \frac{I_i}{2}\left(1 + \cos\left(\frac{2\pi n \Delta L}{\lambda}\right)\right)$$

Plotting the output intensity as a function of phase difference between the interfering waves is shown in Figure 2. The used $I_i = 5$ W/cm^2.

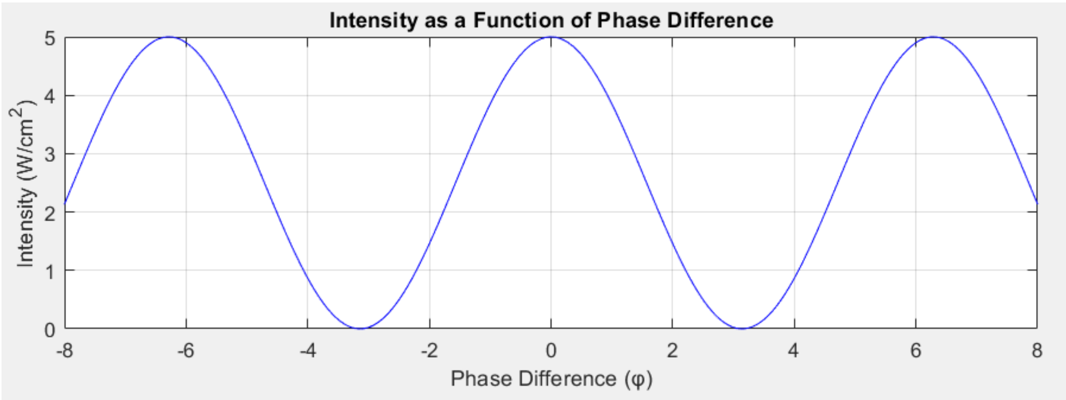


Figure 2. Mach-Zander Intensity Plot as a Function of Phase Difference

The plots show the constant intensity, representing the sum of the two waves intensities, $(I_1 + I_2 = I_i)$, divided by two (the effect of the output Y-splitters), being modulated by the interference term, the cosine term. The two waves add constructively, when the phase difference is a multiple of 2π, and destructively for other phase differences. When the is a multiple of π the intensities cancel exactly, and the output intensity goes to zero.

Physically, Figure 2 plot represents, the solutions of ΔL, and λ (one parameter varied at a time) for which constructive interference occurs, by equating the phase to 2π, as given below.

$$n\Delta L = m\lambda$$

Plotting intensity as a function of λ with $\Delta L = 1.5$ mm, normalized to the maximum intensity therefore produces the transmission plot of the Mach-Zander Interferometer, shown in Figure 3, in dB and linear scale.

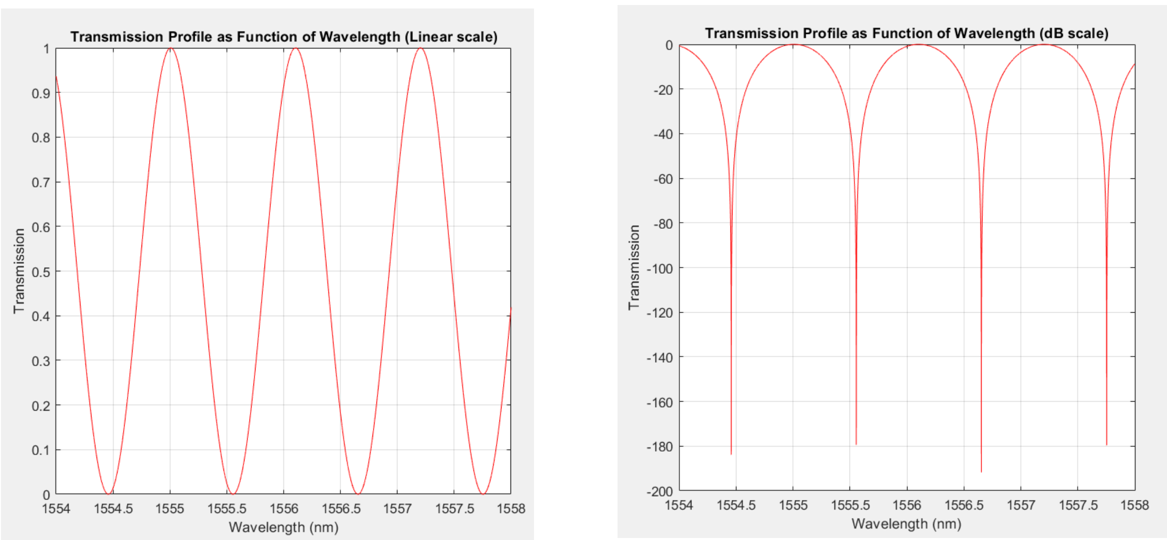


Figure 3. Mach-Zander Transmission Plot as a Function of Wavelength, linear and dB scale.

The considered equation for the output intensity is only however, a special case of the generalization, where the intensity of the input light is not divided equally among the two arms. Letting I_1 and I_2 be the lights intensities going through arms one and two, respectively, then the output intensity is as given below, based on [2]. This can be derived using the I_T equation.

$$I_o = I_1 + I_2 + 2\sqrt{I_1 I_2} \cdot \cos(\Delta\varphi_1 - \Delta\varphi_2)$$

$$I_o = I_1 + I_2 + 2\sqrt{I_1 I_2} \cdot \cos\left(\frac{2\pi n \Delta L}{\lambda}\right)$$

Factoring in the halving of the intensity of the light at the output Y-splitters, and assuming a P factor, representing the percentage of light intensity going into the first arm, the equation becomes:

$$I_o = I_1 + I_2 + 2\sqrt{I_1 I_2} \cdot \cos\left(\frac{2\pi n \Delta L}{\lambda}\right)$$

$$I_o = \frac{1}{2} \cdot \left(I_i \cdot P + I_i \cdot (1 - P) + 2I_i\sqrt{P \cdot (1 - P)} \cdot \cos\left(\frac{2\pi n \Delta L}{\lambda}\right)\right)$$

$$I_o = \frac{I_i}{2} \cdot \left(1 + 2\sqrt{P \cdot (1 - P)} \cdot \cos\left(\frac{2\pi n \Delta L}{\lambda}\right)\right)$$

The plotted output intensity as a function of phase for various P is shown in Figure 4. The reason why the consideration of the generalized case is crucial, because it can be seen that in the non-ideal case of the intensities in both arms not being equal, what is referred to as the modulation depth (equation below [2]), is affected. When P is 0.5, M = 1, and as P increases or decreases below those values, the modulation depth decreases, and the intensities don't cancel exactly at phases that are multiples of π.

$$M = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

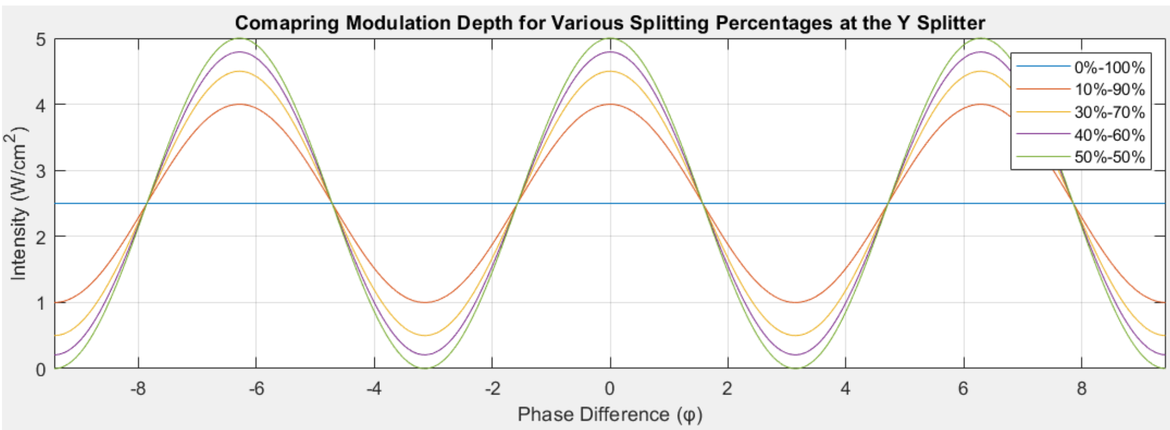


Figure 4. Mach-Zander I_o as a function of phase difference for various splitting percentages at the Y-splitter.

Accordingly, the transmission profile is changed, instead of the transmission going to very low values of – dBs (Figure 3) instead it ranges from approximately -14 to -45 dB for various splitting percentages as shown in Figure 5.

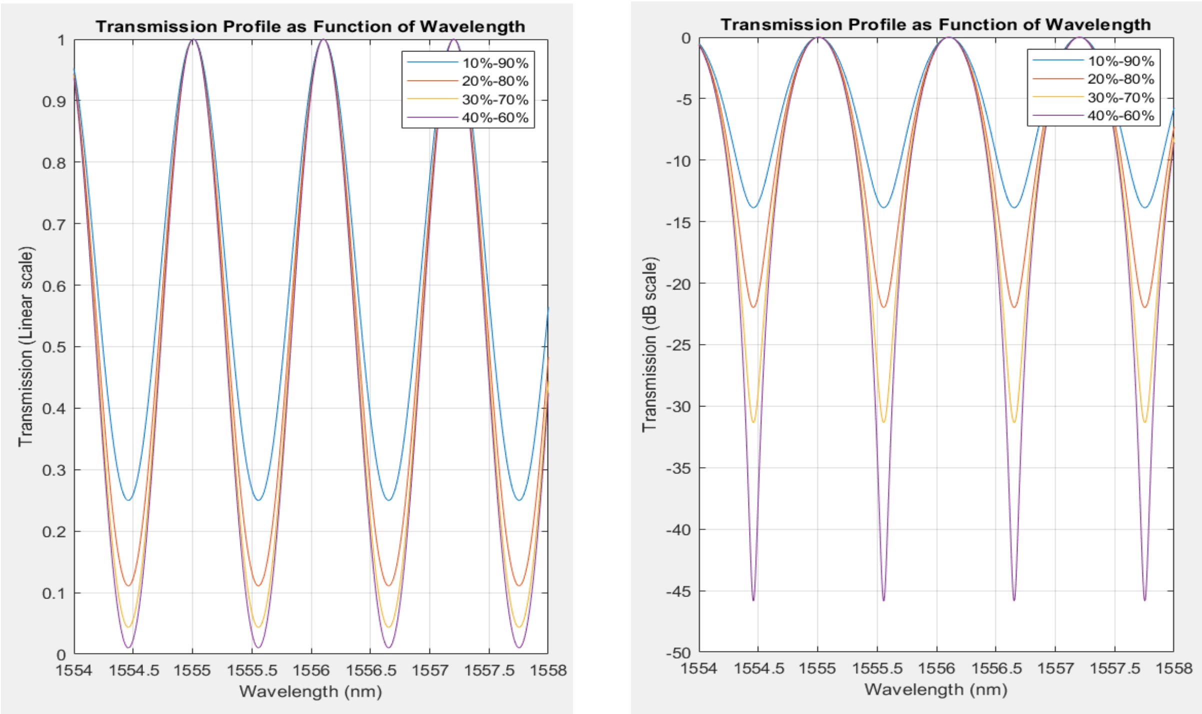


Figure 5. Mach-Zander Transmission Plot as a Function of Wavelength for various splitting percentages at the Y-splitter, linear and dB scale.

Moreover, polarization change due to long loops of the fibre can result in losses, degrading modulation depth. Hence, Y-splitters of 40%-60% characterization would yield the results of those of potentially 10%-90%. This presents therefore the expected modulation effect on the source spectrum in the experimental results. Moreover, the spacing between each consecutive wavelength at which constructive interference occurs is described by the free spectral range (FSR), discussed in the next section.

FSR and FSR Change

It can be shown that the spacing between each consecutive spectral wavelength and frequency at which constructive interference occurs is referred to as free spectral range (FSR). Its expression can be derived from seeing that a peak occurs at consecutive wavelengths where the accumulated phase (Δφ) is equal to 2π, marking a period. This is used to derive the expression for the FSR in wavelength as given below, note that the used refractive index is n_g, the group index [3].

$$\frac{\Delta\varphi}{\Delta\lambda} = \left|\frac{\partial\varphi}{\partial\lambda}\right| = \left|\frac{\partial}{\partial\lambda}\left[\frac{2\pi n_g L_i}{\lambda}\right]\right| = \frac{2\pi n_g L_i}{\lambda^2} = \frac{2\pi}{\lambda^2/n_g L_i}$$

Therefore,

$$FSR_{\lambda} = \Delta\lambda = \frac{\lambda_0^2}{n_g \cdot L_i}$$

The same logic is used to obtain the FSR in frequency as derived is below.

$$\frac{\Delta\varphi}{\Delta f} = \left|\frac{\partial\varphi}{\partial f}\right| = \left|\frac{\partial}{\partial f}\left[\frac{2\pi n_g L_i \cdot f}{c}\right]\right| = \frac{2\pi}{c/n_g L_i}$$

$$FSR_f = \Delta f = \frac{c}{n_g \cdot L_i}$$

The fringe pattern against exhibits wavelength has a chirp effect, where the spacing between the peaks increases linearly with the wavelength. In the contrary, to against frequency, where is constant. FSR is dependent on (n_g) and the difference in the length of the arms of the MZ interferometer (ΔL).

One of the most common causes in the change of these parameters include normal temperature fluctuations in an experimental setup, or proximity of a relatively warmer object such as the hand. The thermal expansion coefficient, $\left(\frac{\partial \Delta L}{\partial T}\right)$, of optical fibre (SMF-28) is 0.55 $\mu m/m \cdot ℃$ [4], while the thermo-optics coefficient, $\left(\frac{\partial n_g}{\partial T}\right)$, describing the rate of change in the refractive index as a function of temperature is $0.811 \times 10^{-5}\ ℃^{-1}$ at 1550 nm wavelength [5], which maybe assumed constant around nearby wavelength ranges.

In its general form, the difference between the phase accumulation between the electric fields of the two arms can be expressed as $\Delta\varphi = \frac{2\pi n_1}{\lambda} \cdot L_1 - \frac{2\pi n_2}{\lambda} \cdot L_2$, and the denominator of FSR_{λ} equation is $n_1 L_1 - n_2 L_2$; using this equation would allow for a more accurate account of temperature change locally around a single arms, which is more likely. In this report, however, we assume the simplified case of temperature change effecting all sections of the interferometer equally.

Therefore, we assume thermal expansion, affects both arms equally and ΔL remains unchanged, while the change in the group index is assumed to be equal in both arms, and a change in its value changes the new FSR. Also, since thermal expansion is lower than thermo-optics coefficient, the assumption made should not result in strong deviation from experimental expectations.

Assuming the laboratory temperature is 20 degrees Celsius, then the proximity of the hand near the interferometer can cause a temperature increase of 10 degrees Celsius around the interferometer. The calculated change in group index is as shown below.

$$\Delta n_g = n_g - n_{g,o} = \frac{\partial n_g}{\partial T} \cdot \Delta T$$

$\Delta n_g = (0.811 \times 10^{-5}) \cdot (10) = \mathbf{0.811 \times 10^{-4}}$, or

$$\%\, n_g\ change = \frac{|n_g - n_{g,o}|}{n_{g,0}} \times 100$$

$$\boldsymbol{\sim\%\, n_g\ change = +0.0056\%}$$

Let $\Delta L = 1.5$ mm, $\lambda_0 = 1569$ nm, and $n_g = 1.4606$, then the change in FSR is calculated as shown below.

$$FSR_0 = \frac{(1569nm)^2}{1.4606 \cdot 1.5\ \text{mm}} = 1.12363\ \text{nm}$$

$$FSR_{new} = \frac{(1550nm)^2}{(1.4606+0.811\times10^{-4}) \cdot 1.5\ \text{mm}} = 1.12357\ \text{nm}$$

Therefore, the percentage of change in FSR is expected to be around:

$$\%\, FSR\ change = \frac{|FSR_{new} - FSR_0|}{FSR_0} \times 100$$

$$\boldsymbol{\sim\%\, FSR\ change = 0.0056\%}$$

It is therefore concluded that FSR change is very small to be observed for normal temperature changes. The expression for the new FSR as a function of temperature is derived as shown below. Figure x shows the percent change in FSR as a function temperature. In the next section we discuss phase change and modelling it.

$$FSR_\lambda = \frac{\lambda_0^2}{n_{g,1}L_1 - n_{g,}L_2} = \frac{\lambda_0^2}{n_{2,g} \cdot L_i}$$

$$FSR_{\lambda,new} = \frac{\lambda_0^2}{(n_{2,g} + \Delta n) \times L_i} = \frac{\lambda_0^2}{\left(n_{2,g} + \frac{\partial n_g}{\partial T} \cdot \Delta T\right) \times L_i}$$

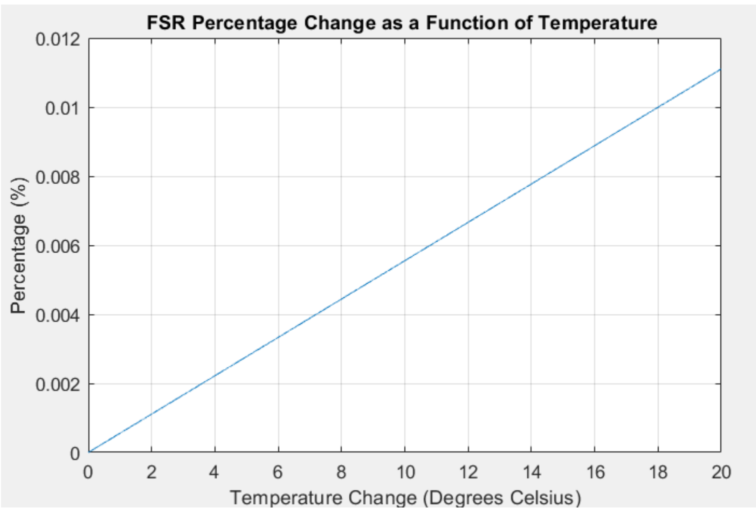


Figure 6. FSR percentage change a function of temperature change, considering only group index changes when the thermo-optics coefficient is $0.811 \times 10^{-5}\,°C^{-1}$.

Phase Change

The phase in the intensity equation represents the relative phase difference between the two interfering waves. Since it is a function of n_g and ΔL which vary as a function of temperature, the relative phase difference also varies with temperature. Assuming linear relations, the new intensity plot is modulated by a constant phase shift, derived as shown below.

$$\Delta\varphi = \frac{2\pi}{\lambda} \cdot \left(n_{1,g}L_1 - n_{2,g}L_2\right) = \frac{2\pi}{\lambda} \cdot \left(n_{2,g} \cdot (L_i + L_2) - n_{2,g}L_2\right)$$

$$\Delta\varphi = \frac{2\pi}{\lambda} \cdot \left(n_{2,g} \cdot L_i\right)$$

$$\Delta\varphi_{new} = \frac{2\pi}{\lambda} \cdot \left[\left(n_{2,g} + \Delta n\right) \cdot (L_i + L_2 + \Delta L) - \left(n_{2,g} + \Delta n\right) \cdot (L_2 + \Delta L)\right]$$

$$\Delta\varphi_{new} = \frac{2\pi}{\lambda} \cdot \left(n_{2,g} + \Delta n\right) \times L_i$$

$$\varphi_{shift} = \Delta\varphi_{new} - \Delta\varphi = \frac{2\pi}{\lambda} \cdot (\Delta n) \times L_i$$

This provides an estimate of phase change as function of temperature change, assuming negligible thermal expansion effects due to justifications mentioned in the FSR change section. Note that fibre paths and loops affecting polarization as well as vibrations in the lab setup also influence phase shift unmodeled here. Moreover, assuming the same case of $\Delta T = 10$ degrees Celsius, $L_i = 1.5$ mm, and $n_g = 1.4606$. The phase change is calculated as $28.2542°$, as shown below.

This depicts how phase is significantly sensitive to temperature change compared to FSR and would dominant the chang in the spectrum observed. Figure 6 shows the phase shifts introduced due to change in temperature in the range from 0 degrees Celsius to 20 degrees Celsius, and with the thermo-optics coefficient is $0.811 \times 10^{-5}\ ℃^{-1}$.

$$\varphi_{shift} = \frac{2\pi \cdot (\,1.5\text{ mm})}{(1550nm)} \cdot (0.811 \times 10^{-5}) \cdot (10) = 28.2542°$$

In the next section we explore an attempt to quantify the change in FSR and phase change in a given spectrum compared against a reference spectrum.

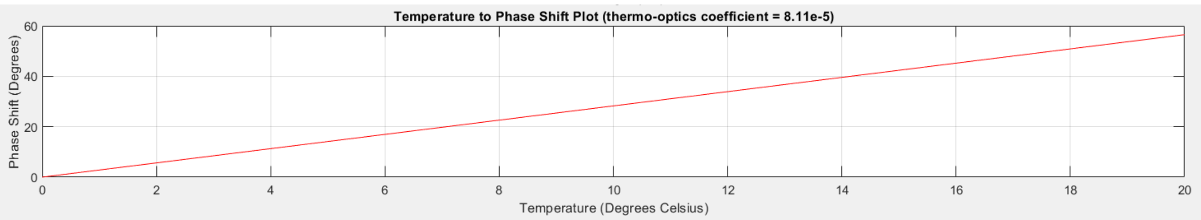


Figure 6. Phase shift in the MZI as a function of temperature change, considering only group index changes when the thermo-optics coefficient is $0.811 \times 10^{-5}°C^{-1}$.

Mathematical Attempt at Quantifying FSR Change and Phase Change

The exploration carried out in this section is to attempt to mathematically formalize a method of quantifying the exact difference in FSR and phase shift between two spectrums. It was proposed that if the peaks of the fringe patterns of a given spectrum can be expressed as a function of the peaks of the fringes in a reference spectrum modulated by the FSR difference and phase shift, in the expression shown below.

$$\lambda_n = h\left(\lambda_{n,ref}, \delta_{FSR}, \varphi\right)$$

Depending on the form of the resulting expression, if a linear function of the variables for example, then every measurement made of the peak position of the signal relative to the peak position in the reference signal would increase the number of rows or the equations capturing the system at hand; and as measurements increases it should refine the estimate reducing and allow the system better change of avoiding degeneracy, or one unique value of FSR difference and phase shift.

Using Figure 7, representing two signals of zero shift only a difference in FSR, the derived expression for λ_n as a function of $\lambda_{n,ref}$ and δ_{FSR} is given below. Note that, since this considers only a small range of the spectrum, FSR is assumed constant across the spectrum.

$$\lambda_{1,ref} = \lambda_0 + \frac{FSR_1}{4}$$

$$\boldsymbol{\lambda_{n,ref} = \lambda_0 + (4n+1) \cdot \frac{FSR_1}{4}}$$

$$\lambda_1 = \lambda_0 + \frac{FSR_1'}{4}$$

$$\lambda_1 = \lambda_0 + \frac{FSR_1 - \delta_{FSR}}{4} = \lambda_0 + \frac{FSR_1}{4} - \frac{\delta_{FSR}}{4}$$

$$\lambda_1 = \lambda_{1,ref} - \frac{\delta_{FSR}}{4}$$

$$\boldsymbol{\lambda_n = \lambda_{n,ref} - (4n+1) \cdot \frac{\delta_{FSR}}{4}}$$

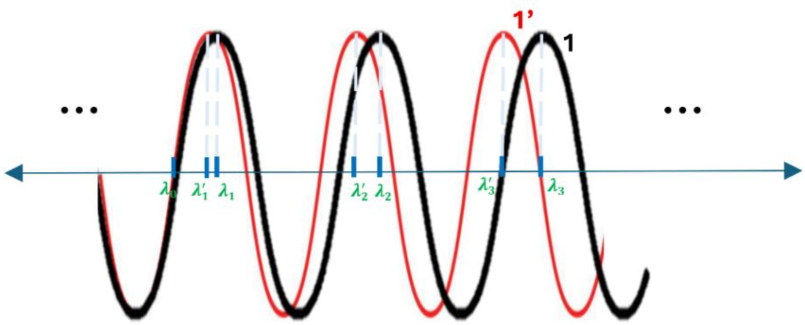


Figure 7. Wavelength spectrum representation of a reference signal and another signal of slightly different interferometric periodicity (lower FSR_λ).

Next, the case of phase shift and FSR change is considered in Figure 8. Note that the red signal experienced a phase shift to the right (delay). The shift in wavelength introduced due to phase shift can be expressed as given below.

$$\lambda_0 = \frac{\varphi}{2\pi} \cdot FSR_1' + \lambda_{0,ref}$$

$$\lambda_0 = \frac{\varphi}{2\pi} \cdot (FSR_1 - \delta_{FSR}) + \lambda_{0,ref}$$

Therefore, the final expression, $\lambda_n = h(\lambda_{n,ref}, \delta_{FSR}, \varphi)$, uses the results of the first considered and accounts for the wavelength shift derived due to phase shift as shown below.

$$\boldsymbol{\lambda_1 = \lambda_{1,ref} - \frac{\delta_{FSR}}{4} + \frac{\varphi}{2\pi} \cdot (FSR_1 - \delta_{FSR})}$$

$$\boldsymbol{\lambda_n = \lambda_{n,ref} - (4n+1) \cdot \frac{\delta_{FSR}}{4} + \frac{\varphi}{2\pi} \cdot (FSR_1 - \delta_{FSR})}$$

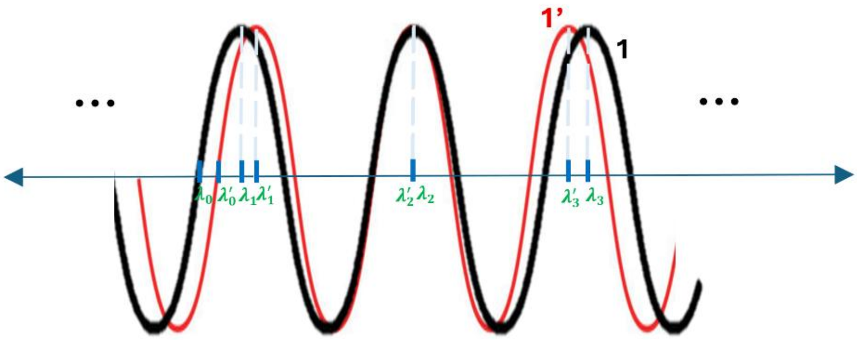


Figure 8. Wavelength spectrum representation of a reference signal and another signal of slightly different interferometric periodicity (lower FSR_λ) and added phase shift.

Now, assume FSR_1 is known, and measurements $\lambda_{n,ref}$ and λ_n can be extracted. Then, for a given data point (λ_n, $\lambda_{n,ref}$), let $a = \lambda_n - \lambda_{n,ref}$. Let $k_1 = -\frac{1}{4}$, $k_2 = -\frac{1}{2\pi}$, and $k_3 = \frac{FSR_1}{2\pi}$, the final expression then becomes of the form:

$$a = k_1 \cdot \varphi + k_2 \cdot \varphi \cdot \delta_{FSR} + k_3 \cdot \delta_{FSR}$$

The expression for a given measurement point can be thought of as contour of a 2D surface function of the form $z = f(\varphi, \delta_{FSR})$, hence since every new measurement results in a different contour of the function rather than more variation allowing for non-degeneracy, it is concluded therefore, that the parameters φ and δ_{FSR} could not be decoupled and quantified, instead in the next section we discuss tools used for the detection of either or both phase and FSR change.

Detectability of FSR and Phase Change

Tools for the detection of the phase and/or FSR change essentially detect a wavelength shift between two signals. A wavelength shift between the two signals can be due to a phase shift or an interferometric periodicity change (FSR). The phase shift alone of (φ) introduces a wavelength shift given by $\frac{\varphi}{2\pi} \cdot (FSR)$, while an FSR change alone introduces a wavelength shift given by $\frac{\delta_{FSR}}{4}$, where δ_{FSR} is the difference in the FSR of the two signals. When both changes occur, the resulting wavelength shift is expressed as $\frac{\delta_{FSR}}{4} + \frac{\varphi}{2\pi} \cdot (FSR_1 - \delta_{FSR})$, where FSR_1 is the interferometric periodicity of the reference signal.

Moreover, for a given wavelength shift between two signals, there effective FSR difference can be determined, in such case, the problem can be restated as the detection of the change in the interferometric periodicity of the signals. To model the problem, assume the reference signal is of the sinusoidal form given by $\cos\left(\frac{2\pi}{FSR_{\lambda,1}} \cdot x_{nm}\right)$, the sensor or examined signal is of the form $\cos\left(\frac{2\pi}{FSR_{\lambda,2}} \cdot x_{nm}\right)$, with $FSR_{\lambda,2}$ being greater than $FSR_{\lambda,1}$ by a small percentage p.

The sum of two signals whose ratios of periods is not a factor of an irrational number can be always expressed as a ratio of integers, and the signals are said to be commensurate signals, as they share a common period. For the example given, the ratio of interferometric periods and the common period are expressed as given below.

$$\frac{FSR_{\lambda,1}}{FSR_{\lambda,2}} = \frac{1+p}{1} = \frac{M}{N}$$

$$T_{Common} = FSR_{\lambda,1} \times N = FSR_{\lambda,2} \times M$$

Let $p = 10^{-3}$, which represent the percentage of difference between the signals, then ratio of integers is 1001/1000, the signals align back together after 1000 cycles of the first signal, or 1001 cycles of the second signal.

As p decreases, the number of cycles required before signals align over increases. Over a limited range of wavelength, the summed signals appear as if are *aperiodic*, with no repeating period within the observed range, which is limited by the source spectrum range.

Figure 9 and 10 illustrates this effect, it shows the superimposed plot of a modulated source spectra by two separate MZIs whose output signals differ in FSR by 10 % or $p = 0.1$, the signals align together four time. Hence, over the observed spectrum, the two signals are said to be commensurate periodic signals.

Figure 10 shows the superimposed spectra for $p = 0.001$. The signals remain close, varying at very similar rates, and do not align back, hence they are said to be aperiodic over the observed plotted spectrum range.

As was shown, summing two periodic signals results in a periodic sum, while summing two aperiodic signals results in an aperiodic sum. The latter is also known as Fourier series synthesis, while the former is known as Fourier transform synthesis.

With signals being unsimilar adequately, we obtain a Fourier transform, varying the signal barely, providing us with no information to draw out; from this we introduce the concept of detectability limits in the presented methods of FSR change detection, including the methods of optical Vernier effect, the FFT, and the cross-correlation tool.

Figure 9. superimposed plot of a modulated source spectra by two MZIs whose output signals differ in FSR by when percentage of difference is 10%

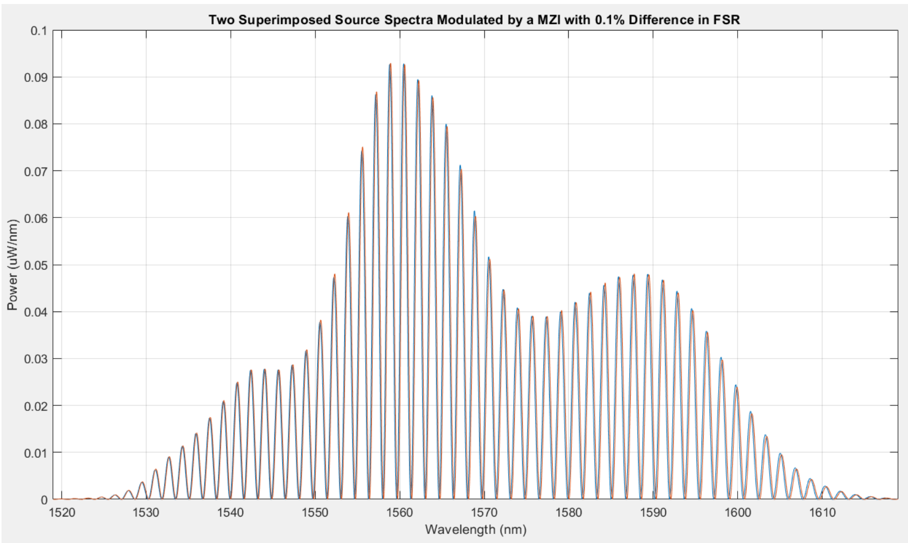


Figure 10. superimposed plot of a modulated source spectra by two MZIs whose output signals differ in FSR by when percentage of difference is 0.1%

The Optical Vernier Effect

The principle of the optical Vernier effect is used in sensing application, where it can, with high sensitivity, detect the small changes of an added wavelength shift between two signals; one is said to be the reference signal, and the other is said to be the sensing signal. Envision a setup of two MZI arranged in *parallel*, the signals, therefore, at the output of each filter system (MZI) are summed through the Y-splitter/coupler before the OSA input port (Figure 11).

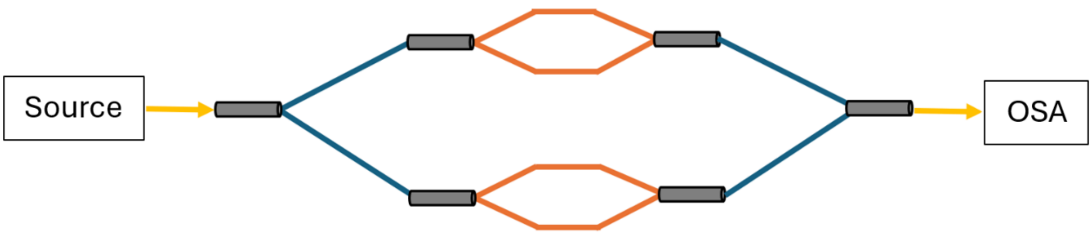


Figure 11. Two Mach-Zehnder interferometers arranged in parallel

Consider two signals with some phase shift and FSR difference, and a corresponding effective FSR difference, $\delta_{FSR} = FSR_1 - FSR_2$; by the trigonometric superposition relationships, their *sum* can be expressed as the multiplication of two signals, as shown below.

$$output = \cos\left(\frac{2\pi}{FSR_1} \cdot x_\lambda\right) + \cos\left(\frac{2\pi}{FSR_2} \cdot x_\lambda\right)$$

Which is equivalent to,

$$output = 2 \cdot \cos\left(\left(\frac{\pi}{FSR_1} - \frac{\pi}{FSR_2}\right) \cdot x_\lambda\right) \times \cos\left(\left(\left(\frac{\pi}{FSR_1} + \frac{\pi}{FSR_2}\right)\right) \cdot x_\lambda\right)$$

There interferometric frequencies can be expressed as v_{low} and v_{high} as shown below,

$$v_{low} = \frac{\pi}{FSR_1} - \frac{\pi}{FSR_2} = \pi \cdot \left(\left|\frac{FSR_2 - FSR_1}{FSR_2 \cdot FSR_1}\right|\right) = \frac{2\pi}{\left(\left|\frac{2 \cdot FSR_2 \cdot FSR_1}{FSR_2 - FSR_1}\right|\right)}$$

$$v_{high} = \frac{\pi}{FSR_1} + \frac{\pi}{FSR_2} = \pi \cdot \left(\frac{FSR_2 + FSR_1}{FSR_2 \cdot FSR_1}\right) = \frac{2\pi}{\left(\frac{2 \cdot FSR_2 \cdot FSR_1}{FSR_2 + FSR_1}\right)}$$

The resulting signal can therefore be viewed as comprising a signal of the lower frequency, v_{low}, also referred to as the *envelope* modulating a signal of higher frequency, v_{high}, also referred to as the *carrier* in amplitude. Figure 12 shows an example of an output signal when $FSR_1 = 1$ nm, $\delta_{FSR} = 0.2$ nm, plotted over a 20 nm spectrum range. The expression for the envelope signal, can be shown to correspond to the period over which the accumulated phase difference between the signals is a multiple of 2π, this explains the peaks pattern observed every full period of the envelope, suggesting r-alignment (Figure 12).

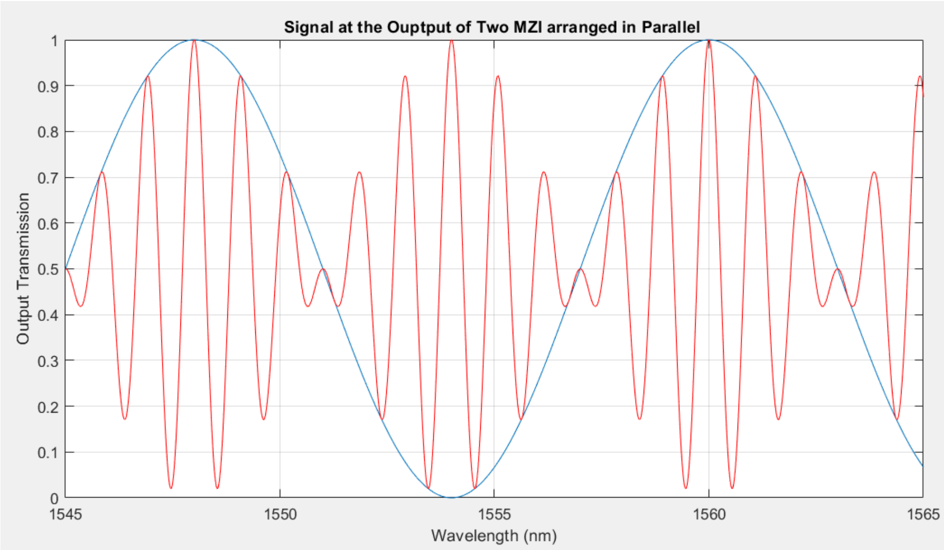


Figure 12. Transmission plot of the output signal of two MZI arranged in parallel, when the reference and sensing signals FSR are 1 nm and 1.2 nm, respectively.

The shape or the spreading of the envelope is a function of the difference between the FSRs of the sensing and reference signals, through the relationship given below. In effect, the FSR of the reference signal is known, and the initial FSR difference between the reference and the sensing signals is made as small as possible.

$$FSR_{envelope} \propto \frac{1}{\delta_{FSR}}$$

A parameter referred to as the *Magnification Factor* (M– factor), given below, describes the enlargement of the envelope FSR as a fraction of the sensing interferometer FSR.

$$M = \frac{FSR_{envelope}}{FSR_2} = \frac{2 \cdot FSR_1}{FSR_2 - FSR_1}$$

When the difference is very small initially, the envelope period extends to large values and hence is effectively seen as a constant over small spectrum ranges. As the parameters around the sensing interferometer changes, the FSR difference becomes larger, the envelope period and the magnification factor decrease inversely with the change.

For that reason, the optical Vernier effect is known for its high sensitivity, as Figure 13 shows, the Magnification factor is plotted against the FSR difference with FSR_1 = 1 nm. The steepness of the M-factor plot is large enough, allowing for accurate mapping of M-factors to small FSR differences, certainly, beyond the minimum detectable change in FSR (arbitrarily marked by the red lines in Figure 13), which is a function of the spectrum as will be discussed in the following subsection. In practice, through curve fitting, the FSR of the envelope can determined, and with the reference signal FSR known, the difference in FSR can determined. The sign of the M-factor indicates change in Sensing FSR with respect to reference; positive M-factor, indicates an increases in the sensing FSR, and vice versa.

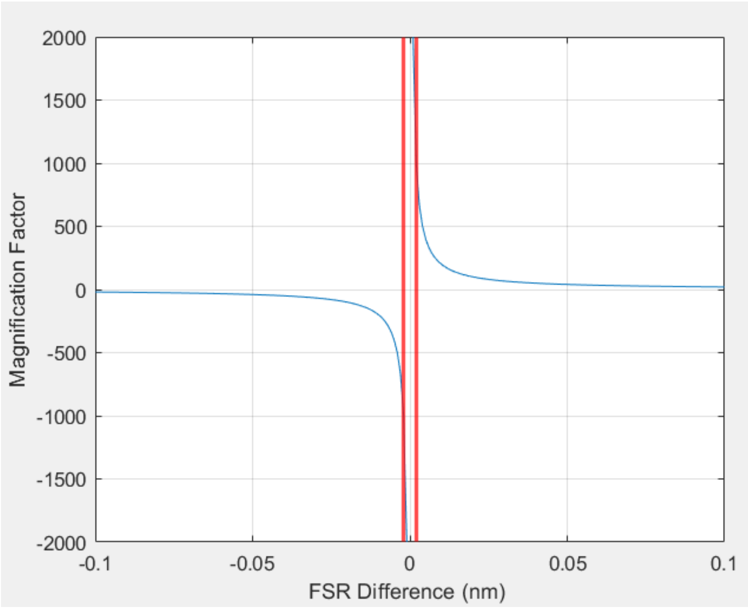


Figure 13. Magnification factor as a function of FSR difference. The red lines mark an arbitrary number as the minimum detectable change in FSR.

Minimum FSR Change Detectability Limit

The minimum detectable change in FSR is dependent on the FSR of the envelop in relation to the spectrum range of the source. The optical Vernier effect analyses the FSR change in the wavelength domain, where through curve fitting, the FSR of the envelope can determined, and with the reference signal FSR known, the difference in FSR can determined. Let S be the spectrum range. Then, for absorbable and determinable envelope, S must be least one-fourth the interferometric period, as given below.

$$S \geq \frac{1}{4} \cdot \frac{2 \cdot FSR_2 \cdot FSR_1}{FSR_2 - FSR_1} = \frac{1}{2} \cdot \frac{(FSR_1 - \delta_{FSR}) \cdot FSR_1}{\delta_{FSR}}$$

Therefore,

$$S = \frac{1}{2} \cdot \frac{\left(FSR_1 - \delta_{FSR,min}\right) \cdot FSR_1}{\delta_{FSR,min}}$$

$$\delta_{FSR,min} = \frac{FSR_1^2}{2S + FSR_1}$$

Or,

$$\% \, min. \, detectable \, FSR \, change = \frac{FSR_1}{2S + FSR_1} \times 100$$

For $FSR_1 = 1.12363$ nm, and $S = 100$ nm, then,

$$\% \, min. \, detectable \, FSR \, change \, (OVE) = 0.5587 \,\%$$

This would require increasing the group refractive index to 1.4686, which would occur at a temperature change of ~984 °C, assuming no change in the conditions and that the thermos-optics coefficients extrapolate linearly to that range.

For that, in the scenario of symmetric heating of the arms, limitations of the detectability of FSR change set by the spectrum range making it seem as if the summed output signals of a single MZI acquired at different temperatures (imitating parallel configurations), are in effect aperiodic exhibiting minimal shape changes, and the optical vernier effect cannot be observed.

Consequently, it is more common to track and study phase shift introduced by temperature changes in a spectrum. Moreover, we can restate the detectability limit viewed from the frequency spectrum as opposed to the wavelength spectrum, presented in the next subsection.

The FFT

It is known that when applying the FFT, the window size, or the wavelength spectrum span limits the resolution of the resulting FFT spectrum, or how closely spaced the frequency bins are, and accordingly, the smallest distinguishable change in FSR. This is shown as follows. The FFT resolution can expressed as given in the relation below.

$$\delta f = f_{k+1} - f_k = \frac{F_s}{L}$$

Where F_s is the sampling frequency, and L is the number of samples obtained from the original spectrum, here is the wavelength spectrum. Moreover, L can be expressed as the ratio of the wavelength spectrum span (S) to the sampling period (T_s) in the wavelength spectrum, which is also $\Delta\lambda$ or the separation between the wavelength bins. Moreover, F_s is $\frac{1}{T_s}$ or $\frac{1}{\Delta\lambda}$, and the frequency resolution is δf can therefore be expressed as:

$$\delta f = \frac{(1/\Delta\lambda)}{(S/\Delta\lambda)} = \frac{1}{S}$$

The above formula can be reasoned through by recalling that the Fourier transform of a rectangular window is a sinc function with a main lobe width being inverse of the window length, $\frac{1}{S}$, thus limiting the spacing between each distinguishable frequency component.

The FFT, the Envelope FSR, and the Minimum FSR Change Detectability Limit

In this section, we derive the equation for the minimum detectable change in FSR when the FFT tool is used, by relating back to interferometric period of the envelope FSR, allowing for the unification of both concepts; and we also show the equation using the common intuition. The expression for the envelope signal, can be can be expressed as a function of the frequency difference between the constitutive signals, as shown below.

$$\cos\left(\frac{2\pi}{FSR_{\lambda,envelope}} \cdot x_\lambda\right) = \cos\left(2\pi \cdot \left(\frac{\Delta f}{2}\right) \cdot x_\lambda\right)$$

Therefore,

$$\frac{\Delta f}{2} = \frac{1}{FSR_{\lambda,envelope}}$$

Or,

$$\Delta f = \frac{1}{FSR_{\lambda,envelope}/2}$$

With FSR_2 expressed as percentage change from FSR_2, the minimum detectable change in FSR can be determined when $\Delta f = 1/S$, the limit set by the observed spectrum range. Meaning half of the envelope period fitting into the spectrum range, S.

Moreover, when the width of the sinc function convolved with the detected frequency components within the signal exceeds the difference between the signal's frequencies (half of the envelop period extends beyond the spectrum range), there frequency components become merged into one component by the windowing spectrum range sinc function. It can be shown that the Δf expression can be re-expressed to match expression below.

$$\delta f = \frac{1}{FSR_{\lambda,o}} - \frac{1}{FSR_{\lambda,o} \times (1 + \%\, min.\, detectable\ FSR\ change)} = \frac{1}{S}$$

Hence,

$$\%\, min.\, detectable\ FSR\ change\ (FFT) = \frac{FSR_{\lambda,o}}{S - FSR_{\lambda,o}}$$

For $FSR_{\lambda,o} = 1.12363$ nm, and $S = 100$ nm, then,

$$\%\, min.\, detectable\ FSR\ change\ (FFT) = 1.1364\%$$

Which is approximately ~152 times greater than the expected change in FSR. Over a 100 nm spectrum range (the considered range above), the down-chirp effect of the interferometric periodicity in wavelength results in an increase of 0.14323 nm in the FSR towards the end of the spectrum compared to its begging. Which results in the output FFT spectrum exhibiting relatively messier effects, including the broadening effect in the sinc of the extracted signal in the direction of lower frequencies, increasing the uncertainty in the actual frequency component detected.

To study this effect and to validate the calculation of the minimum detectable change in FSR, we simulate on a real acquired source spectrum the modulation effect of the two parallel MZIs, when FSR difference between the output of signal of the reference interferometer (reference signal) differs from that of the sensing interferometer (reference signal) by (1) **5% above** the % min. detectable FSR change, (2) **30% below** the % min. detectable FSR change, and (3) **10 ×** % min. detectable FSR change. The 5% in the first case is added to accommodate for rounding errors.

For each considered case, Figures 14-16 show the subplots of the separate outputs of the MZI interferometer superimposed on each other, the algebraic sum over the spectrum and with the envelope superimposed, and finally the FFT spectrum applied on the sum (or simulated output of the two MZIs).

The FSR of the detected signals is acquired using the equation given below. Table 1 summarizes the theoretical and simulated results of the signals FSRs.

$$FSR_{\lambda} = \frac{1}{cycles/nm}$$

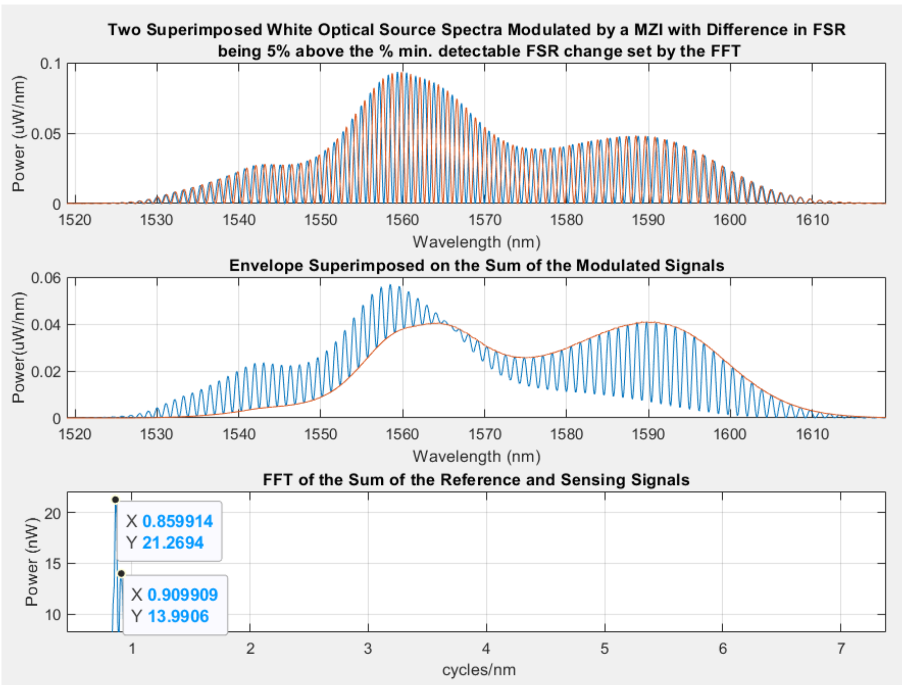


Figure 14. Simulated parallel MZIs when the percent difference between the reference and sensing signals is 5% above the % min. detectable FSR change

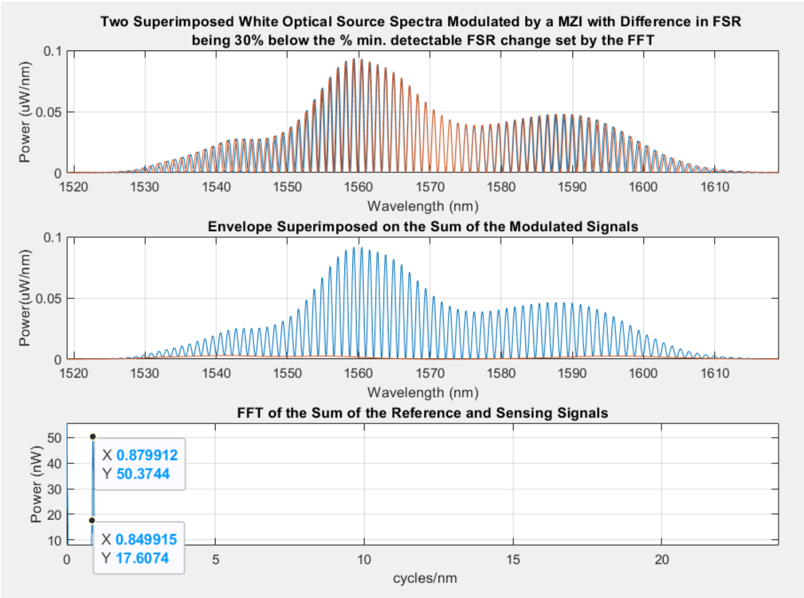


Figure 15. Simulated parallel MZIs when the percent difference between the reference and sensing signals is 30% below the % min. detectable FSR change

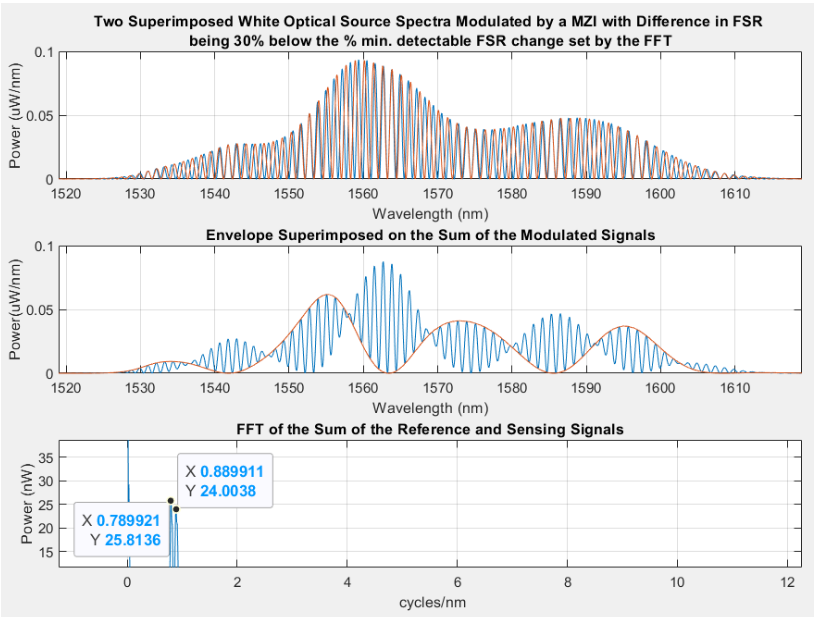


Figure 15. Simulated parallel MZIs when the percent difference between the reference and sensing signals is 10 × % min. detectable FSR change

Table 1. Theoretical vs. Simulation Results of Reference and Sensing Signals FSRs

FSR percentage difference	Reference and Sensing Signals FSRs **(nm)**		Percentage Error **(%)**
	Theoretical Results	FFT Simulation Results	
5% above the % min. detectable FSR change	1.1236	1.0990	2.1894
	1.1370	1.1629	2.2779
30% below the % min. detectable FSR change	1.1236	1.1365	1.1481
	1.1326	1.1766	3.8849
10 × % min. detectable FSR change	1.1236	1.237	0.009
	1.2513	1.2659	1.1667

As expected, in the first case (Figure x), the signals align once, towards the end of the spectrum, it means they are adequately different that they appear as periodic within the observed spectrum range, thus, allowing for tools such as the optical Vernier effect and the FFT to discriminate among the signals.

From the second subplot, this is indicated by at least half of the period of the envelope fits within the spectrum range; this is not observed in the second case (Figure x), where signals are not adequately dissimilar. In the third case (Figure x), the signals align multiple times over the spectrum range, and it shown multiple periods of the envelope fitted.

Moreover, as expected, the percentage error decreased as the % FSR difference between the summed signals increased. While it was expected that for the second case, the FFT would not be able to distinguish the two signals, it separated the signals with an error of ~4%, which remain a poor performance.

The third case achieved an accuracy as low as 0.009% for identifying the first signal, and ~1% to separate the second signal. The first case, the limiting case, achieved an accuracy that is in-between the two cases, ~2%. Note that rounding errors would change these percentages slightly. These examples served to allow for the visualizations of the output results, expectations, and to validate the calculations made for the OVE and the FFT tools. In the next section, we present the last FSR detection tool, the cross-correlation.

Cross-Correlation

Signals sharing a common period exhibits a periodic cross-correlation patterns at the number of cycles for which they align. Said in another perspective, and the observation of this pattern is limited by the amount of FSR difference between the signals. Figures x and x shows examples of the cross-correlation plot between two periodic and aperiodic signals. The expression for the number of alignments within a given range, is given below, influencing detectability limits. Derivation is left for the reader.

$$\# \, alignment = \frac{S}{T_{common}} = \frac{S}{FSR_{envelope}}$$

Under the assumption that FSR change is negligible with in the observed spectrum range, then, the wavelength lags at which they first align can be used to calculate for the phase shift between the two signals, as given below.

$$\varphi = \frac{2\pi \cdot \lambda_{lag}}{FSR_1}$$

This is discussed and tested in the methodology section with simulations plots. In the sext section, we discuss the usage of FFT for the extraction of the FSR_1, as well as determining the maximum representable frequency component, which is indicative is the largest possible L which can produce non-aliased fringe patterns.

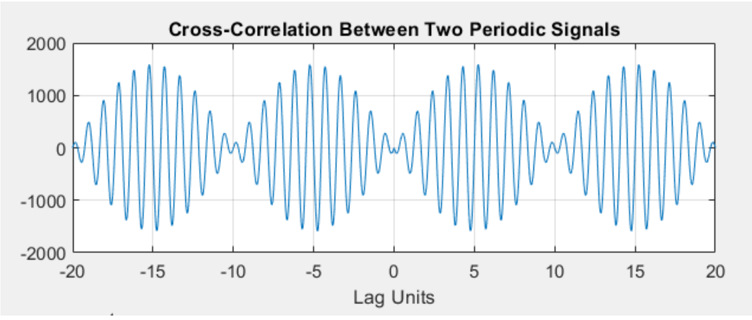


Figure x. Cross-correlation plot between two periodic signals, cos($2\pi x$) and cos($2\pi(1.1)x$)

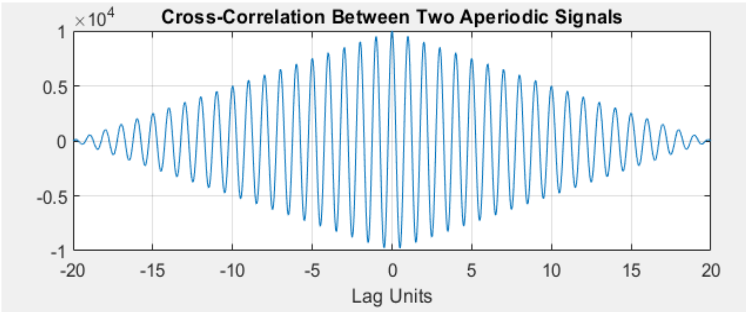


Figure x. Cross-correlation plot between two periodic signals, cos($2\pi x$) and cos($2\pi(1.01)x$)

Non-Aliasing Fringe Patterns Corresponding to Maximum Arms Mismatch

From the previous section, it was concluded that the wavelength spectrum span controlled the resolution of the resulting FFT spectrum hence how closely spaced the frequency bins are, affecting the minimum detectible change in FSR extracted through the FFT. The other parameter of the FFT spectrum is the sampling frequency, determining the maximum representable frequency component in the applied spectrum to be $\frac{F_s}{2}$.

For a given spectrum interferometric periodicity, the FFT detected frequency component is its inverse, and the arm mismatch length (ΔL) can be related to F_s as:

$$f = \frac{1}{FSR_\lambda} < \frac{F_s}{2}, \text{ or } 2\Delta\lambda < FSR_\lambda$$

Therefore,

$$2\Delta\lambda = \frac{\lambda_0^2}{n_g \Delta L_{max}}$$

$$\Delta L_{max} = \frac{1}{2\Delta\lambda} \cdot \frac{\lambda_0^2}{n_g}$$

For $\Delta\lambda = 0.01$ nm, $n_g = 1.4606$, $\lambda_0 = 1550$ nm.

$$\boldsymbol{\Delta L_{max} = 82.244\ \mathrm{mm}}$$

This takes into account the limitations set by the FFT and applies only under the assumption that the source coherence length exceeds this value for observable fringes. Thus far, we discussed the how a temperature change of 10 Celsius can negligibly change the FSR but introduces a considerable phase shift in the spectrum. We also showed how observing that small FSR change requires a source of 30 times the spectrum range of that used.

Through the optical vernier effect and FFT methods we also consolidated that given the fixed spectrum range, minimum detectable FSR change requires unrealizable change in temperature for an observable change. This summarizes theoretical expectations, in the next section, we discuss the methodology of the experiment, including components and equipment used, the setup and procedure, and finally presents MATLAB simulation of phase shift and FSR change on the source spectrum. On the results and analysis section, simulation results are analysed against experimental results.

Methodology

For the purpose of analysing the spectral changes due to thermal fluctuations in the laboratory setup over a period of time, the methodology carried out is detailed in this section. In the following subsection, we list the components and equipment used.

Equipment, and Components

Listed below are the components and equipment used and their specifications:

- Optical Spectrum Analyzer: YOKOGAWA AQ6370C (Figure 9)

Operates in the telecom range of 600 nm to 1700 nm, has a wavelength resolution of 0.02 nm, and a sensitivity of -90 dBm, hence can detect signals in the pico-W range.

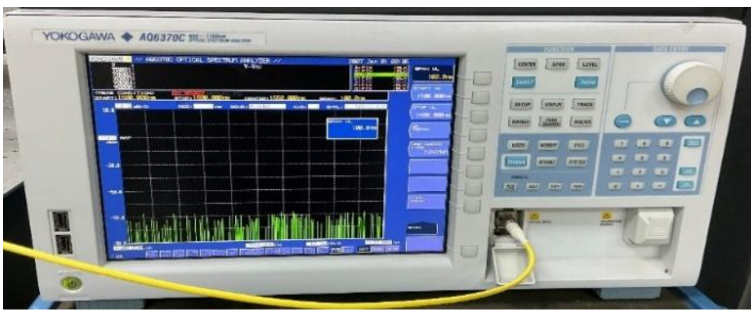


Figure 9. Optical spectrum analyzer, YOKOGAWA AQ6370C

- C & L Band ASE730 Light Source (Figure 10)

Operates in the wavelength range of 1530 nm to 1610 nm and delivers + 15 dBm over the operating wavelength range.

Figure 10. C & L Band ASE730 Light Source

- **(x4)** Fiber Optic Adapter (Figure 11)

Figure 11. Fiber Optic Adapter

- **(x2)** Y-Splitter (Figure 12)

Figure 12. Y-Splitter

- **(x2)** Fiber Arms, Patch Cables (Figure 13)

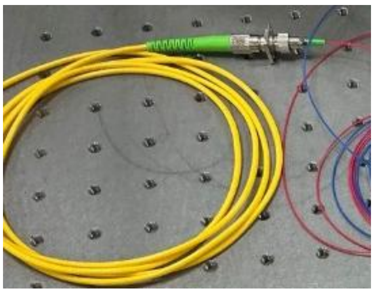


Figure 13. Patch Cables

- NI GPIB-USB-HighSpeed Interface Adapter Controller (Figure 14)

Figure 14. NI GPIB-USB-HighSpeed Interface Adapter

Setup and Procedure

The Mach-Zehender interferometer was built following Figure 1, as shown in Figure 15. The first Y-splitter was connected to the source spectrum then connected to the interferometer arms through the fibre optics adapter; the arms were then connected to the second Y-splitter through the two other fibre optics adapters. Finally, one branch at the output of the second Y-splitter was connected to the optical spectrum analyser.

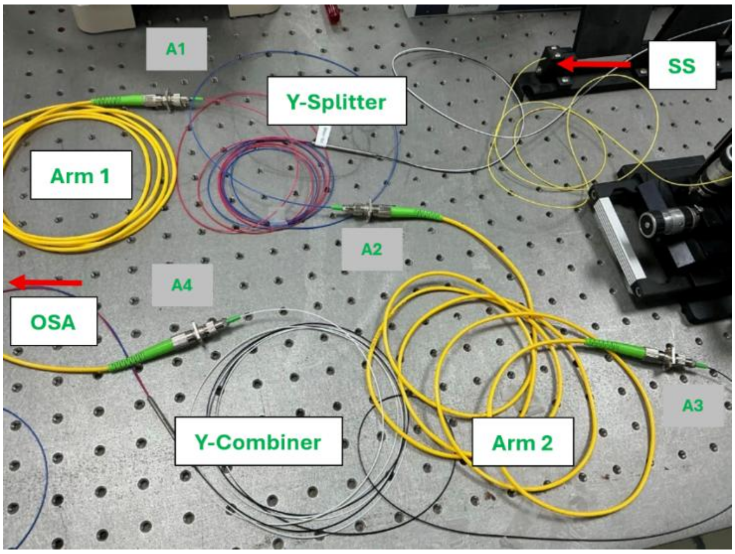


Figure 15. Experimental configuration of the Mach-Zehnder interferometer

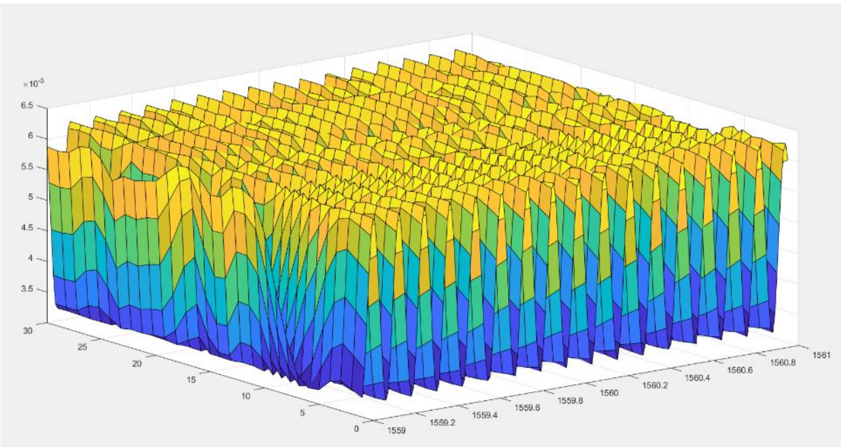
The obtained fringe patterns over a 10 nm spectrum range are shown in Figure 16. The figure indicates that the mismatch between the two arms is not negligible. Reducing the spectrum range to only 10 nm allowed for the observation of the changes in the patterns over different sweeps of the spectrum, to be more trackable.

For that, we export the spectrum measurement divided into 10 nm ranges as csv for analysis. We also acquire 30 readings every 3 seconds for enough changes to be captured. This requires interfacing the spectrum analyser to the laptop, the steps and codes are given in Appendix A and Appendix B, respectively.

Moreover, to simulate changes in FSR and phase changes on the unmodulated source spectrum, we connect the source directly to the OSA and acquire the measurements. In the next section, we show the determination of the most sensitive range of the spectrum of which carry the further analysis on. We then show the usage of the FFT on the selected range for the accurate determination of the OPD or mismatch between the arms. This is then used to accurately simulate FSR and phase changes from a reference spectrum.

Figure 16. Optical spectrum analyser results at the output of the second Y-Splitter, showing the resulting fringe patters.

The 3D surface of the collected data (wavelength, #Reading, Power)



MATLAB Simulation

In this section, we illustrate the usage of FFT for the extraction of the arm mismatch length. We also simulate various temperature changes and determine the corresponding phase shift and % FSR change on the spectrum, used to compare the actual exported experimental results.

Moreover, MATLAB simulations are carried out on the most sensitive range, whose most responsive to small fluctuation sin temperature change. The determination of this range is done by plotting the variation of the power measurement of a single wavelength across multiple readings. The wavelengths point in a given range experiencing the strongest fluctuation in power measurement was determined to be the most responsive.

Appendix C shows the obtained plots as described, from which it was determined that the **1555 nm to 1565 nm** range should be used. The code used for obtaining the plot is given in Appendix D. The FFT and simulations are performed on this range, as shown in the next subsection.

Arm Mismatch Length Extraction

To determine the arm mismatch length, we employ the FFT. The obtained the plot is shown in Figure 17. The code used to obtain the plot is given in Appendix D. The first frequency component provides a solution for L_i as given below.

The higher frequency component corresponds to signals interfering with an OPD that 10 times the arm mismatch length, this could be due to reflections introduced at different parts of the system, resulting in its travel of longer distances before summed with the other arm's wave. The dc term corresponds to the dc term present in the MZI intensity equation.

$$L_i = \frac{\lambda_0^2 \times f}{n_g} = \frac{(1560\ nm)^2 \times 8.29171\ cycles/nm}{1.4606} = \mathbf{13.8154\ mm}$$

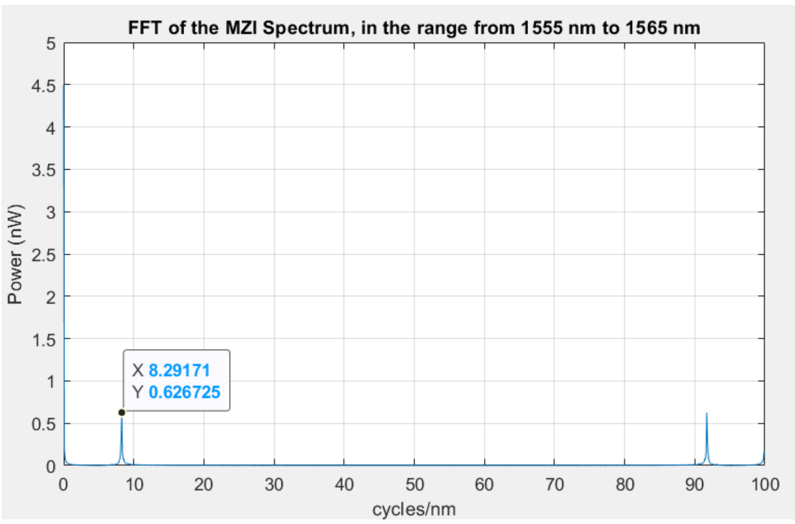


Figure 17. FFT frequency spectrum of the MZI for the range 1555 n m to 1565 nm

Simulated Temperature Change & Phase Shift and FSR change

In this section, we simulate temperature changes on the acquired source spectrum over the 1559.5 nm to 1560.5 nm range (reduced for clearer plots) modulated by a MZI with $L_i = 13.8154$ mm, to match the experimental parameters. Figures x and x show the simulated fringe patterns with the % FSR change and phase shift also displayed for each temperature change.

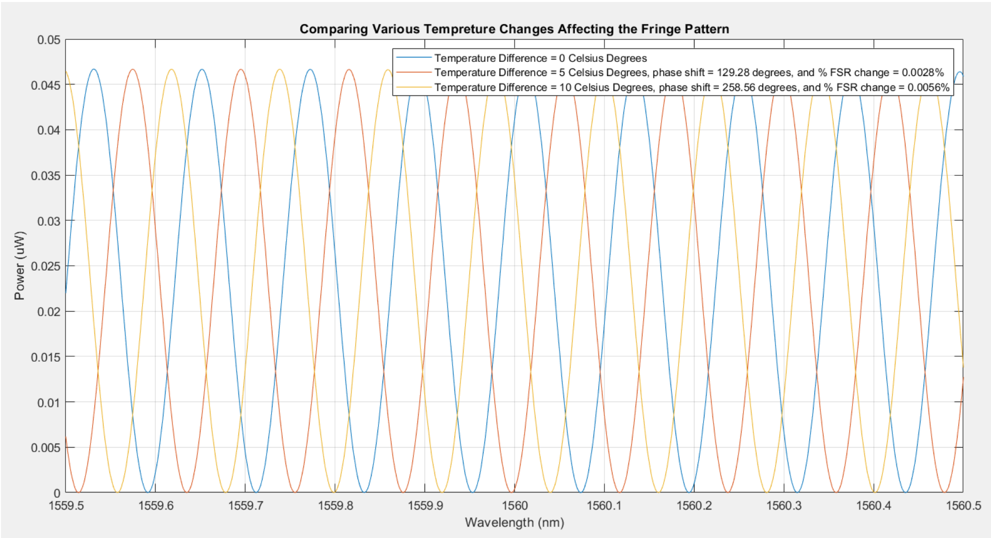


Figure 17. Simulated temperature changes, 5 ℃ *and* 10 ℃*, on the source spectrum by a MZI with* $L_i = 13.8154$ *nm.*

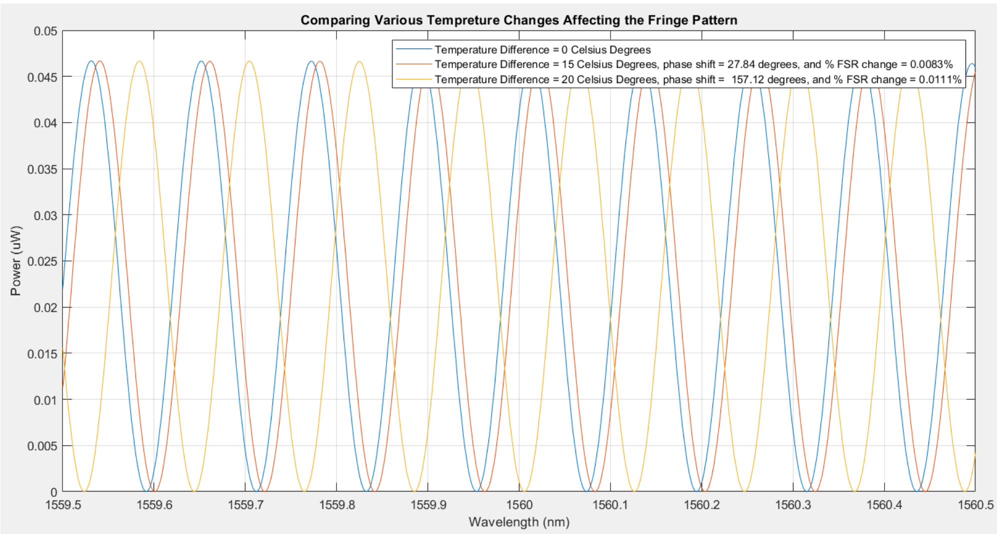


Figure 17. Simulated temperature changes, 15 ℃ *and* 20 ℃*, on the source spectrum by a MZI with* $L_i = 13.8154$ *nm.*

The % FSR change determined for the temperature change in the range from 0 ℃ to 20 ℃ reaches as high as 0.011%, that is still below the threshold of detectability by all the methods explored. As for the phase shift, it ranged from values of 0 to 360 for lower temperatures and unwraps back to the same range for higher temperatures.

This can be verified from the phase shift v. temperature change plot shown in Figure x, where after the 14°C tempreture change point, the phase goes back to zero. In the next section, we compare the acquired results againt the simulation results and apply the detectability tools presented.

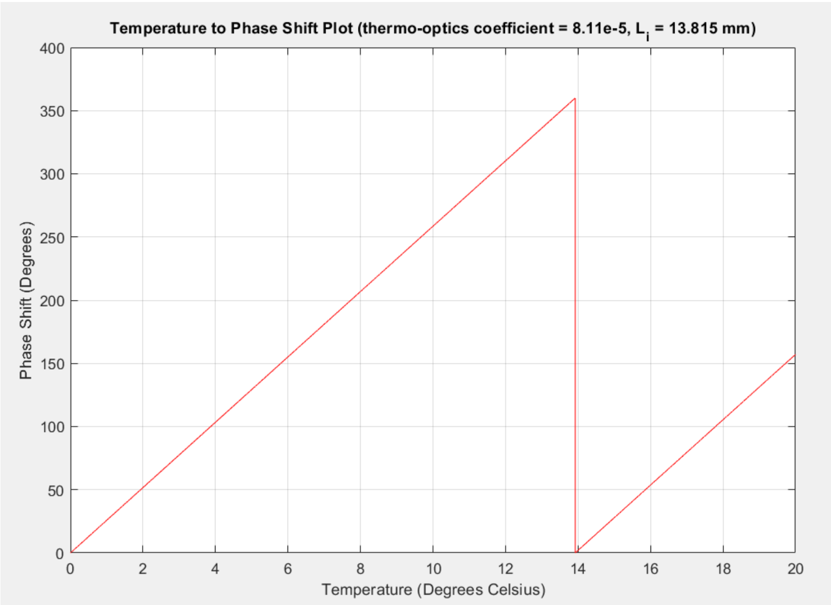


Figure 17. Phase shift as a function of temperature change when MZI path different is 13.8154 mm

Results Analysis

Figure x shows the acquired spectrum results at various instances (0, after 27 sec., after 1 min. and 25 sec. from the first acquired signal). The reader can verify that the "original signal" and second signal acquired closely matches the simulated results (Figures x) for a $\Delta T = 0$ °C and $\Delta T = 5$ °C, respectively. While the third signal acquired seems to be fall in the range of temperature change of 5 °C to 10 °C. The full captured signals over the spectrum range S = 10 nm is shown in Figure x.

As expected, no periodicity pattern is observed among the captured signals, for with the speculated temperature changes in the captured signals is below the threshold when S is 100 nm, let alone when S = 10 nm in the current case. In the next subsection, we explore information's that can be extracted from the spectrum using the cross-correlation tool and validate theoretical expectations.

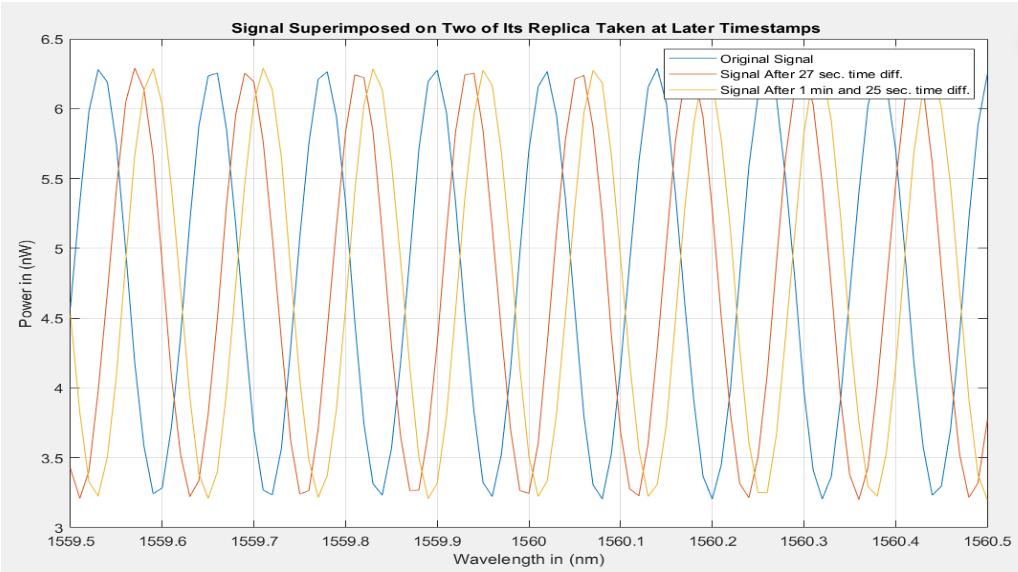


Figure x. Acquired spectrum results at various timestamps, S = 2 nm

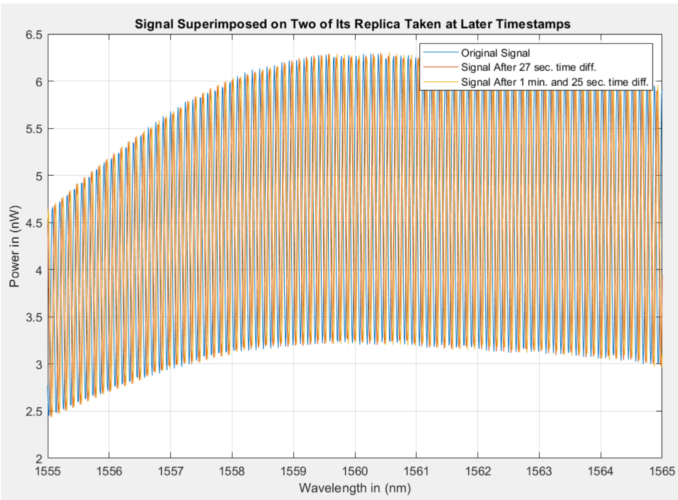


Figure x. Acquired spectrum results at various timestamps, S = 10 nm.

Cross-Correlation Tool

The plots of the cross-correlation between the two captured signals and the original signal are shown in Figures x and x, with the wavelength at which the cross-correlation peak occurs marked. As expected, for

$$\varphi_1 = \frac{2\pi \cdot \lambda_{lag,1}}{FSR_\lambda} = \frac{2\pi \cdot 0.04\ nm}{\left(\frac{(1560\ nm)^2}{1.4606 \cdot 13.8154\ \mathrm{mm}}\right)}$$

$$\varphi_1 = 2.0839 \text{ radians or } \mathbf{112.40\ degrees}$$

From the phase shift versus temperature plot, this corresponds to **4.348** °C temperature change.

$$\varphi_2 = \frac{2\pi \cdot \lambda_{lag,2}}{FSR_\lambda} = \frac{2\pi \cdot 0.06\ nm}{\left(\frac{(1560\ nm)^2}{1.4606 \cdot 13.8154\ \mathrm{mm}}\right)}$$

$$\varphi_2 = 3.1258 \text{ radians or } \mathbf{179.10\ degrees}$$

This corresponds to 4.348 °C temperature change **6.928** °C.

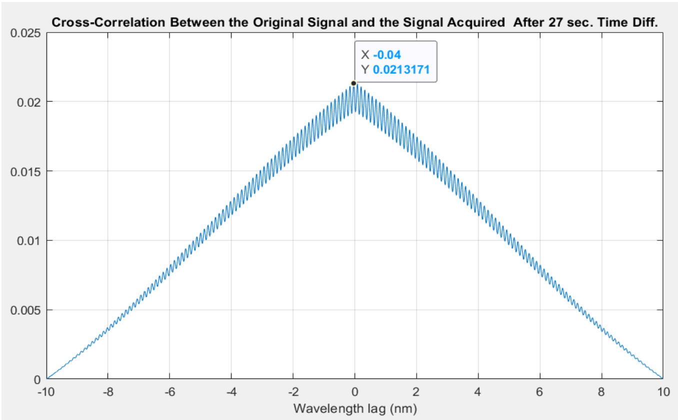


Figure x. Cross-Correlation Between the Original Signal and the Signal Acquired After 1 min. and 25 sec. Time Diff.")

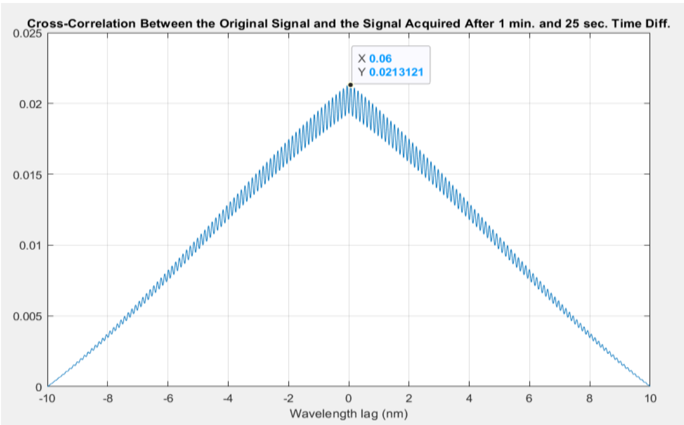
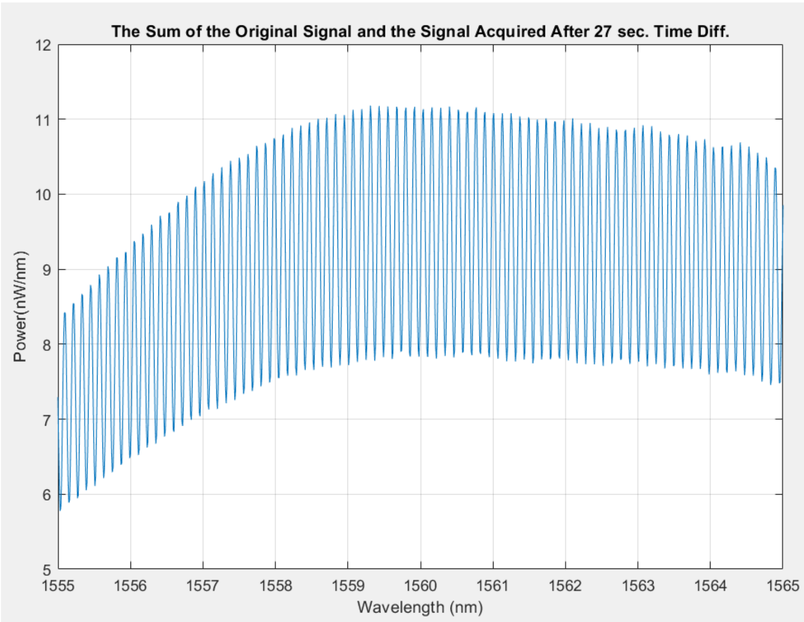
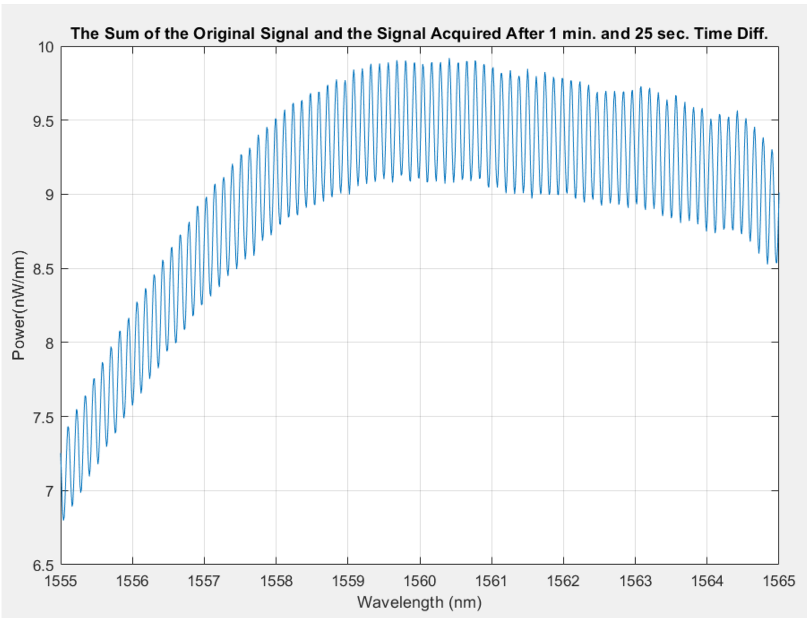


Figure x. Cross-Correlation Between the Original Signal and the Signal Acquired After 1 min. and 25 sec. Time Diff.")

The Optical Vernier Effect Tool

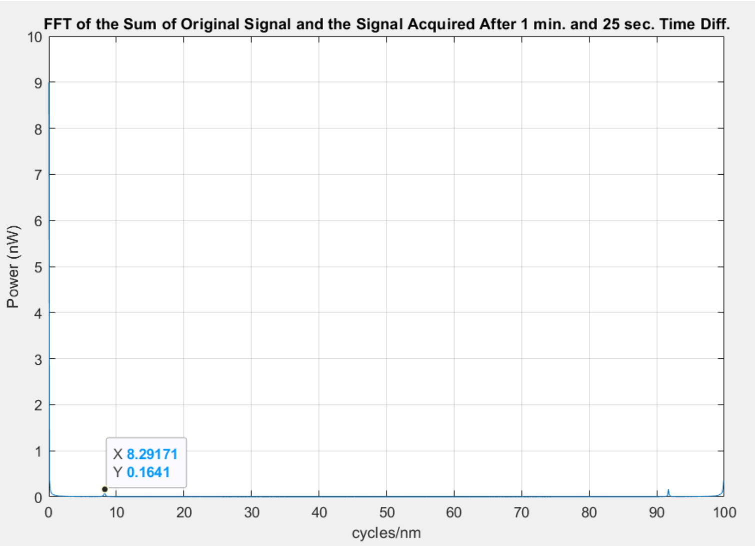
(1) Sum of the two signals, observe an envelope? Note, if yes, attempt to fit one, mention S relation again to how well we see an envelope. Any observed difference in envelope ? why fitting an envelope is hard, relate to **modulation depth**, what could affect modulation depth,

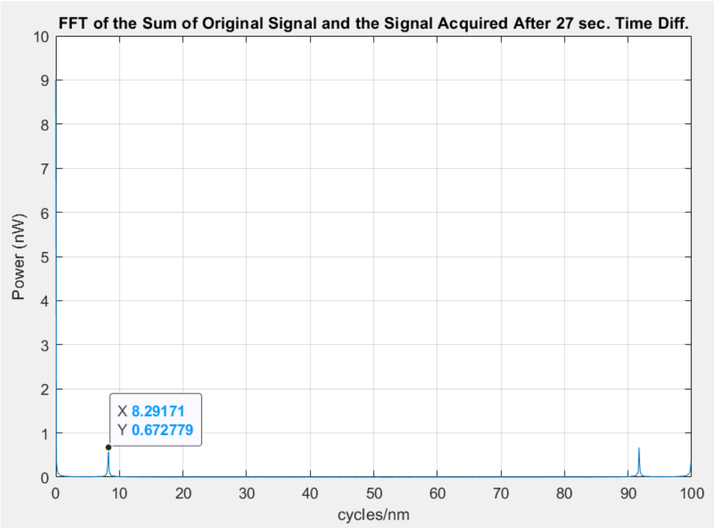




The FFT Tool

(2) FFT plot of the summed signals, again simulating parallel configuration , showing one detected signal, add the fact that S is even reduced resolution is reduced as wll





Comparison of Wavelength Shifts Detection Tools

Application: Battery Temperature Estimation

In this section, the application of a Mach-Zehnder interferometer in determination of a battery's temperature is studied. The application setup is emulated by heating a can wrapped around by fibre, the other arm is connected in the same manner to increase the stability of the system. The described setup diagram is shown in Figure x. The purpose of this exploration is to define a reliable and reproducible relationship between temperature change and the changes in the fringes of MZI spectrum.

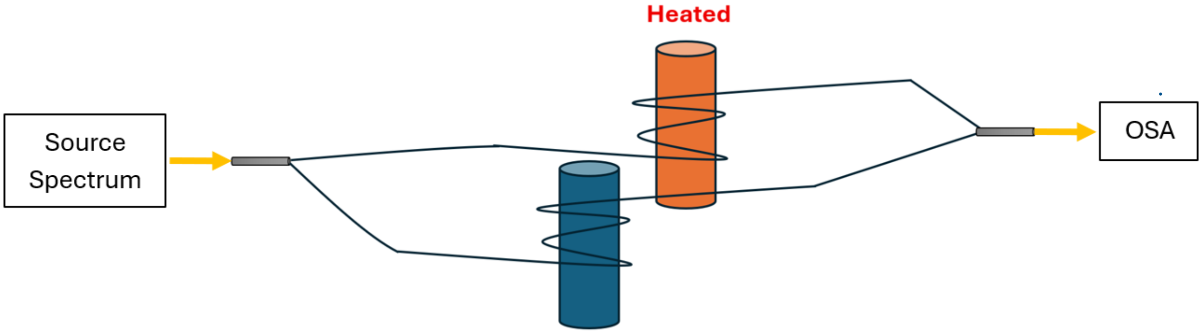


Figure x. Mach-Zehnder interferometer setup in an emulated battery temperature estimation application

In this setup, since one arms is only heated, thermal expansion becomes non-negligible, and the effect becomes amplified by the change in the refractive index in one arm. These direct us to re-consider some of the assumptions made in the earlier sections of the report, where in this application, ignoring the FSR might be misleading. We express the FSR in the form shown below, to accommodate for separable changes in n and L experienced by one arm only.

$$FSR = \frac{\lambda_0^2}{n_1 L_1 - n_2 L_2}$$

Before heating one arm (the can), the FSR can be expressed as shown below.

$$FSR_0 = \frac{\lambda_0^2}{n_1 L_1 - n_2 L_2} = \frac{\lambda_0^2}{n_2 \cdot (L_i + L_2) - n_2 L_2} = \frac{\boldsymbol{\lambda_0^2}}{\boldsymbol{n_2 \cdot L_i}}$$

When there is temperature change, the equation becomes as shown below,

$$FSR_{new} = \frac{\lambda_0^2}{(n_2 + \Delta n) \cdot (L_i + L_2 + \Delta L) - n_2 L_2}$$

$$FSR_{new} = \frac{\lambda_0^2}{n_2 \cdot L_i + n_2 \cdot \Delta L + \Delta n \cdot L_i + \Delta n \cdot L_2 + \Delta n \cdot \Delta L}$$

With L_2, Δn, and ΔL defined as given below.

$$L_2 = L_{c,straight} + 2\pi N r$$

$$\Delta n = \frac{\partial n_g}{\partial T} \cdot \Delta T$$

$$\Delta L = \frac{\partial L}{\partial T} \cdot (L_i + L_2) \cdot \Delta T$$

The change in FSR in percentage is expressed as given below.

$$\%\ FSR\ change = \frac{n_2 \cdot \Delta L + \Delta n \cdot L_i + \Delta n \cdot L_2 + \Delta n \cdot \Delta L}{n_2 \cdot L_i + n_2 \cdot \Delta L + \Delta n \cdot L_i + \Delta n \cdot L_2 + \Delta n \cdot \Delta L} \times 100$$

The same logic applies for the added phase shift. When there is no temperature change, the relative phase difference between the electric fields of both arms is as given below.

$$\Delta\varphi = \frac{2\pi}{\lambda} \cdot (n_1 L_1 - n_2 L_2) = \frac{2\pi}{\lambda} \cdot (\boldsymbol{n_2 \cdot L_i})$$

When there is a change of temperature, the new relative phase difference is as given below.

$$\Delta\varphi_{new} = \frac{2\pi}{\lambda} \cdot (n_2 \cdot L_i + n_2 \cdot \Delta L + \Delta n \cdot L_i + \Delta n \cdot L_2 + \Delta n \cdot \Delta L)$$

The added phase shift is therefore determined to be as given below.

$$\varphi_{shift} = \Delta\varphi_{new} - \Delta\varphi$$

$$\varphi_{shift} = \frac{2\pi}{\lambda} \cdot (n_2 \cdot \Delta L + \Delta n \cdot L_i + \Delta n \cdot L_2 + \Delta n \cdot \Delta L)$$

The plots for the percentage change of FSR and added phase shift against temperature change for two L_i values are shown in Figures x, and x. The MATLAB code used to generate the plots is given in Appendix D. While the theoretical plots might be counter intuitive, with FSR change reaching as high as nearly100% when N = 10 and L_i = 80 um, it matches mathematical expectations.

(1) the number of loops amplifying the thermal expansion effect, and (2) when deriving the FSR as a function of temperature, as shown below, it is shown that the smaller the initial L_i between the arms, the more significant is the effect of temperature through thermal expansion and the thermo-optics coefficient.

$$\frac{\partial FSR}{\partial T} = \lambda_0^2 \cdot \frac{\partial}{\partial T}(n_2 \cdot L_i)^{-1}$$

$$\frac{\partial FSR}{\partial T} = \frac{-\lambda_0^2}{(n_2 \cdot L_i)^2} \cdot \left[l_i \cdot \frac{\partial n_2}{\partial T} + n_2 \cdot \frac{\partial L_i}{\partial T}\right]$$

In the next section, we present the experimental setup and steps used to investigate and validate the theoretical results.

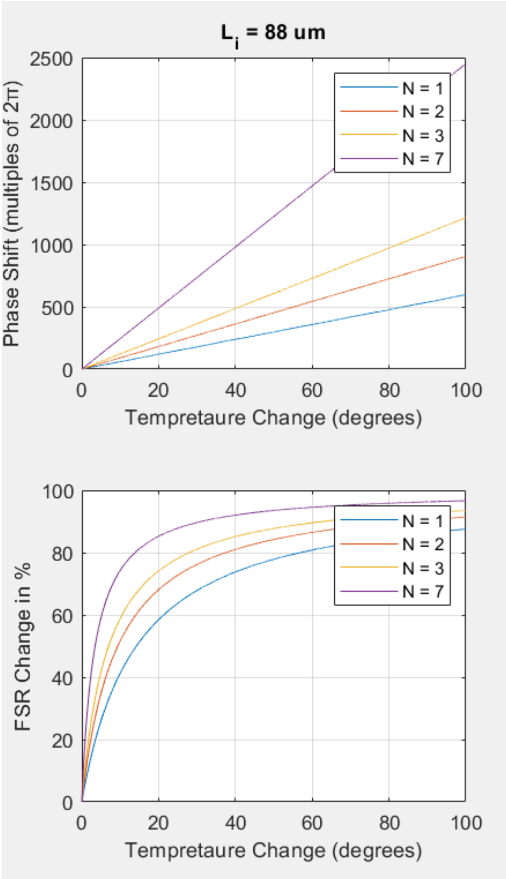


Figure x. % FSR change and phase shift plots as a function of temperature. OPD = 80 um

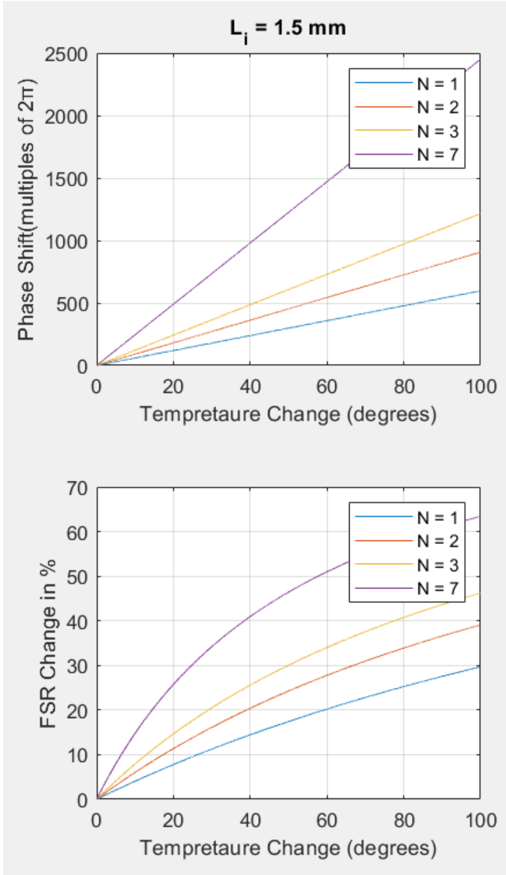
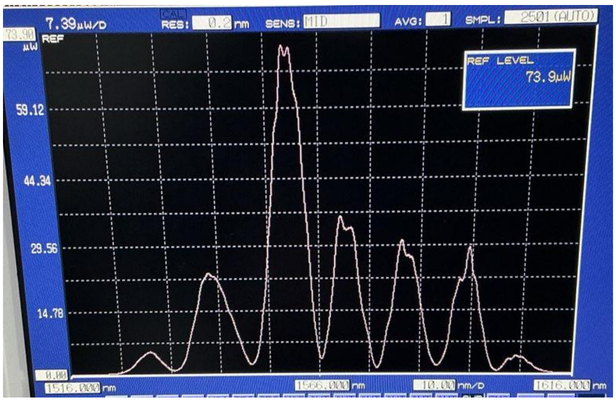


Figure x. % FSR change and phase shift plots as a function of temperature. OPD = 1.5 mm

Experimental Implementation and Results

The implemented setup is shown in Figure x. The spectrum analyser results showing the fringe pattern is shown in Figure x. Since the fringes are more spread out, with only a few oscillations present over the spectrum range, using the FFT to extract the OPD and for tracking FSR change may yield erroneous results. Instead, we attempt to fit the simulated MZI modulating the acquired direct source spectrum with the acquired experimental spectrum.



This is done as follows. We recognize new FSR and phase shift can be expressed as given below.

$$FSR_{new} = \frac{\lambda_0^2}{n_2 \cdot L_i \cdot (1 + \%\, n_2 L_i\ change)}$$

$$\varphi_{shift} = \frac{2\pi}{\lambda} \cdot (\%\, n_2 L_i\ change)$$

With,

$$\%\, n_2 L_i\ change = n_2 \cdot \Delta L + \Delta n \cdot L_i + \Delta n \cdot L_2 + \Delta n \cdot \Delta L$$

We, therefore, can fine tune the percentage change in optical path difference, until a reasonable fit is observed. The resulting modulated spectrum is as given below. Moreover, this allows us to determine the change in FSR and added phase shift in the spectrum.

$$I(\lambda) = S(\lambda) \times \frac{I_i}{2}\left(1 + \cos\left(\frac{2\pi \cdot n_2 \cdot L_i \cdot (1 + \%\, n_2 L_i\ change)}{\lambda}\right)\right)$$

We begin by acquiring the spectrum measurements when there is no change in temperature to estimate the initial mismatch in length between the arms. Figure x shows the fitted simulation plot with the experimentally obtained spectrum. The change in the optical path difference, $\%\, n_2 L_i\ change$, needed to be fine tuned to 0.3% to obtain the fitting presented, resulting in a 0.3% change in FSR and a phase shift of 88,47 degrees form the original plot where nL is exactly $n_g \times L_i$.

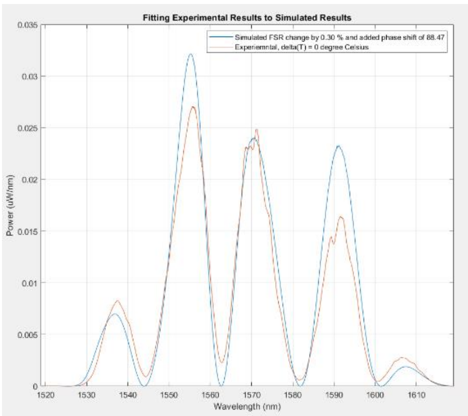


Figure x. Fitted experimental results to simulated results when $\Delta T = \sim 0°C$

Moreover, the estimated arm mismatch was estimated as given below. Note that since the spectrum ranges from 1519 nm to 1619 nm, $\lambda_0 = 1569$ nm. The fitted spectra to the acquired experimental spectra for various temperature changes are shown in Figures x – x. The developed MATLAB code for this implementation is given in Appendix D.

$$\boldsymbol{L_i = 88\ \mu m}$$

$$FSR_0 = \frac{(1569\ \text{nm})^2}{1.4606 \times 88\ \mu m} = \mathbf{19.153\ nm}$$

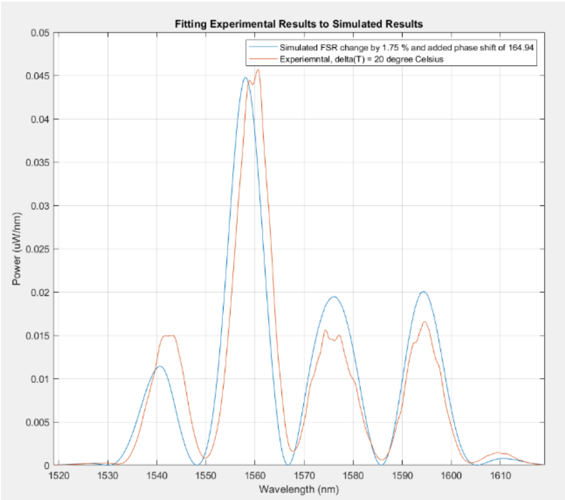


Figure x. Fitted experimental results to simulated results when $\Delta T = 20$℃

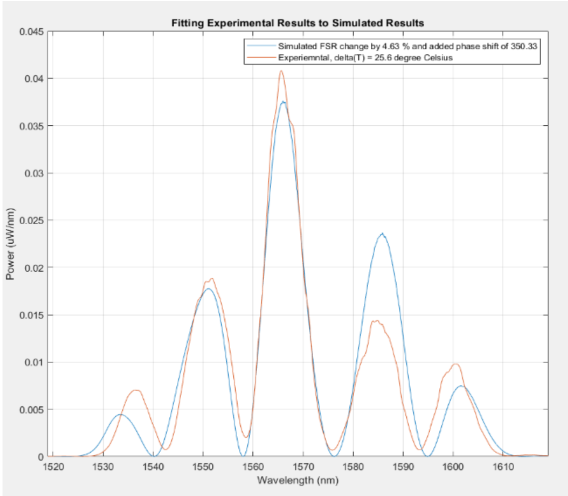


Figure x. Fitted experimental results to simulated results when $\Delta T = 25.6$℃

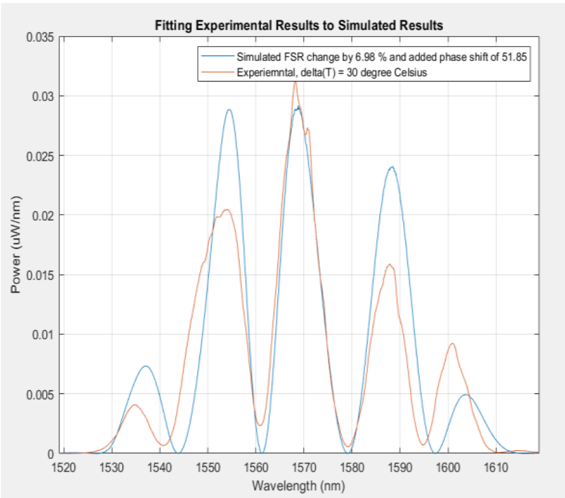


Figure x. Fitted experimental results to simulated results when $\Delta T = 30$℃

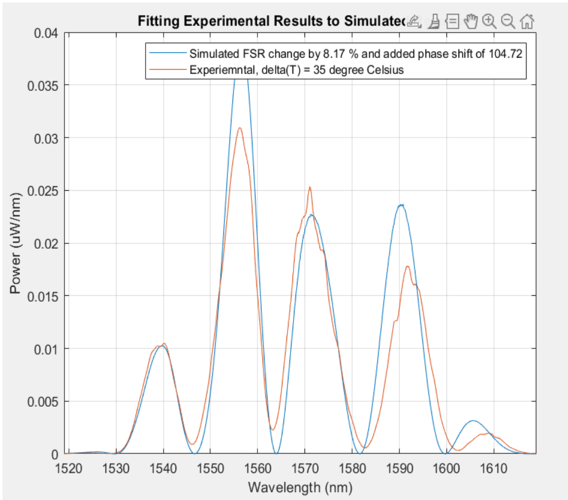


Figure x. Fitted experimental results to simulated results when $\Delta T = 35$℃

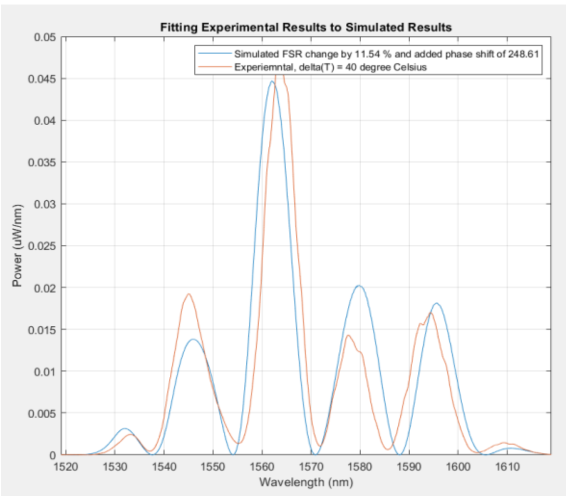


Figure x. Fitted experimental results to simulated results when $\Delta T = 40$℃

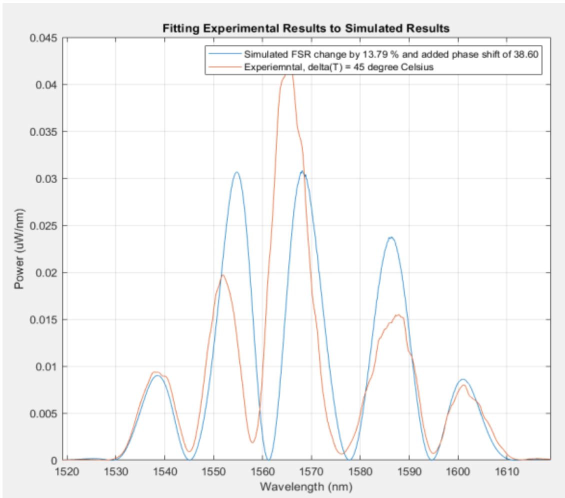


Figure x. Fitted experimental results to simulated results when $\Delta T = 45$℃

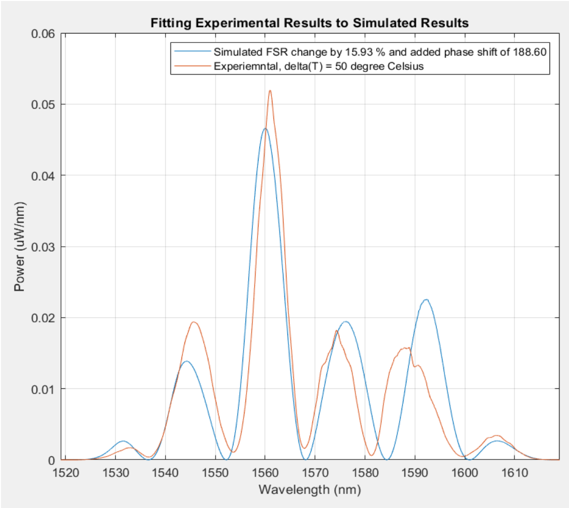


Figure x. Fitted experimental results to simulated results when $\Delta T = 50°C$

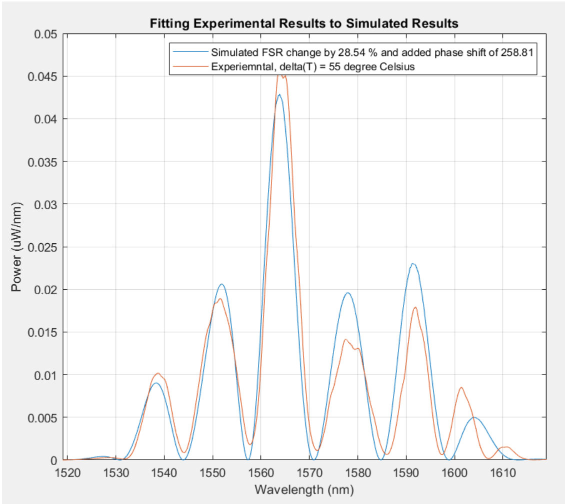


Figure x. Fitted experimental results to simulated results when $\Delta T = 55°C$

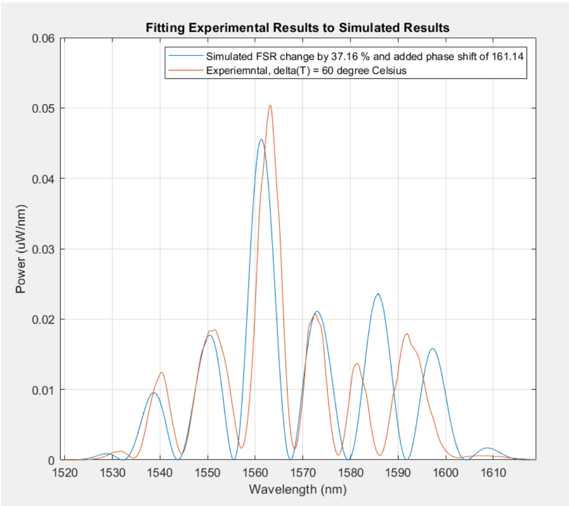


Figure x. Fitted experimental results to simulated results when $\Delta T = 60°C$

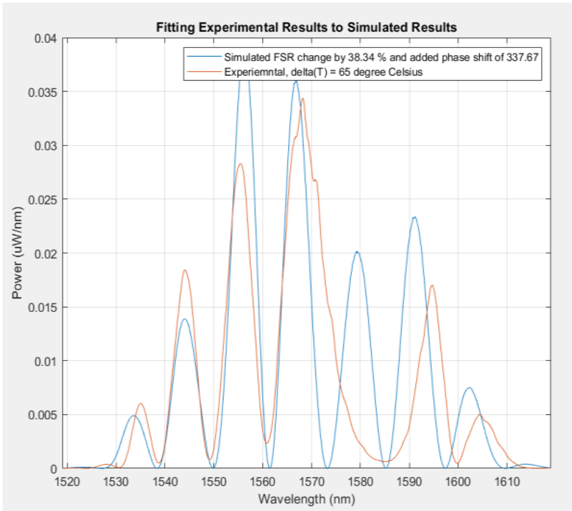


Figure x. Fitted experimental results to simulated results when $\Delta T = 65°C$

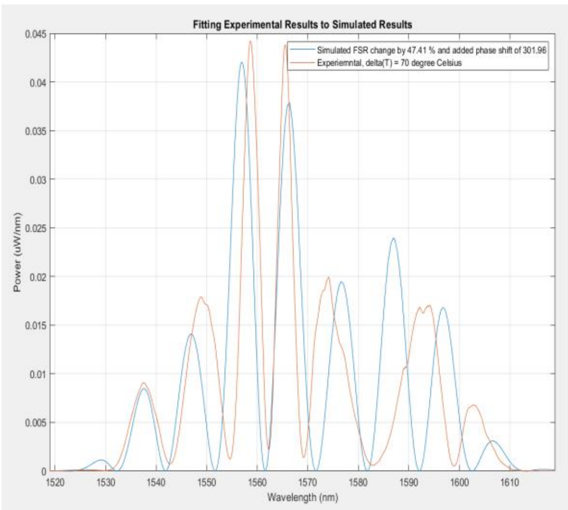


Figure x. Fitted experimental results to simulated results when $\Delta T = 70°C$

Table 1 presents a summary of the experimental results, including the fine-tuned % $n_2 L_i$ $change$, the corresponding calculated percentage chang in FSR, the added phase shift, and the calculated new FSR value. Figure x

Table 1. Summary of Experimental Results from Figures x - x

$\boldsymbol{\Delta T}$ (°C)	% $\boldsymbol{n_2 L_i}$ $\boldsymbol{change}$	% $\boldsymbol{FSR}$ $\boldsymbol{change}$	$\boldsymbol{\varphi_{shift}}$ (degrees)	$\boldsymbol{FSR_{new}}$ (nm)
~0	0.3	0.3	88.47	19.210
2	1.78	1.75	164.94	19.488
5	4.85	4.63	350.33	20.040
10	7.5	6.89	51.45	20.473
15	8.9	8.17	104.72	20.718
20	13.05	11.54	248.61	21.363
25	16	13.79	38.6	21.794
30	19.95	15.93	188.6	22.204
35	39.95	28.54	258.81	24.619
40	59.12	37.1	161.14	26.259
45	62.18	38.34	337.67	26.496
50	90.135	47.41	301.96	28.233

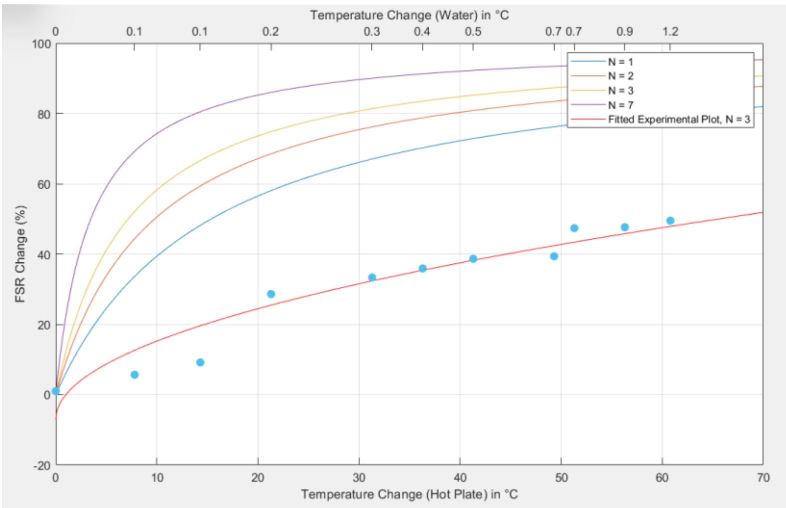
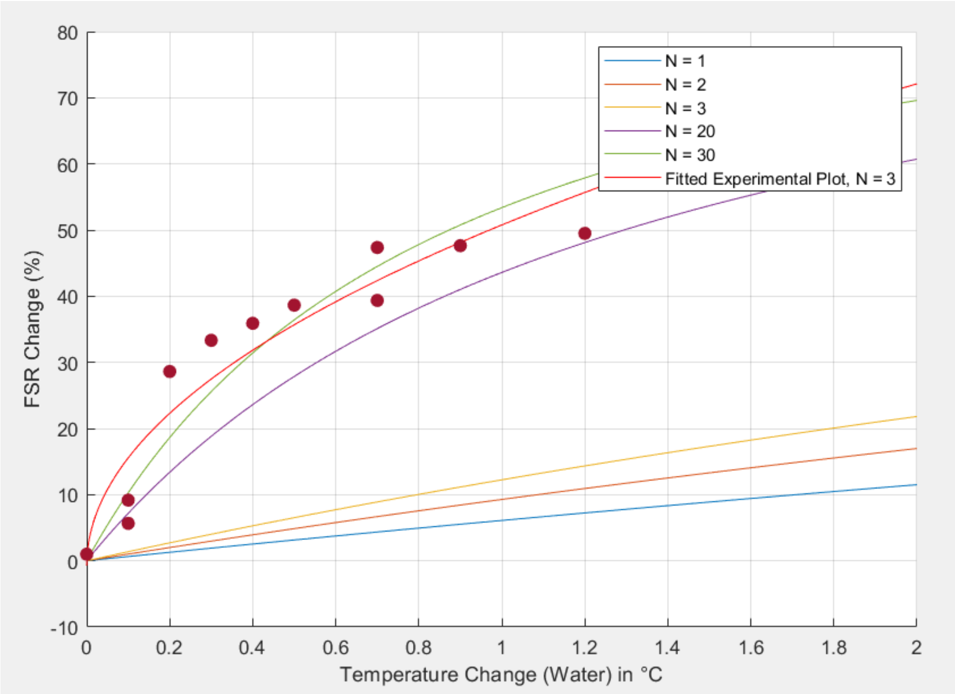
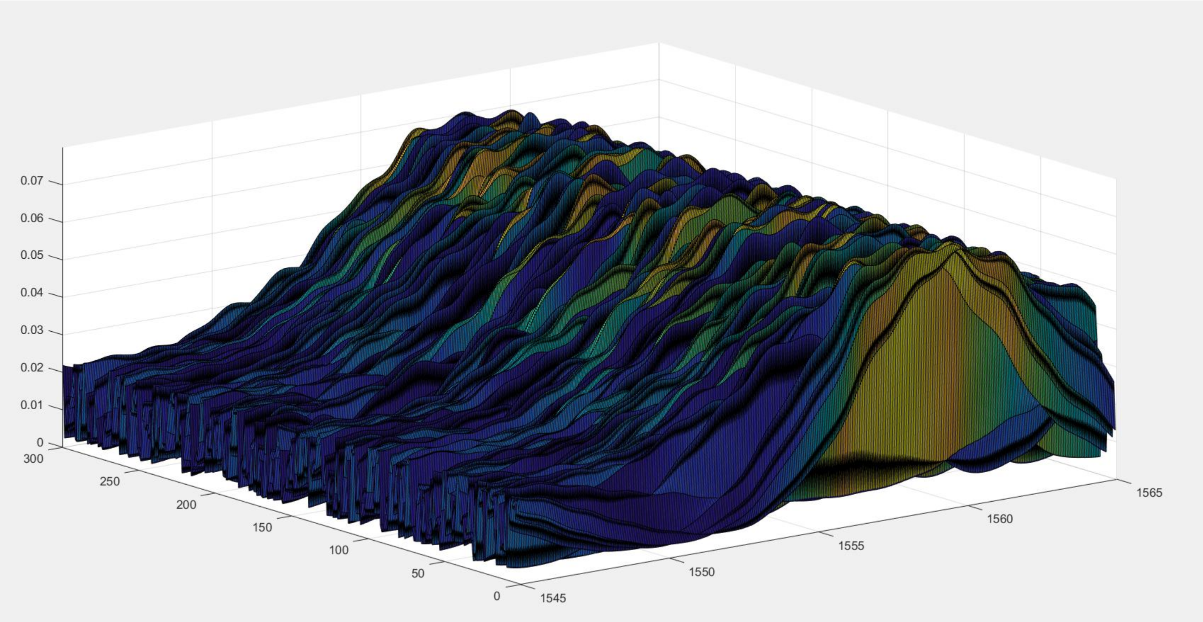


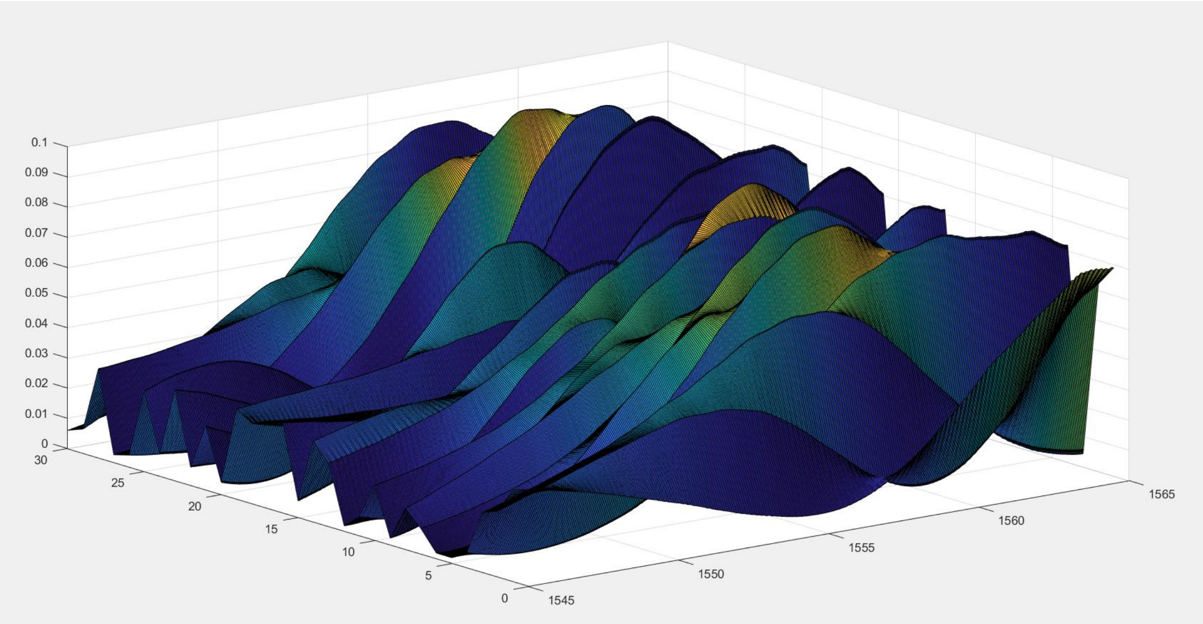
Figure x. Fitted experimental data to a square root curve.



Modelling of Heat Transfer

The 3D surface of the collected data (wavelength, #Reading, Power) when temperature is varied





Critical Analysis and Future Work

What worked well, What did not work and why, How you would improve it next time

References

[1]. Saleh and Malvin Carl Teich, *Fundamentals of Photonics*. John Wiley & Sons, 2019.

[2]. Lukas Chrostowski and M. Hochberg, *Silicon photonics design: from devices to systems*. Cambridge: Cambridge University Press, 2016.

....

Appendices

Appendix A – Interfacing the Laptop with the Spectrum Analyzer

To export the spectrum analyser measurements for analysis, the laptop was interfaced with the Yokogawa AQ6370C spectrum analyser using a NI GPIB-USB-HS adapter. This was accomplished through the following steps:

1. Connect the NI GPIB-USB-HS to the laptop USB port and to the AQ6370C's **GP-IB1** port.
2. Install NI-488.2
3. Install NI-VISA
4. Install NI Measurement & Automation Explorer (NI-MAX)
5. Restart the laptop
6. On the OSA front panel, set the "REMOTE INTERFACE" as "GP-IB" (Figure A1).

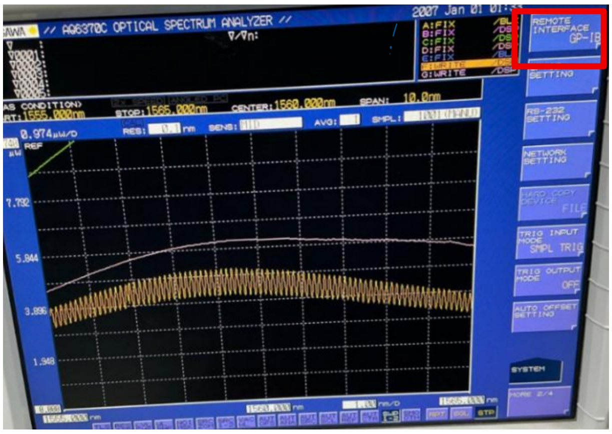


Figure A1. YOKOGAWA AQ6370C spectrum analyser front panel, showing Remote sensing set to GP-IB

7. To verify that the laptop is detecting the NI GPIB-USB-HS adapter, open *Device Manager*, the adapter should be shown under "NI GPIB Interfaces" (Figure A2).

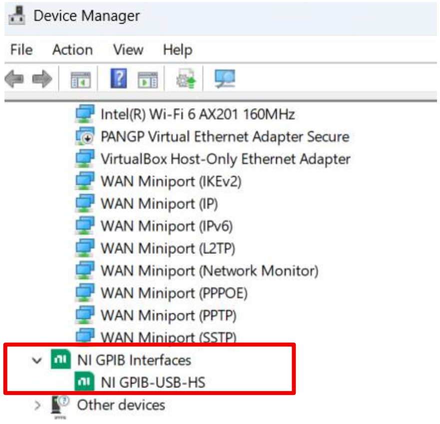


Figure A2. Device Manager window, showing the NI adapter under NI GPIB Interfaces.

8. In the list of applications, search NI MAX, right click, and run as administrator.
9. In the right side of the "My System – NI MAX", expand the "Devices and Interfaces", and the adapter should be successfully detected as shown in Figure A3.

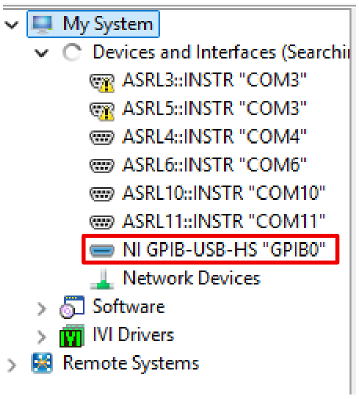


Figure A3. NI MAX window, showing the adapter under Devices and Interfaces.

10. Right click on the NI adapter option, "NI GPIB-USB-HS "GPIB0"""", and then select "Scan for Devices". The spectrum analyser, Yokogawa AQ6370C, should appear under GPIB0, asshown in Figure A4. The 1 in "GPIB0::1::INST" indicates the GPIB address set in the OSA front panel. This successfully confirms the detection of the spectrum analyser.
11. To communicate with the Yokogawa AQ6370C via Python, install Python from https://www.python.org/downloads/
12. On Windows PowerShell, run the following line of code to create a folder named "aq6370c_logger".

```
mkdir aq6370c_logger
```

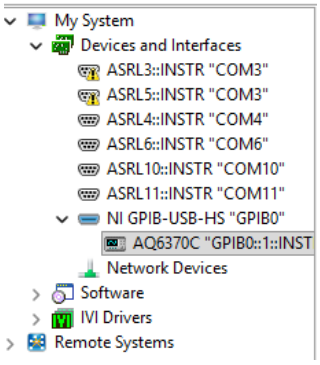


Figure A4. NI MAX window, showing Yokogawa AQ6370C under GPIB0

13. To move the created files and virtual environments into the created folder, run the following code:

```
cd aq6370c_logger
```

14. Create a virtual environment using the following line:

```
python -m venv .venv
```

15. If faced with an error when running the previous line, run the following line of code, allowing the running of the code as a normal user not administrator:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

16. To activate the virtual environment, enter:

```
.\.venv\Scripts\activate
```

17. Install the following libraries using the following code:

```
pip install pyvisa numpy pandas
```

18. To create a file named “test”, execute:

```
cd C:\Users\hp\aq6370c_logger
notepad test.py
```

19. This will open a notepad page, paste the code given in Appendix B.
20. Change the “ADDR = GPIB0::**1**::INSTR”, referring to the GPI0 instrument address (**1**) as applicable. Also, edit the logged trace as required through the changing the variable “TRACE_SCPI = "TRG"”.
21. The code records the spectral readings of the selected trace at a regular interval. Change the total number of acquired readings through “N_READINGS”, and the time interval between each recording through “INTERVAL_S”.
22. Save the file and close the notepad page.
23. To run the code, run the following line of code:

```
python test.py
```

24. At the end of running the code, two csv files are saved, one with the spectral measurement’s readings, and the timestamps corresponding to each spectral measurement.

What needs to be done:

Take the fiber , heat it , track temp using coupler

Let frequency be very high,

Do it multiple of times, do we see consistency in placement